ZEREF SPARK — Assembly Guide

Chibi companion robot on a floating pedestal with a crystal-reading station · v0.4 (all decisions made, firmware zs-0.4.0)

DESIGN TARGET — not printed or test-fitted | CONCEPT DESIGN / NOT BUILT HARDWARE

System logic, architecture and concept: Cory Davis (NavisWORLD) — Reality Bridge / Beast Box (“the brain lives on the host; models are mouths”). CAD geometry, renders and this guide were **AI-generated** from Cory's logic (CadQuery). Parts marked **PX** are **third-party modules** shown as simplified proxy shapes — measure your purchased boards. Height target 314 mm (robot) · ≈371 mm with pedestal. Every dimension and electrical value marked ESTIMATE is unverified.

1. Safety — read before you start

Adult supervision is required for every soldering, LiPo, heat-set and UV step.

LiPo battery (BT1)

- Use only a **protected** 1S cell (PCM board under the wrap). Never puncture, bend, crush or screw into it; mount with foam tape only.
- Charge only through the PowerBoost (U1) — never a bare supply. Never charge unattended or on a flammable surface; a LiPo-safe bag is recommended for first charges.
- Check polarity of the JST-PH lead before connecting (no industry standard). A reversed cell can destroy U1 instantly.
- Swollen, hot, hissing or smelling cell → disconnect (if safe), move outside onto sand/concrete, do not reuse.
- Keep F2 (optional) and the cell PCM in place; never bypass them.

UV-A 365 nm (D5)

- 365 nm is barely visible, so your eyes don't blink or look away from it. **Never look at the UV LED, lit or not,** and never run it with the shield off. The reed interlock cuts its current in hardware, and firmware also requires shield sense + AS7341 ACK.
- Firmware limits: UV on ≤ 2 s per reading, ≥ 5 s off between readings, a watchdog forces it off, and it stays off at boot.
- For bench work with the shield off, power the UV LED **only** through a current-limited supply ≤ 20 mA, wear UV-blocking safety glasses (EN 166 / ANSI Z87 + UV400), and keep skin out of the beam.
- Check the LED's datasheet photobiological risk group (IEC 62471). Use a low-power 20 mA part, not a high-power "UV torch" LED.
- Keep the station out of reach of young children. Cory's family: Caleb should not be able to open the shield while the robot is powered. Use a screw-retained shield if needed (open decision).
- Fluorescence work is low-power only. Don't use the station to cure resin or sterilise anything.

Soldering

- Ventilate / use a fume extractor; lead-free solder recommended; wash hands after.
- Iron rests in its stand; the tip is ~350 °C. Wear eye protection when clipping leads.
- Never solder on a powered circuit or with the LiPo connected.

Heat-set inserts

- Insert tip ≈ 220 °C (PLA) / 245 °C (PETG). The brass stays hot for over a minute — use tweezers, never fingers.
- Push straight; if an insert tilts, reheat and correct immediately.

Electrical

- USB-C supply: 5 V / 3 A. Do not connect the DevKit's own USB 5 V to the robot while 5V_SYS is powered (back-feed). The USB pigtail carries D+/D−/GND only.

- Before first power-on of each stage, measure 5V↔GND and 3V3↔GND resistance: > 1 kΩ (after caps charge).

2. Tools and consumables

Tool	Used for
FDM printer (≥ 200 × 200 mm bed), PETG + PLA (+ clear resin optional)	all printed parts
Temperature-controlled soldering iron, fine tip, lead-free solder, flux, desoldering braid	wiring, passives
Heat-set insert tip for M3	ZS-F01 inserts
Multimeter (V, Ω, continuity, diode test)	every check
Hex/Phillips drivers (M2, M2.5, M3), small pliers, flush cutters	fasteners
Wire strippers 22–30 AWG, crimp tool for JST-PH / JST-SH (or pre-crimped leads)	harness
Heat-shrink 1.5–6 mm, Kapton tape, foam tape, hot-glue gun, CA glue, clear silicone	insulation, mounting
Hobby knife, files, 400–1000 grit paper, isopropyl alcohol	cleanup, window cleaning
UV-blocking safety glasses (UV400, EN 166 / ANSI Z87)	any bench UV work
LiPo-safe bag, fume extractor	safety
Computer with VS Code + PlatformIO, USB-C data cable (RB-009), 5 V / 3 A USB-C supply	firmware, power
PTFE white reference disc (ZS-S12) and dark plug ZS-605	calibration

3. Bill of materials (complete)

Mechanical, module and printed rows first, then every electrical/passive part (v0.3). Full electrical detail: `docs/BOM_electrical.csv`.

ID	Part	Qty	Type	Material / source	Location in robot	Function	Status	Notes
RB-001	ESP32-S3 DevKitC-1 or compatible	1	Purchased (Reality Bridge)	Espressif or compatible	Torso – back face of ZS-303 cradle (USB end down)	MCU / Wi-Fi / USB – the nervous system	Required	Proxy 26x14x69 mm ESTIMATE incl. headers; verify the purchased board. Retained by rails + 2 zip ties
RB-002	ICM-42688-P breakout	1	Purchased (Reality Bridge)	breakout	Torso – ZS-304 sensor sled (z~100 near body centre)	6-axis IMU (motion / tilt / tap)	Required	Proxy 23x18x4 ESTIMATE
RB-003	BH1750 breakout	1	Purchased (Reality Bridge)	breakout	Right ear pod (ZS-202) behind 8 mm window in ZS-205	Ambient light	Required	Proxy 18x16x4 ESTIMATE; LED-ring light leak – firmware must sample with pod LEDs off
RB-004	BME280 breakout	1	Purchased (Reality Bridge)	breakout	Torso – lower sensor sled behind front belly vents	Temp / humidity / pressure	Required	Kept away from ESP32 heat; readings will still be biased by internal heat (ESTIMATE)
RB-005	MAX30102 breakout	1	Purchased (Reality Bridge)	breakout	Torso – behind chest core centre (looks out through ZS-307)	Optical waveform – NOT MEDICAL	Optional	Proxy 22x18x5 ESTIMATE; 11 mm light pipe may be too long for a usable signal – open decision
RB-006	INMP441 breakout	1	Purchased (Reality Bridge)	breakout	Left ear pod (ZS-201) behind mic holes in ZS-204	I2S microphone	Required	Proxy 23x15x4 ESTIMATE; L/R pin to GND
RB-007	NEO-M9N-class GNSS breakout	1	Purchased (Reality Bridge)	breakout	Head – back face of ZS-103 near crown	GNSS	Optional	Proxy 30x25x6 ESTIMATE; indoor reception will be poor
RB-008	M3 x 8 mm screws	8	Fastener (Reality Bridge)	steel	4 torso split bosses + 4 head split bosses	Close shells (clamp bulkheads + captured pegs)	Reference	Used through back-half clearance bosses into front-half heat-set inserts

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RB-009	USB-C data/power cable	1	Purchased (Reality Bridge)		Pedestal USB-C port to host / 5 V supply	Power + data to Beast Box host	Required	Use a 5 V / 3 A capable cable + supply
ZS-101	Head shell front	1	Printed	PETG or PLA matte navy	Head	Helmet front + visor ledge + accent-pipe holes	Design target	4 x M3 heat-set inserts
ZS-102	Head shell back	1	Printed	PETG or PLA matte navy	Head	Helmet back + speaker grille/ring	Design target	
ZS-103	Head bulkhead	1	Printed	PETG	Head split plane	Mount plate: face carrier / amp / GNSS / optional LiPo	Design target	Clamped between head shells
ZS-104	Visor	1	Printed or thermoformed	smoked translucent PETG (or 1.5 mm smoked PETG sheet vacuum-formed)	Head front	Black glossy face; eyes glow through	Design target	Glue / silicone onto ZS-101 ledge
ZS-105	Face carrier	1	Printed	PETG black	Behind visor	Holds eye LED strips + light pipes	Design target	2 x 2.5 mm PT screws into ZS-103
ZS-106	Eye light pipe (happy arc)	2	Printed	clear PETG / clear resin	Behind visor	Cyan happy-closed-eye arcs	Design target	
ZS-107	Accent light pipe	2	Printed	clear resin	Top of head	Magenta accent lights	Design target	Flange retained inside head
ZS-108	Neck coupler	1	Printed	PETG	Neck	Captured between both shell pairs; 14 mm harness bore	Design target	
ZS-201	Ear pod L (mic)	1	Printed	PETG matte navy	Left side of head	Houses INMP441 + LED ring; flanged peg captured by head split	Design target	
ZS-202	Ear pod R (light)	1	Printed	PETG matte navy	Right side of head	Houses BH1750 + LED ring	Design target	
ZS-203	Pod rim diffuser	2	Printed	translucent white/violet PETG	Pod outer rim	Violet rim light	Design target	
ZS-204	Pod cap L (mic holes)	1	Printed	PETG gloss black	Left pod face	Mic port array	Design target	Snap/glue into rim
ZS-205	Pod cap R (light window)	1	Printed	PETG gloss black + clear window	Right pod face	BH1750 window	Design target	
ZS-206	Ear (rabbit)	2	Printed	PETG matte navy	On top of pods	Tall ears with inner-face slot row	Design target	1 x M3 x 10 self-tap each from inside pod
ZS-301	Torso shell front	1	Printed	PETG matte navy	Torso	Egg front + chest hole + BME vents	Design target	4 x M3 heat-set inserts

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ZS-302	Torso shell back	1	Printed	PETG matte navy	Torso	Egg back + ESP32 vents	Design target	
ZS-303	Torso bulkhead / ESP32 cradle	1	Printed	PETG	Torso split plane	ESP32 rails on back; sensor sled on front	Design target	
ZS-304	Sensor sled	1	Printed	PETG	Torso front of bulkhead	IMU + BME280 carrier	Design target	2 x M2 x 6 self-tap
ZS-305	Chest core housing	1	Printed	PETG black	Chest	Cup + optical baffle; LED ring and diffuser load from front	Design target	
ZS-306	Chest ring diffuser	1	Printed	translucent white PETG	Chest	Cyan ring glow	Design target	Press/glue
ZS-307	Chest centre lens	1	Printed	clear (if MAX30102) or violet translucent	Chest centre	Magenta/violet centre / optical window	Design target	
ZS-401	Arm	2	Printed	PETG matte navy	Shoulders	Stubby arm + mitten hand; flanged peg captured by torso split	Design target	Static (no servo) in this revision
ZS-402	Leg	2	Printed	PETG matte navy	Hips	Stubby leg + rounded foot; flanged peg captured	Design target	Static
ZS-501	Pedestal (rock hex) v0.2	1	Printed	PLA dark grey/black 15% infill (station walls 100%)	Base	Floating base + crystal station cup / window seat / groove / pogo + reed pockets / driver bay / harness channels	Design target	Now 190 x 164.5 x 45 mm (ESTIMATE); 3 x 16 mm felt pads
ZS-502	Support tube	1	Purchased + cut	clear acrylic tube 12 OD x 8 ID x 75 mm	Pedestal to torso	Floating support + 4-wire USB harness	Design target	Stiffness unverified – see open decisions
ZS-E01	WS2812B 12-LED ring (~37 mm OD)	3	Purchased		Chest + both pods	Cyan chest ring / violet pod rims	Proposed	OD 37 / ID 23 ESTIMATE – verify ring you buy
ZS-E02	WS2812B strip segment (5 LEDs)	2	Purchased		Face carrier behind eye pipes	Eye glow	Proposed	
ZS-E03	WS2812B 2-LED accent board	1	Purchased/custom		Head crown inside	Magenta accents	Proposed	
ZS-E04	74AHCT125 (or SN74AHCT1G125) level shifter	1	Purchased		Torso harness	3.3 V -> 5 V LED data	Proposed	
ZS-E05	MAX98357A I2S amp breakout	1	Purchased		Head bulkhead back	Speaker amp	Proposed	Needs firmware addition (I2S1)

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ZS-E06	28 mm 8 ohm ~1 W speaker	1	Purchased		Head back shell ring	Voice / sounds from Beast Box host	Proposed	28 x 6 mm ESTIMATE
ZS-E07	USB-C receptacle breakout (with 5.1 k CC pull-downs)	1	Purchased		Pedestal back pocket	Power + data entry	Proposed	
ZS-F01	M3 heat-set insert (short)	8	Fastener	brass	Front shell bosses		Proposed	4.0 mm pilot ESTIMATE – match your insert
ZS-F02	M3 x 10 self-tapping (plastic)	2	Fastener		Ears to pods		Proposed	
ZS-F03	M2 x 6 self-tapping	2	Fastener		Sensor sled		Proposed	
ZS-F04	2.5 mm PT screw x 12	2	Fastener		Face carrier posts		Proposed	
ZS-F05	Zip tie 2.5 mm	2	Fastener		ESP32 cradle		Proposed	
ZS-W01	Silicone wire 28 AWG assorted + JST-SH/PH connectors	1 set	Wiring		Neck + pods + tube	Harness	Proposed	~14 conductors through 14 mm neck bore
ZS-109	Battery tray	1	Printed	PETG	Head bulkhead back	Holds 503450 LiPo (slides in from top)	Design target	2 x M2 x 6 into ZS-103
ZS-B01	1S LiPo 503450 ~1000 mAh WITH protection PCM (DW01-class)	1	Purchased		ZS-109 tray	Untethered power	Approved	503450 = 5.0 x 34 x 50 mm nominal; must have built-in protection (over-charge / over-discharge / over-current / short); JST-PH 2-pin matching PowerBoost polarity – CHECK polarity
ZS-B02	Adafruit PowerBoost 1000C (MCP73871 charger + load-sharing power path + TPS61090 5.2 V boost)	1	Purchased	Adafruit #2465	Head bulkhead back (x 12..48 z 158..181)	Charge + power path + 5 V boost	Approved	~36 x 23 x 5 mm without USB-A (ESTIMATE; Adafruit lists 23 x 45 x 10 with USB-A); 1 A boost output limit; MCP73871 datasheet DS20002090; 2 x M2.5/M2 screws
ZS-B03	SS12D00-class slide switch	1	Purchased		Head back shell slot (x~30 z~176)	PowerBoost EN to GND = OFF	Approved	Signal-level switch only (EN pin)
ZS-B04	Resistors 100 k + 100 k (0.1%) + 100 nF	1 set	Purchased		Torso harness	Battery voltage divider to GPIO5 (ADC1_CH4)	Approved	
ZS-601	Station LED plug	1	Printed	white PETG (mixing cavity white or PTFE-tape lined)	Pedestal underside at station	Holds white / 365 nm / 940 nm LED board under window	Design target	2 x M3 x 8 self-tap

ID	Part	Qty	Type	Material / source	Location in robot	Function	Status	Notes
ZS-602	Light shield (removable)	1	Printed	opaque black PETG or PLA (check with torch for pinholes; paint inside matte black)	Over crystal cup	Ambient shade + carries AS7341; skirt in groove = light trap	Design target	Lift 5 mm then slide forward (removal path checked: 4.2 mm to torso)
ZS-603	Sensor mount disc	1	Printed	matte black PETG	Inside shield top	AS7341 board + 12 mm aperture + optional 20 mm UV-cut filter recess	Design target	2 x M2 x 5
ZS-604	Driver bay cover	1	Printed	PLA	Pedestal bottom	Closes MOSFET driver bay	Design target	2 x M2 x 6 self-tap
ZS-605	Dark-reference plug	1	Printed	opaque black PLA (100% infill)	Crystal cup (instead of crystal)	Stray-light / dark-reference check	Design target	
ZS-S01	AS7341 11-channel spectral sensor breakout (Adafruit #4698 or equivalent)	1	Purchased	ams-OSRAM AS7341 datasheet DS000504	ZS-603 in shield looking down	Crystal transmission / fluorescence raw channel counts F1-F8 (415-680 nm) + Clear + NIR (910 nm) + flicker	Approved	I2C 0x39 (fixed); board ~25.6 x 17.8 mm ESTIMATE; alternative AS7343 (14 ch) – verify its address and register map in its own ams-OSRAM datasheet before swapping
ZS-S02	Borosilicate (or fused-silica) window disc 29.6 x 2 mm	1	Purchased		Cup bottom window seat	Passes 365 nm UV + visible + 940 nm	Approved	Do NOT use standard acrylic – it absorbs much of 365 nm
ZS-S03	UV-cut longpass filter disc 20 x 2 mm (e.g. ~420 nm longpass or UV-blocking acrylic)	1	Purchased		ZS-603 recess	Blocks 365 nm leak into sensor in fluorescence mode	Optional	Remove for transmission mode
ZS-S04	White LED 3.5 x 3.5 mm (high-CRI preferred)	1	Purchased		Station LED board	Transmission source	Approved	~20 mA (ESTIMATE – set by resistor)
ZS-S05	365 nm UV-A LED 3.5 x 3.5 mm low-power (~20 mA)	1	Purchased		Station LED board	Fluorescence excitation	Approved	Check LED datasheet photobiological risk group (IEC 62471); see UV safety in ASSEMBLY.md
ZS-S06	940 nm IR LED 3.5 x 3.5 mm	1	Purchased		Station LED board	NIR transmission	Optional	
ZS-S07	Station LED board 20 x 20 mm (perfboard or custom)	1	Purchased/custom		ZS-601 pocket	Carries ZS-S04/05/06	Proposed	

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ZS-S08	Driver board: 3 x AO3400A logic-level N-MOSFET + 3 x 100 R gate + 3 x 100 k pull-down + LED resistors (white 100 R / UV 68 R / IR 180 R at 5 V ESTIMATE)	1	Purchased/custom		Pedestal driver bay	Low-side LED switches from GPIO 1 / 2 / 21 (v0.3)	Approved	Recompute resistors from the real LED Vf
ZS-S09	Glass reed switch ~12 mm (normally open)	1	Purchased		Pedestal reed slot (x -32)	UV HARDWARE interlock (in series with UV LED) + shield sense	Approved	Glass – handle carefully
ZS-S10	Neodymium magnet 6 x 2 mm	1	Purchased		Shield flange pocket above reed	Closes reed only when shield is seated	Approved	
ZS-S11	4-pin magnetic pogo connector pair (receptacle + pads)	1	Purchased		Pedestal pocket (x 29..34) / shield flange	AS7341 VIN GND SDA SCL to removable shield	Approved	~2.54 mm pitch ESTIMATE – pocket sized 5 x 8 mm
ZS-S12	PTFE white reference disc 25.4 x 3 mm (virgin PTFE sheet)	1	Purchased		Crystal cup (calibration only)	White / diffuse-transmission reference	Approved	Keep clean; store in bag
ZS-W02	30 AWG silicone wire (11 conductors through tube) + twisted D+/D- pair	1 set	Wiring		ZS-502 tube	USB + station harness	Proposed	
J1	USB-C 16-pin receptacle breakout (5.1k CC pull-downs on board)	1	Electrical / passive (v0.3 electrical)	Connector breakout 5 V / 3 A —	Pedestal rear	CC 5.1k = [OB]; only VBUS/GND/D+/D− used	ADD (module)	Sheet A — see docs/BOM_electrical.csv & docs/schematics/
D1	USBLC6-2SC6	1	Electrical / passive (v0.3 electrical)	ESD array SOT-23-6 5.25 V working —	Pedestal, next to J1	on D+/D−; VBUS pin to VBUS	ADD	Sheet A — see docs/BOM_electrical.csv & docs/schematics/
D2	SMBJ5.0A	1	Electrical / passive (v0.3 electrical)	TVS diode SMB (DO-214AA) 5.0 V standoff, 600 W pk —	Pedestal, after J1	clamps hot-plug spikes on VBUS	ADD	Sheet A — see docs/BOM_electrical.csv & docs/schematics/
F1	PTC resettable fuse 2.0 A hold / ~4 A trip	1	Electrical / passive (v0.3 electrical)	Polyfuse 1812 6 V —	Pedestal	VBUS peak ≈1.87 A (charger limit 1.8 A + station LEDs 60 mA)	ADD · ESTIMATE	Sheet A — see docs/BOM_electrical.csv & docs/schematics/
C1	10 µF	1	Electrical / passive (v0.3 electrical)	MLCC X5R 0805 10 V ±10%	Pedestal after F1	VBUS local decoupling	ADD	Sheet A — see docs/BOM_electrical.csv & docs/schematics/

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C2	47 µF	1	Electrical / passive (v0.3 electrical)	Polymer / low-ESR electrolytic radial 6.3 mm 10 V ±20%	Head, at U1 USB pad	absorbs cable inductance at charger input	ADD	Sheet A — see docs/BOM_electrical.csv & docs/schematics/
F2	PTC 2.0 A hold	1	Electrical / passive (v0.3 electrical)	Polyfuse 1812 6 V —	Battery lead	backup to cell PCM	ADD (optional) · ESTIMATE	Sheet A — see docs/BOM_electrical.csv & docs/schematics/
SW1	SS12D00 SPDT slide switch	1	Electrical / passive (v0.3 electrical)	Switch through-hole 2.54 mm 50 V / 0.5 A —	Head rear	EN to GND = OFF (switch closed = OFF). Charging still works when OFF.	ADD	Sheet A — see docs/BOM_electrical.csv & docs/schematics/
C3	220 µF	1	Electrical / passive (v0.3 electrical)	Polymer / low-ESR electrolytic radial 6.3 mm 10 V ±20%	At U1 5V out	boost output reservoir for LED + amp peaks	ADD	Sheet A — see docs/BOM_electrical.csv & docs/schematics/
C4	100 nF	1	Electrical / passive (v0.3 electrical)	MLCC X7R 0603 16 V ±10%	At U1 5V out	HF decoupling	ADD	Sheet A — see docs/BOM_electrical.csv & docs/schematics/
C6	10 µF	1	Electrical / passive (v0.3 electrical)	MLCC X5R 0805 10 V ±10%	At U0 5V pin	DevKit input decoupling	ADD	Sheet A — see docs/BOM_electrical.csv & docs/schematics/
R1	100 kΩ	1	Electrical / passive (v0.3 electrical)	Thick film 0603 0.1 W ±1%	Torso harness	divider top: VBAT → VBAT_SENSE	ADD	Sheet A — see docs/BOM_electrical.csv & docs/schematics/
R2	100 kΩ	1	Electrical / passive (v0.3 electrical)	Thick film 0603 0.1 W ±1%	Torso harness	Vadc = VBAT·R2/(R1+R2) = VBAT/2 → 1.50–2.10 V; I = 4.2/200k = 21 µA	ADD	Sheet A — see docs/BOM_electrical.csv & docs/schematics/
C5	100 nF	1	Electrical / passive (v0.3 electrical)	MLCC X7R 0603 16 V ±10%	At GPIO5	ADC sample-hold reservoir; τ = 50k·100n = 5 ms	ADD	Sheet A — see docs/BOM_electrical.csv & docs/schematics/
R3	2.2 kΩ (DNP if breakouts' parallel pull-ups ≤ 3.3 kΩ)	1	Electrical / passive (v0.3 electrical)	Thick film 0603 0.1 W ±1%	Torso sled	SDA0 pull-up; tr = 0.847·R·C = 0.847·2.2k·100pF = 186 ns ≤ 300 ns (400 kHz)	ADD / DNP	Sheet B — see docs/BOM_electrical.csv & docs/schematics/
R4	2.2 kΩ (DNP rule as R3)	1	Electrical / passive (v0.3 electrical)	Thick film 0603 0.1 W ±1%	Torso sled	SCL0 pull-up; sink current 3.3/2.2k = 1.5 mA ≤ 3 mA	ADD / DNP	Sheet B — see docs/BOM_electrical.csv & docs/schematics/
C7	100 nF at U3 IMU	1	Electrical / passive (v0.3 electrical)	MLCC X7R 0603 16 V ±10%	U3	add only if breakout lacks it	OB (usually) / ADD	Sheet B — see docs/BOM_electrical.csv & docs/schematics/
C8	100 nF at U4 BME280	1	Electrical / passive (v0.3 electrical)	MLCC X7R 0603 16 V ±10%	U4		OB / ADD	Sheet B — see docs/BOM_electrical.csv & docs/schematics/
C9	100 nF at U5 BH1750	1	Electrical / passive (v0.3 electrical)	MLCC X7R 0603 16 V ±10%	U5		OB / ADD	Sheet B — see docs/BOM_electrical.csv & docs/schematics/

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C10	100 nF at U6 MAX30102	1	Electrical / passive (v0.3 electrical)	MLCC X7R 0603 16 V ±10%	U6	MAX30102 LED pulses need local cap	OB / ADD	Sheet B — see docs/BOM_electrical.csv & docs/schematics/
C11	10 µF at U7 GNSS	1	Electrical / passive (v0.3 electrical)	MLCC X5R 0805 10 V ±10%	U7	GNSS acquisition current bursts	ADD (optional)	Sheet B — see docs/BOM_electrical.csv & docs/schematics/
R5	33 Ω	1	Electrical / passive (v0.3 electrical)	Thick film 0603 0.1 W ±1%	Near U0	I2S0 BCLK series damping (long pod run)	ADD (optional)	Sheet C — see docs/BOM_electrical.csv & docs/schematics/
R6	33 Ω	1	Electrical / passive (v0.3 electrical)	Thick film 0603 0.1 W ±1%	Near U0	I2S0 WS series damping	ADD (optional)	Sheet C — see docs/BOM_electrical.csv & docs/schematics/
R7	22 kΩ	1	Electrical / passive (v0.3 electrical)	Thick film 0603 0.1 W ±1%	At U9 SD	boot-mute pull-down: V = 5·22k/(1M+22k) = 0.108 V below 0.16 V shutdown threshold (assumes 1M [OB] pull-up — verify on your board)	ADD · ESTIMATE	Sheet C — see docs/BOM_electrical.csv & docs/schematics/
C12	100 µF	1	Electrical / passive (v0.3 electrical)	Aluminium electrolytic radial 5 mm 10 V ±20%	At U9 VIN	class-D supply reservoir	ADD	Sheet C — see docs/BOM_electrical.csv & docs/schematics/
C13	10 µF	1	Electrical / passive (v0.3 electrical)	MLCC X5R 0805 10 V ±10%	At U9 VIN		ADD	Sheet C — see docs/BOM_electrical.csv & docs/schematics/
U2	SN74AHCT1G125 (or 1 gate of 74AHCT125)	1	Electrical / passive (v0.3 electrical)	Logic buffer SOT-23-5 / DIP-14 VCC 5 V —	Torso	3.3 V → 5 V level shift (VIH = 2.0 V at 5 V)	ADD	Sheet D — see docs/BOM_electrical.csv & docs/schematics/
C14	100 nF	1	Electrical / passive (v0.3 electrical)	MLCC X7R 0603 16 V ±10%	At U2		ADD	Sheet D — see docs/BOM_electrical.csv & docs/schematics/
R8	330 Ω (300–500 Ω range)	1	Electrical / passive (v0.3 electrical)	Thick film 0603 0.1 W ±1%	At first LED DIN	damps ringing / protects first WS2812 input	ADD	Sheet D — see docs/BOM_electrical.csv & docs/schematics/
C15	1000 µF	1	Electrical / passive (v0.3 electrical)	Aluminium electrolytic radial 8 mm 10 V ±20%	Chest ring 5V/GND	LED bulk — inrush & surge protection	ADD	Sheet D — see docs/BOM_electrical.csv & docs/schematics/
C16	100 µF	1	Electrical / passive (v0.3 electrical)	Aluminium electrolytic radial 5 mm 10 V ±20%	Pod ring L		ADD	Sheet D — see docs/BOM_electrical.csv & docs/schematics/
C17	100 µF	1	Electrical / passive (v0.3 electrical)	Aluminium electrolytic radial 5 mm 10 V ±20%	Pod ring R		ADD	Sheet D — see docs/BOM_electrical.csv & docs/schematics/

ID	Part	Qty	Type	Material / source	Location in robot	Function	Status	Notes
C18	100 µF	1	Electrical / passive (v0.3 electrical)	Aluminium electrolytic radial 5 mm 10 V ±20%	Eye boards		ADD	Sheet D — see docs/BOM_electrical.csv & docs/schematics/
LED1–48	WS2812B (100 nF per LED on ring PCBs)	48	Electrical / passive (v0.3 electrical)	Smart LED 5050 5 V, 60 mA max each —	Chest 12, eyes 2×5, pods 2×12, crown 2	48·60 mA·8% brightness cap = 0.23 A peak (cap 10→8 % on 2026-11-09	D1)	OB (ring modules) · ESTIMATE
J2	4-pin magnetic pogo pair	1	Electrical / passive (v0.3 electrical)	Connector 2.54 mm pitch 2 A per pin (typ.) —	Pedestal ↔ shield	VBUS, GND, SDA1_P, SCL1_P	ADD · ESTIMATE	Sheet E — see docs/BOM_electrical.csv & docs/schematics/
R9	4.7 kΩ	1	Electrical / passive (v0.3 electrical)	Thick film 0603 0.1 W ±1%	Pedestal driver bay	SDA1 pull-up; in parallel with [OB] 10k → 3.2k	ADD	Sheet E — see docs/BOM_electrical.csv & docs/schematics/
R10	4.7 kΩ	1	Electrical / passive (v0.3 electrical)	Thick film 0603 0.1 W ±1%	Pedestal driver bay	SCL1 pull-up	ADD	Sheet E — see docs/BOM_electrical.csv & docs/schematics/
R11	100 Ω	1	Electrical / passive (v0.3 electrical)	Thick film 0603 0.1 W ±1%	Pedestal	SDA1 series (hot-plug current limit)	ADD	Sheet E — see docs/BOM_electrical.csv & docs/schematics/
R12	100 Ω	1	Electrical / passive (v0.3 electrical)	Thick film 0603 0.1 W ±1%	Pedestal	SCL1 series	ADD	Sheet E — see docs/BOM_electrical.csv & docs/schematics/
R13	10 kΩ (DNP if board has INT pull-up)	1	Electrical / passive (v0.3 electrical)	Thick film 0603 0.1 W ±1%	Pedestal	AS_INT open-drain pull-up	ADD / DNP	Sheet E — see docs/BOM_electrical.csv & docs/schematics/
D3	TPD4E05U06	1	Electrical / passive (v0.3 electrical)	ESD array USON-10 / SOT 5.5 V working —	At J2	ESD on exposed pogo pins	ADD	Sheet E — see docs/BOM_electrical.csv & docs/schematics/
C19	10 µF	1	Electrical / passive (v0.3 electrical)	MLCC X5R 0805 10 V ±10%	At J2 (pedestal)	pogo VIN bulk	ADD	Sheet E — see docs/BOM_electrical.csv & docs/schematics/
C20	100 nF	1	Electrical / passive (v0.3 electrical)	MLCC X7R 0603 16 V ±10%	At J2		ADD	Sheet E — see docs/BOM_electrical.csv & docs/schematics/
Q1	AO3400A (white)	1	Electrical / passive (v0.3 electrical)	N-MOSFET logic level SOT-23 30 V, 5.7 A, Rds 33 mΩ @ 4.5 V —	Driver board ZS-S08	Vgs(th) ≤ 1.45 V → fully on at 3.3 V for 20 mA	ADD	Sheet E — see docs/BOM_electrical.csv & docs/schematics/
Q2	AO3400A (UV)	1	Electrical / passive (v0.3 electrical)	N-MOSFET SOT-23 30 V —	Driver board		ADD	Sheet E — see docs/BOM_electrical.csv & docs/schematics/
Q3	AO3400A (IR, optional)	1	Electrical / passive (v0.3 electrical)	N-MOSFET SOT-23 30 V —	Driver board	gate on GPIO21 (v0.3)	ADD	Sheet E — see docs/BOM_electrical.csv & docs/schematics/
R14	100 Ω	1	Electrical / passive (v0.3 electrical)	Thick film 0603 0.1 W ±1%	Driver board	Q1 gate series	ADD	Sheet E — see docs/BOM_electrical.csv & docs/schematics/

ID	Part	Qty	Type	Material / source	Location in robot	Function	Status	Notes
R15	100 Ω	1	Electrical / passive (v0.3 electrical)	Thick film 0603 0.1 W ±1%	Driver board	Q2 gate series	ADD	Sheet E — see docs/BOM_electrical.csv & docs/schematics/
R16	100 Ω	1	Electrical / passive (v0.3 electrical)	Thick film 0603 0.1 W ±1%	Driver board	Q3 gate series	ADD	Sheet E — see docs/BOM_electrical.csv & docs/schematics/
R17	100 kΩ	1	Electrical / passive (v0.3 electrical)	Thick film 0603 0.1 W ±1%	Driver board	Q1 gate pull-down (LED OFF at boot / floating)	ADD	Sheet E — see docs/BOM_electrical.csv & docs/schematics/
R18	100 kΩ	1	Electrical / passive (v0.3 electrical)	Thick film 0603 0.1 W ±1%	Driver board	Q2 gate pull-down	ADD	Sheet E — see docs/BOM_electrical.csv & docs/schematics/
R19	100 kΩ	1	Electrical / passive (v0.3 electrical)	Thick film 0603 0.1 W ±1%	Driver board	Q3 gate pull-down	ADD	Sheet E — see docs/BOM_electrical.csv & docs/schematics/
D4	White LED 3 mm / 3528 (ZS-S04)	1	Electrical / passive (v0.3 electrical)	LED 3 mm Vf ≈3.0 V @ 20 mA —	Station LED board		ADD · ESTIMATE Vf	Sheet E — see docs/BOM_electrical.csv & docs/schematics/
D5	365 nm UV-A LED (ZS-S05)	1	Electrical / passive (v0.3 electrical)	LED 3535 Vf ≈3.6 V @ 20 mA —	Station LED board	check IEC 62471 risk group	ADD · ESTIMATE Vf	Sheet E — see docs/BOM_electrical.csv & docs/schematics/
D6	940 nm IR LED (ZS-S06, optional)	1	Electrical / passive (v0.3 electrical)	LED 3 mm Vf ≈1.3 V @ 20 mA —	Station LED board		ADD (optional) · ESTIMATE Vf	Sheet E — see docs/BOM_electrical.csv & docs/schematics/
R20	100 Ω	1	Electrical / passive (v0.3 electrical)	Thick film 0805 0.125 W ±1%	LED board	(5.0−3.0)/0.020 = 100 Ω → 20 mA; P = $0.02^2 \cdot 100$ = 0.04 W	ADD · ESTIMATE	Sheet E — see docs/BOM_electrical.csv & docs/schematics/
R21	68 Ω	1	Electrical / passive (v0.3 electrical)	Thick film 0805 0.125 W ±1%	LED board	(5.0−3.6)/0.020 = 70 Ω → E12 68 Ω → 20.6 mA; P = 0.029 W	ADD · ESTIMATE	Sheet E — see docs/BOM_electrical.csv & docs/schematics/
R22	180 Ω	1	Electrical / passive (v0.3 electrical)	Thick film 0805 0.125 W ±1%	LED board	(5.0−1.3)/0.020 = 185 Ω → 180 Ω → 20.6 mA; P = 0.076 W	ADD · ESTIMATE	Sheet E — see docs/BOM_electrical.csv & docs/schematics/
S1	Glass reed switch NO ~12 mm (ZS-S09)	1	Electrical / passive (v0.3 electrical)	Reed switch axial ≥100 V / 0.5 A —	Pedestal reed slot	in SERIES with UV LED supply: VBUS → S1 → UV_5V (hardware interlock)	ADD	Sheet E — see docs/BOM_electrical.csv & docs/schematics/
R23	12 kΩ	1	Electrical / passive (v0.3 electrical)	Thick film 0603 0.1 W ±1%	Driver board	SHIELD_SENSE = 5·20k/32k = 3.13 V (HIGH) when S1 closed; I = 0.16 mA	ADD	Sheet E — see docs/BOM_electrical.csv & docs/schematics/

ID	Part	Qty	Type	Material / source	Location in robot	Function	Status	Notes
R24	20 kΩ	1	Electrical / passive (v0.3 electrical)	Thick film 0603 0.1 W ±1%	Driver board	pulls SHIELD_SENSE to 0 V when S1 open. If VBUS = 5.25 V → 3.28 V (≤ 3.6 V abs max OK)	ADD	Sheet E — see docs/BOM_electrical.csv & docs/schematics/
C21	10 µF	1	Electrical / passive (v0.3 electrical)	MLCC X5R 0805 10 V ±10%	LED board	LED switching decoupling	ADD	Sheet E — see docs/BOM_electrical.csv & docs/schematics/

4. Print plan

Part	Material	Orientation / notes
ZS-101/102 head shells	matte navy PETG, 3–4 walls, 0.2 mm	split face down on bed, tree supports inside
ZS-301/302 torso shells	matte navy PETG	split face down
ZS-103/303 bulkheads, ZS-304 sled, ZS-105 face carrier	PETG	flat
ZS-104 visor	smoked translucent PETG, or vacuum-form 1.5 mm smoked PETG over a printed buck	convex side up, fine layers / sand + clear coat for gloss
ZS-106/107 light pipes, ZS-307 lens	clear resin (best) or clear PETG	polish faces
ZS-203/306 diffusers	translucent white PETG (violet/cyan come from LEDs)	flat
ZS-201/202 pods, ZS-204/205 caps	navy / gloss black PETG	outer face up
ZS-206 ears	navy PETG	lying on the back face, inner-face slots up
ZS-401 arms, ZS-402 legs	navy PETG	peg up, supports under hands/feet
ZS-501 pedestal	dark grey/black PLA 15 % infill	bottom down; optional stone-texture paint
ZS-502 tube	buy 12 OD × 8 ID acrylic, cut 75 mm	deburr; see open decision on stiffness
ZS-109 battery tray	PETG	lips up
ZS-601 station LED plug	white PETG	flange down; line the cavity with PTFE tape or white paint
ZS-602 light shield	**opaque** black PETG/PLA, 4 walls, 100 % top	top down; hold a torch inside in a dark room to check pinholes; paint inside matte black
ZS-603 sensor mount, ZS-605 dark plug	matte black, 100 % infill	flat
ZS-604 driver-bay cover	PLA	flat

Per-part settings (from the parts catalog)

Part	Name	Material / colour	Orientation / settings	Qty
ZS-101	Head shell, front	PETG matte navy #161C33	Split face down; 0.2 mm, 4 walls, 20% gyroid, tree supports inside	1
ZS-102	Head shell, back	PETG matte navy #161C33	Split face down; 0.2 mm, 4 walls, 20%, tree supports	1
ZS-103	Head bulkhead (mount plate)	PETG grey	Flat; 0.2 mm, 3 walls, 30%	1
ZS-104	Visor	Smoked translucent PETG (or vacuum-formed 1.5 mm sheet)	Convex side up; 0.12 mm, 100%, sand + clear coat	1
ZS-105	Face carrier (eye LEDs)	PETG black	Plate down; 0.2 mm, 3 walls	1
ZS-106	Eye light pipe, left (happy arc)	Clear resin or clear PETG	Flat face down; resin 0.05 mm or PETG 100%	1
ZS-106	Eye light pipe, right (happy arc)	Clear resin or clear PETG	Flat face down; resin 0.05 mm or PETG 100%	1
ZS-107	Crown accent light pipe, left	Clear resin	Flange down; 100%	1
ZS-107	Crown accent light pipe, right	Clear resin	Flange down; 100%	1

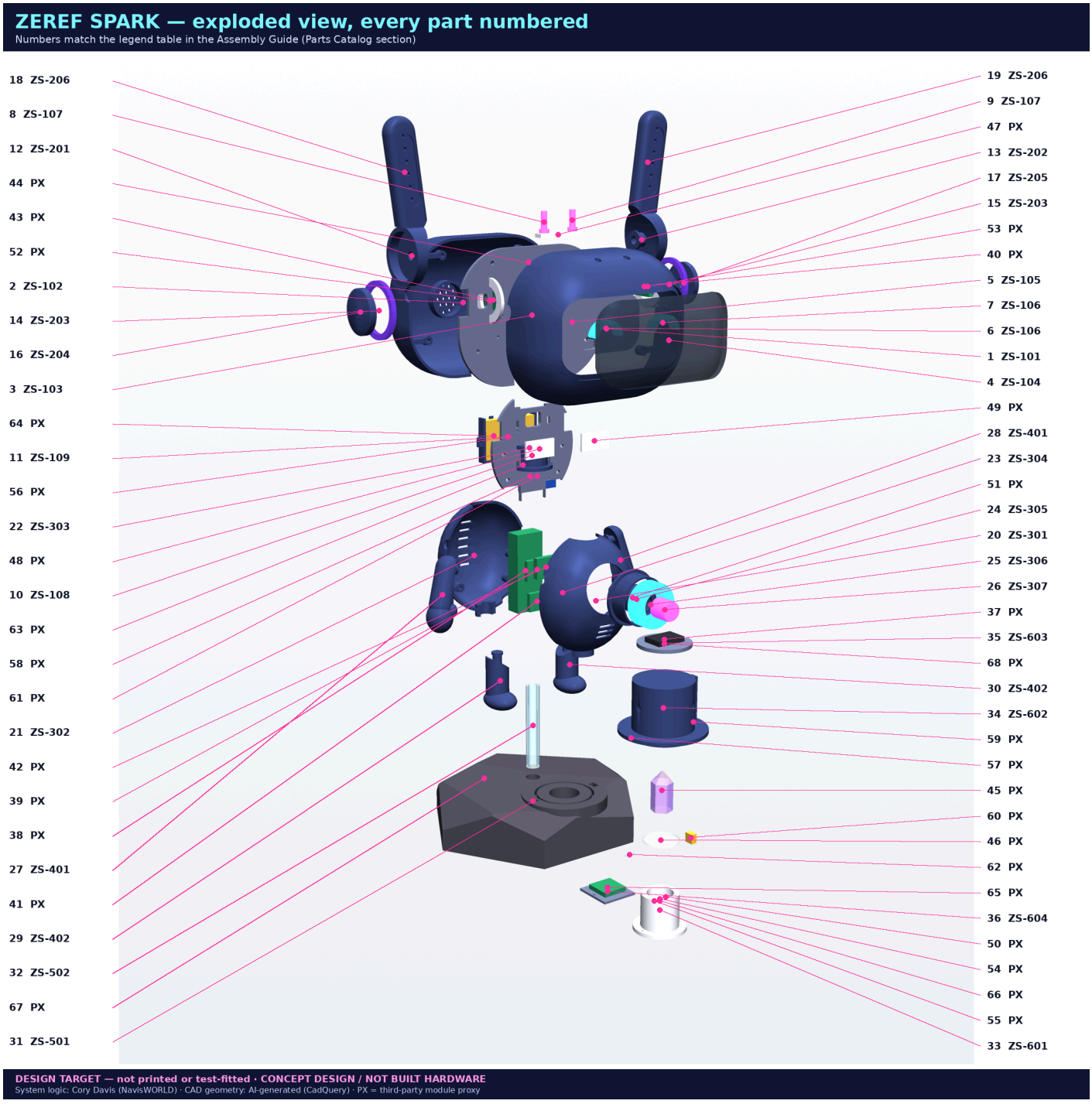
Part	Name	Material / colour	Orientation / settings	Qty
ZS-108	Neck coupler	PETG black	Upright; 0.2 mm, 100%	1
ZS-109	Battery tray	PETG grey	Back down, lips up; 0.2 mm, 3 walls	1
ZS-201	Ear pod, left (mic)	PETG matte navy #161C33	Outer face up; 0.2 mm, 4 walls, supports under peg	1
ZS-202	Ear pod, right (light)	PETG matte navy #161C33	Outer face up; 0.2 mm, 4 walls	1
ZS-203	Pod rim diffuser, left	Translucent white PETG (violet from LEDs)	Flat; 0.2 mm, 100%	1
ZS-203	Pod rim diffuser, right	Translucent white PETG	Flat; 0.2 mm, 100%	1
ZS-204	Pod cap, left (mic holes)	PETG gloss black	Outer face up; 0.12 mm	1
ZS-205	Pod cap, right (light window)	PETG gloss black + clear window	Outer face up; 0.12 mm	1
ZS-206	Rabbit ear, left	PETG matte navy #161C33	Lying on back face, slots up; 0.16 mm, 4 walls	1
ZS-206	Rabbit ear, right	PETG matte navy #161C33	Lying on back face, slots up; 0.16 mm	1
ZS-301	Torso shell, front	PETG matte navy #161C33	Split face down; 0.2 mm, 4 walls, supports	1
ZS-302	Torso shell, back	PETG matte navy #161C33	Split face down; 0.2 mm, 4 walls	1
ZS-303	Torso bulkhead / ESP32 cradle	PETG grey	Flat, rails up; 0.2 mm, 30%	1
ZS-304	Sensor sled (IMU + BME280)	PETG grey	Flat; 0.2 mm	1
ZS-305	Chest core housing	PETG black	Flange down; 0.16 mm, 100%	1
ZS-306	Chest ring diffuser	Translucent white PETG (cyan from LEDs)	Flat; 100%	1
ZS-307	Chest centre lens	Clear (if MAX30102) or violet translucent	Dome up; resin or PETG 100%	1
ZS-401	Arm + mitten hand, left	PETG matte navy #161C33	Peg up; 0.16 mm, supports under hand	1
ZS-401	Arm + mitten hand, right	PETG matte navy #161C33	Peg up; 0.16 mm	1
ZS-402	Leg + foot, left	PETG matte navy #161C33	Peg up; 0.16 mm, supports under foot	1
ZS-402	Leg + foot, right	PETG matte navy #161C33	Peg up; 0.16 mm	1
ZS-501	Pedestal (rock hex) + station body	PLA dark grey/black; 15% infill, station walls 100%	Bottom down; 0.28 mm, 3 walls	1
ZS-502	Support tube (bought, cut)	Clear acrylic tube 12 OD x 8 ID x 75	Cut + deburr (not printed)	1
ZS-601	Station LED plug	White PETG; cavity PTFE-tape lined	Flange down; 0.2 mm, 100%	1
ZS-602	Light shield (removable)	Opaque black PETG/PLA; inside matte black	Top down; 4 walls, 100% top	1
ZS-603	Sensor mount disc (AS7341)	Matte black PETG	Flat; 100%	1

Part	Name	Material / colour	Orientation / settings	Qty
ZS-604	Driver bay cover	PLA black	Flat; 0.2 mm	1
ZS-605	Dark-reference plug	Opaque black PLA 100%	Knob up; 100%	1

Test-print first: one leg peg + socket, one pod peg, one split boss with insert, the neck coupler and the shield-groove ring. Adjust clearances before printing full shells.

5. Parts catalog — every part labeled and shown

Exploded view, every part numbered



#	Part no.	Name	Type
1	ZS-101	Head shell, front	printed (AI-generated CAD)
2	ZS-102	Head shell, back	printed (AI-generated CAD)
3	ZS-103	Head bulkhead (mount plate)	printed (AI-generated CAD)

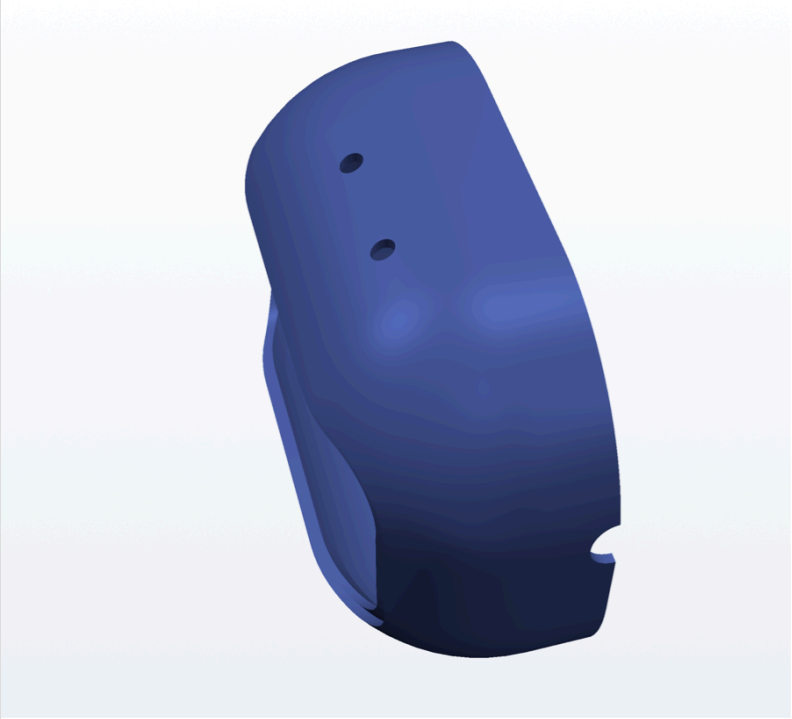
#	Part no.	Name	Type
4	ZS-104	Visor	printed (AI-generated CAD)
5	ZS-105	Face carrier (eye LEDs)	printed (AI-generated CAD)
6	ZS-106	Eye light pipe, left (happy arc)	printed (AI-generated CAD)
7	ZS-106	Eye light pipe, right (happy arc)	printed (AI-generated CAD)
8	ZS-107	Crown accent light pipe, left	printed (AI-generated CAD)
9	ZS-107	Crown accent light pipe, right	printed (AI-generated CAD)
10	ZS-108	Neck coupler	printed (AI-generated CAD)
11	ZS-109	Battery tray	printed (AI-generated CAD)
12	ZS-201	Ear pod, left (mic)	printed (AI-generated CAD)
13	ZS-202	Ear pod, right (light)	printed (AI-generated CAD)
14	ZS-203	Pod rim diffuser, left	printed (AI-generated CAD)
15	ZS-203	Pod rim diffuser, right	printed (AI-generated CAD)
16	ZS-204	Pod cap, left (mic holes)	printed (AI-generated CAD)
17	ZS-205	Pod cap, right (light window)	printed (AI-generated CAD)
18	ZS-206	Rabbit ear, left	printed (AI-generated CAD)
19	ZS-206	Rabbit ear, right	printed (AI-generated CAD)
20	ZS-301	Torso shell, front	printed (AI-generated CAD)
21	ZS-302	Torso shell, back	printed (AI-generated CAD)
22	ZS-303	Torso bulkhead / ESP32 cradle	printed (AI-generated CAD)
23	ZS-304	Sensor sled (IMU + BME280)	printed (AI-generated CAD)
24	ZS-305	Chest core housing	printed (AI-generated CAD)
25	ZS-306	Chest ring diffuser	printed (AI-generated CAD)
26	ZS-307	Chest centre lens	printed (AI-generated CAD)
27	ZS-401	Arm + mitten hand, left	printed (AI-generated CAD)
28	ZS-401	Arm + mitten hand, right	printed (AI-generated CAD)
29	ZS-402	Leg + foot, left	printed (AI-generated CAD)
30	ZS-402	Leg + foot, right	printed (AI-generated CAD)
31	ZS-501	Pedestal (rock hex) + station body	printed (AI-generated CAD)
32	ZS-502	Support tube (bought, cut)	printed (AI-generated CAD)
33	ZS-601	Station LED plug	printed (AI-generated CAD)
34	ZS-602	Light shield (removable)	printed (AI-generated CAD)
35	ZS-603	Sensor mount disc (AS7341)	printed (AI-generated CAD)
36	ZS-604	Driver bay cover	printed (AI-generated CAD)
37	PX	AS7341 11-ch spectral breakout (ZS-S01)	third-party module proxy
38	PX	ESP32-S3 DevKitC-1 (RB-001)	third-party module proxy
39	PX	ICM-42688-P IMU breakout (RB-002)	third-party module proxy
40	PX	BH1750 light breakout (RB-003)	third-party module proxy
41	PX	BME280 breakout (RB-004)	third-party module proxy
42	PX	MAX30102 breakout (RB-005, optional, non-medical)	third-party module proxy
43	PX	INMP441 I2S mic breakout (RB-006)	third-party module proxy
44	PX	NEO-M9N-class GNSS (RB-007, optional)	third-party module proxy
45	PX	Crystal sample (example only)	third-party module proxy
46	PX	Window disc 29.6 x 2 (ZS-S02)	third-party module proxy
47	PX	Crown accent LED board, 2x WS2812B (ZS-E03)	third-party module proxy

#	Part no.	Name	Type
48	PX	Eye LED strip L, 5x WS2812B (ZS-E02)	third-party module proxy
49	PX	Eye LED strip R, 5x WS2812B (ZS-E02)	third-party module proxy
50	PX	940 nm IR LED (ZS-S06, optional)	third-party module proxy
51	PX	WS2812B 12-LED ring, chest (ZS-E01)	third-party module proxy
52	PX	WS2812B 12-LED ring, left pod (ZS-E01)	third-party module proxy
53	PX	WS2812B 12-LED ring, right pod (ZS-E01)	third-party module proxy
54	PX	365 nm UV-A LED (ZS-S05) — EYE HAZARD	third-party module proxy
55	PX	White LED (ZS-S04)	third-party module proxy
56	PX	1S LiPo 503450 ~1000 mAh, protected (ZS-B01)	third-party module proxy
57	PX	Magnet 6 x 2 mm (ZS-S10)	third-party module proxy
58	PX	MAX98357A I2S amp breakout (ZS-E05)	third-party module proxy
59	PX	Pogo pads 4-pin (ZS-S11)	third-party module proxy
60	PX	Pogo receptacle 4-pin (ZS-S11)	third-party module proxy
61	PX	Adafruit PowerBoost 1000C (ZS-B02)	third-party module proxy
62	PX	Reed switch (ZS-S09) — UV interlock	third-party module proxy
63	PX	Slide switch EN (ZS-B03)	third-party module proxy
64	PX	28 mm 8 ohm speaker (ZS-E06)	third-party module proxy
65	PX	Station driver board, Q1-Q3 (ZS-S08)	third-party module proxy
66	PX	Station LED board 20x20 (ZS-S07)	third-party module proxy
67	PX	USB-C receptacle breakout (ZS-E07)	third-party module proxy
68	PX	UV-cut filter 20 x 2 (ZS-S03, optional)	third-party module proxy

Part cards

Each card shows part number, name, material, print orientation, quantity, fasteners and sub-assembly. Printed parts = AI-generated CAD; PX = third-party module proxy (envelope ESTIMATE).

ZS-101
Head shell, front



TYPE
PRINTED / MADE PART — AI-generated CAD geometry

FILE
ZS-101_head_shell_front

MATERIAL / COLOUR
PETG matte navy #161C33

PRINT ORIENTATION / SETTINGS
Split face down; 0.2 mm, 4 walls, 20% gyroid, tree supports inside

QTY (THIS FILE)
1

FASTENERS / FIXING
4x M3 heat-set inserts (ZS-F01); receives 4x RB-008

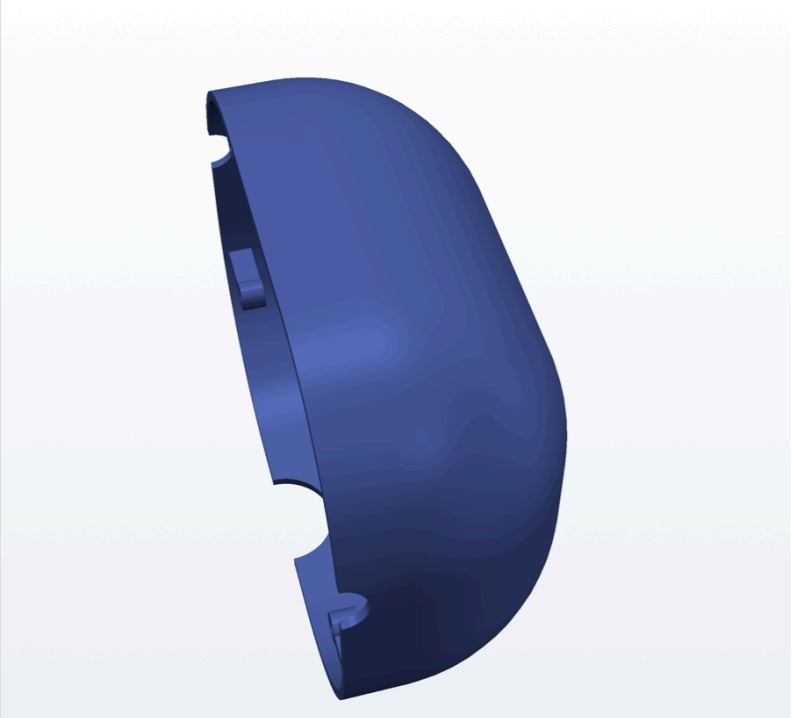
SUB-ASSEMBLY
head

ENVELOPE W X D X H
150.0 x 58.7 x 118.0 mm (ESTIMATE, CAD)

SOLID VOLUME
65.9 cm3 (CAD, before infill)

DESIGN TARGET — not printed or test-fitted · CONCEPT DESIGN / NOT BUILT HARDWARE
System logic: Cory Davis (NavisWORLD) · CAD geometry: AI-generated (CadQuery) · PX = third-party module proxy

ZS-102
Head shell, back



TYPE
PRINTED / MADE PART — AI-generated CAD geometry

FILE
ZS-102_head_shell_back

MATERIAL / COLOUR
PETG matte navy #161C33

PRINT ORIENTATION / SETTINGS
Split face down; 0.2 mm, 4 walls, 20%, tree supports

QTY (THIS FILE)
1

FASTENERS / FIXING
4x RB-008 M3x8 (clearance bosses); speaker glued in ring

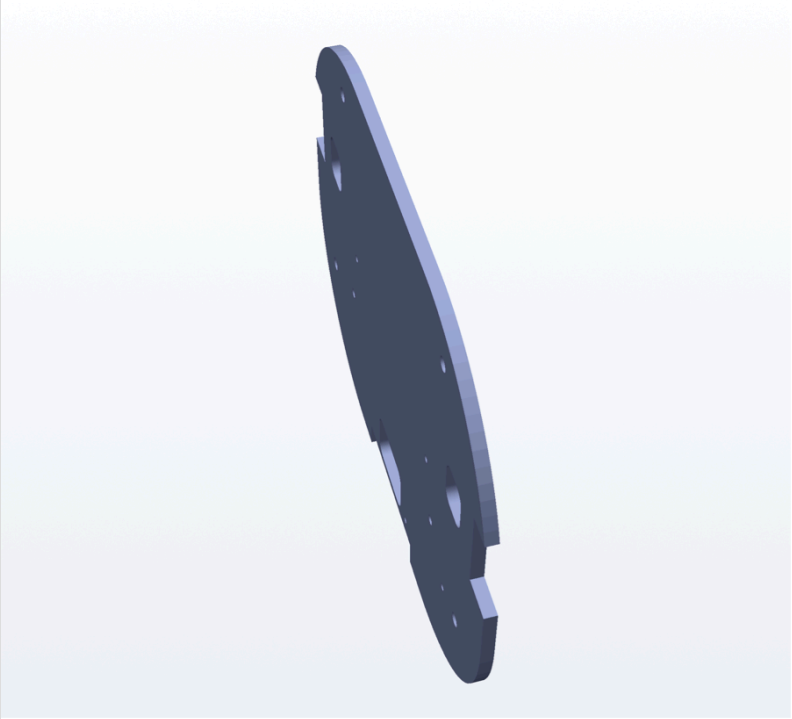
SUB-ASSEMBLY
head

ENVELOPE W X D X H
150.0 x 64.0 x 118.0 mm (ESTIMATE, CAD)

SOLID VOLUME
82.6 cm3 (CAD, before infill)

DESIGN TARGET — not printed or test-fitted · CONCEPT DESIGN / NOT BUILT HARDWARE
System logic: Cory Davis (NavisWORLD) · CAD geometry: AI-generated (CadQuery) · PX = third-party module proxy

ZS-103
Head bulkhead (mount plate)



TYPE
PRINTED / MADE PART — AI-generated CAD geometry

FILE
ZS-103_head_bulkhead

MATERIAL / COLOUR
PETG grey

PRINT ORIENTATION / SETTINGS
Flat; 0.2 mm, 3 walls, 30%

QTY (THIS FILE)
1

FASTENERS / FIXING
Clamped by 4x RB-008; 2x 2.5 PT (face carrier), 2x M2 (tray), 2x M2 (PowerBoost)

SUB-ASSEMBLY
head

ENVELOPE W X D X H
144.6 x 3.0 x 112.6 mm (ESTIMATE, CAD)

SOLID VOLUME
39.9 cm3 (CAD, before infill)

DESIGN TARGET — not printed or test-fitted · CONCEPT DESIGN / NOT BUILT HARDWARE
System logic: Cory Davis (NavisWORLD) · CAD geometry: AI-generated (CadQuery) · PX = third-party module proxy

ZS-104
Visor

TYPE
PRINTED / MADE PART — AI-generated CAD geometry

FILE
ZS-104_visor

MATERIAL / COLOUR
Smoked translucent PETG (or vacuum-formed 1.5 mm sheet)

PRINT ORIENTATION / SETTINGS
Convex side up; 0.12 mm, 100%, sand + clear coat

QTY (THIS FILE)
1

FASTENERS / FIXING
Clear silicone / CA onto ZS-101 ledge

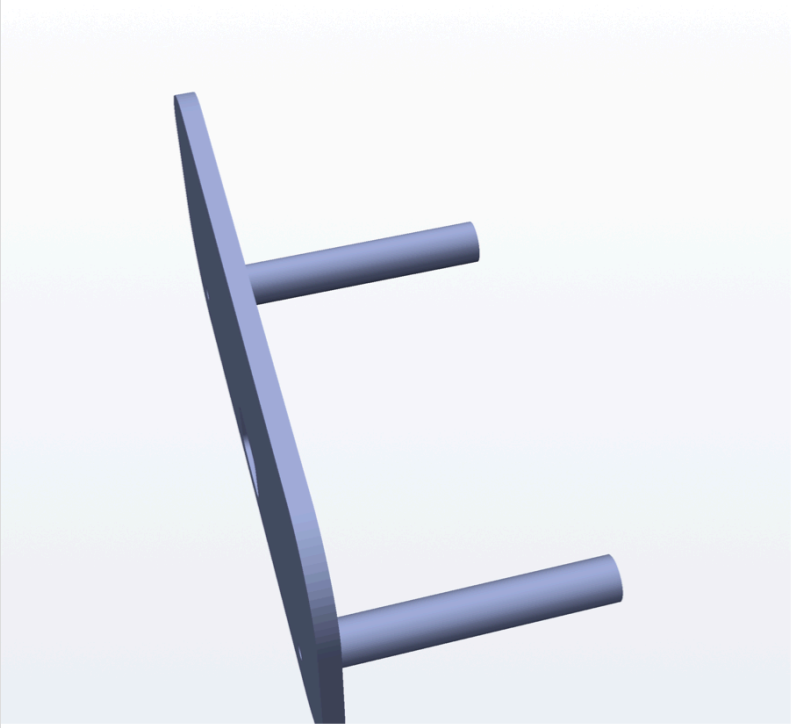
SUB-ASSEMBLY
head

ENVELOPE W X D X H
123.7 x 19.9 x 65.7 mm (ESTIMATE, CAD)

SOLID VOLUME
30.4 cm3 (CAD, before infill)

DESIGN TARGET — not printed or test-fitted · CONCEPT DESIGN / NOT BUILT HARDWARE
System logic: Cory Davis (NavisWORLD) · CAD geometry: AI-generated (CadQuery) · PX = third-party module proxy

ZS-105
Face carrier (eye LEDs)



TYPE
PRINTED / MADE PART — AI-generated CAD geometry

FILE
ZS-105_face_carrier

MATERIAL / COLOUR
PETG black

PRINT ORIENTATION / SETTINGS
Plate down; 0.2 mm, 3 walls

QTY (THIS FILE)
1

FASTENERS / FIXING
2x 2.5 mm PT x12 (ZS-F04) into ZS-103

SUB-ASSEMBLY
head

ENVELOPE W X D X H
100.0 x 46.5 x 46.0 mm (ESTIMATE, CAD)

SOLID VOLUME
15.9 cm3 (CAD, before infill)

DESIGN TARGET — not printed or test-fitted · CONCEPT DESIGN / NOT BUILT HARDWARE
System logic: Cory Davis (NavisWORLD) · CAD geometry: AI-generated (CadQuery) · PX = third-party module proxy

ZS-106
Eye light pipe, left (happy arc)

TYPE
PRINTED / MADE PART — AI-generated CAD geometry

FILE
ZS-106_eye_lightpipe_L

MATERIAL / COLOUR
Clear resin or clear PETG

PRINT ORIENTATION / SETTINGS
Flat face down; resin 0.05 mm or PETG 100%

QTY (THIS FILE)
1

FASTENERS / FIXING
Glue onto eye LED strip

SUB-ASSEMBLY
head

ENVELOPE W X D X H
27.3 x 8.5 x 13.0 mm (ESTIMATE, CAD)

SOLID VOLUME
1.2 cm3 (CAD, before infill)

DESIGN TARGET — not printed or test-fitted · CONCEPT DESIGN / NOT BUILT HARDWARE
System logic: Cory Davis (NavisWORLD) · CAD geometry: AI-generated (CadQuery) · PX = third-party module proxy

ZS-106
Eye light pipe, right (happy arc)

TYPE
PRINTED / MADE PART — AI-generated CAD geometry

FILE
ZS-106_eye_lightpipe_R

MATERIAL / COLOUR
Clear resin or clear PETG

PRINT ORIENTATION / SETTINGS
Flat face down; resin 0.05 mm or PETG 100%

QTY (THIS FILE)
1

FASTENERS / FIXING
Glue onto eye LED strip

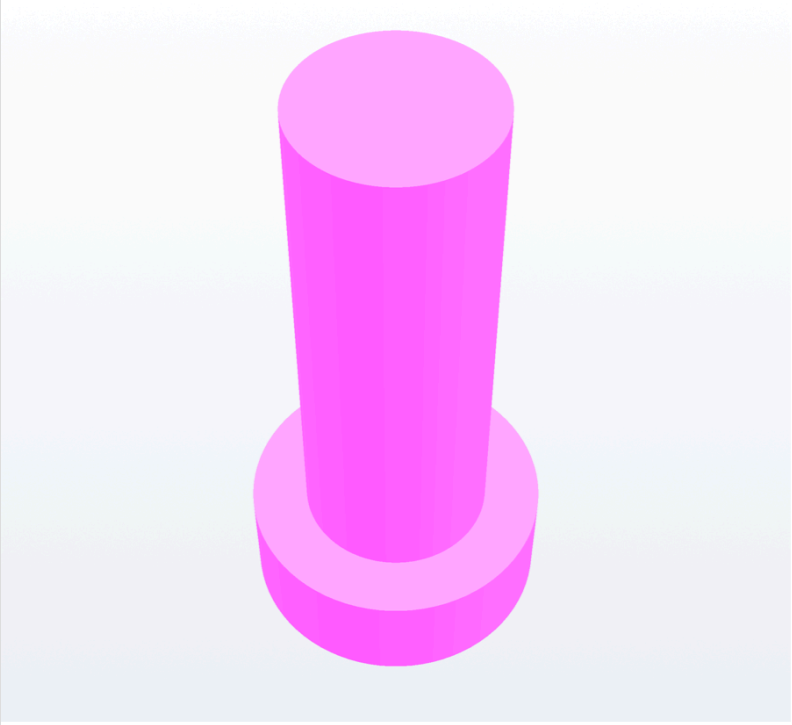
SUB-ASSEMBLY
head

ENVELOPE W X D X H
27.3 x 8.5 x 13.0 mm (ESTIMATE, CAD)

SOLID VOLUME
1.2 cm3 (CAD, before infill)

DESIGN TARGET — not printed or test-fitted · CONCEPT DESIGN / NOT BUILT HARDWARE
System logic: Cory Davis (NavisWORLD) · CAD geometry: AI-generated (CadQuery) · PX = third-party module proxy

ZS-107
Crown accent light pipe, left



TYPE
PRINTED / MADE PART — AI-generated CAD geometry

FILE
ZS-107_accent_lightpipe_L

MATERIAL / COLOUR
Clear resin

PRINT ORIENTATION / SETTINGS
Flange down; 100%

QTY (THIS FILE)
1

FASTENERS / FIXING
Flange captured inside head

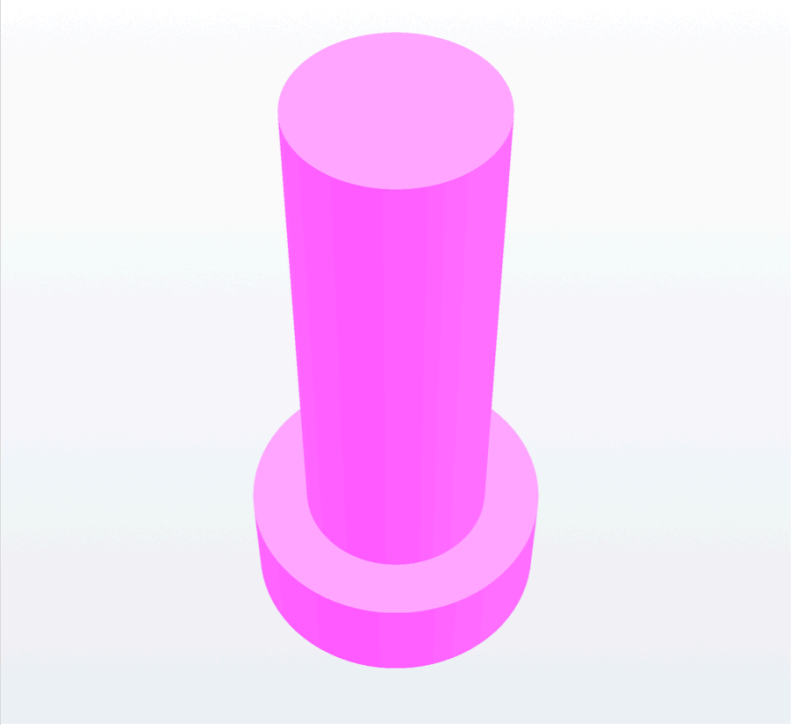
SUB-ASSEMBLY
head

ENVELOPE W X D X H
8.0 x 8.0 x 17.0 mm (ESTIMATE, CAD)

SOLID VOLUME
0.4 cm3 (CAD, before infill)

DESIGN TARGET — not printed or test-fitted · CONCEPT DESIGN / NOT BUILT HARDWARE
System logic: Cory Davis (NavisWORLD) · CAD geometry: AI-generated (CadQuery) · PX = third-party module proxy

ZS-107
Crown accent light pipe, right



TYPE
PRINTED / MADE PART — AI-generated CAD geometry

FILE
ZS-107_accent_lightpipe_R

MATERIAL / COLOUR
Clear resin

PRINT ORIENTATION / SETTINGS
Flange down; 100%

QTY (THIS FILE)
1

FASTENERS / FIXING
Flange captured inside head

SUB-ASSEMBLY
head

ENVELOPE W X D X H
8.0 x 8.0 x 17.0 mm (ESTIMATE, CAD)

SOLID VOLUME
0.4 cm3 (CAD, before infill)

DESIGN TARGET — not printed or test-fitted · CONCEPT DESIGN / NOT BUILT HARDWARE
System logic: Cory Davis (NavisWORLD) · CAD geometry: AI-generated (CadQuery) · PX = third-party module proxy

ZS-108
Neck coupler

TYPE
PRINTED / MADE PART — AI-generated CAD geometry

FILE
ZS-108_neck_coupler

MATERIAL / COLOUR
PETG black

PRINT ORIENTATION / SETTINGS
Upright; 0.2 mm, 100%

QTY (THIS FILE)
1

FASTENERS / FIXING
Captured by both shell pairs (no screws)

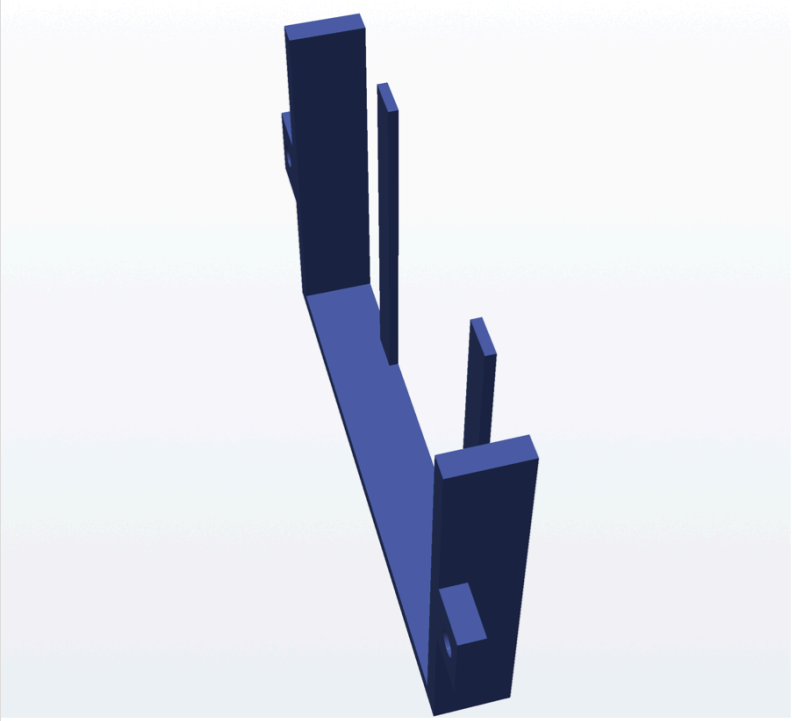
SUB-ASSEMBLY
head

ENVELOPE W X D X H
34.0 x 34.0 x 21.0 mm (ESTIMATE, CAD)

SOLID VOLUME
10.2 cm3 (CAD, before infill)

DESIGN TARGET — not printed or test-fitted · CONCEPT DESIGN / NOT BUILT HARDWARE
System logic: Cory Davis (NavisWORLD) · CAD geometry: AI-generated (CadQuery) · PX = third-party module proxy

ZS-109
Battery tray



TYPE
PRINTED / MADE PART — AI-generated CAD geometry

FILE
ZS-109_battery_tray

MATERIAL / COLOUR
PETG grey

PRINT ORIENTATION / SETTINGS
Back down, lips up; 0.2 mm, 3 walls

QTY (THIS FILE)
1

FASTENERS / FIXING
2x M2x6 into ZS-103

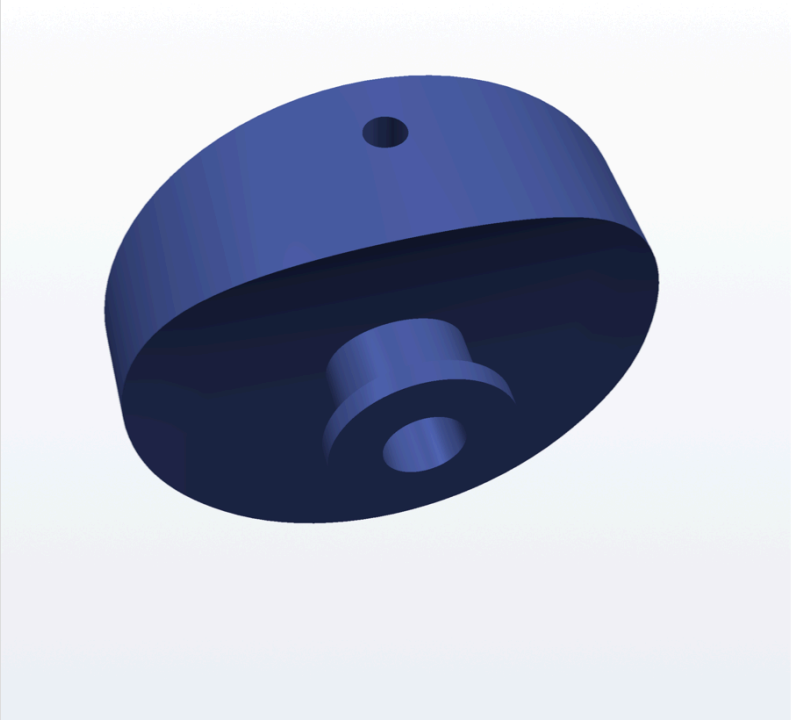
SUB-ASSEMBLY
power

ENVELOPE W X D X H
66.0 x 7.5 x 38.2 mm (ESTIMATE, CAD)

SOLID VOLUME
2.5 cm3 (CAD, before infill)

DESIGN TARGET — not printed or test-fitted · CONCEPT DESIGN / NOT BUILT HARDWARE
System logic: Cory Davis (NavisWORLD) · CAD geometry: AI-generated (CadQuery) · PX = third-party module proxy

ZS-201
Ear pod, left (mic)



TYPE
PRINTED / MADE PART — AI-generated CAD geometry

FILE
ZS-201_ear_pod_L_mic

MATERIAL / COLOUR
PETG matte navy #161C33

PRINT ORIENTATION / SETTINGS
Outer face up; 0.2 mm, 4 walls, supports under peg

QTY (THIS FILE)
1

FASTENERS / FIXING
Flanged peg captured by head split; 1x M3x10 PT up into ear

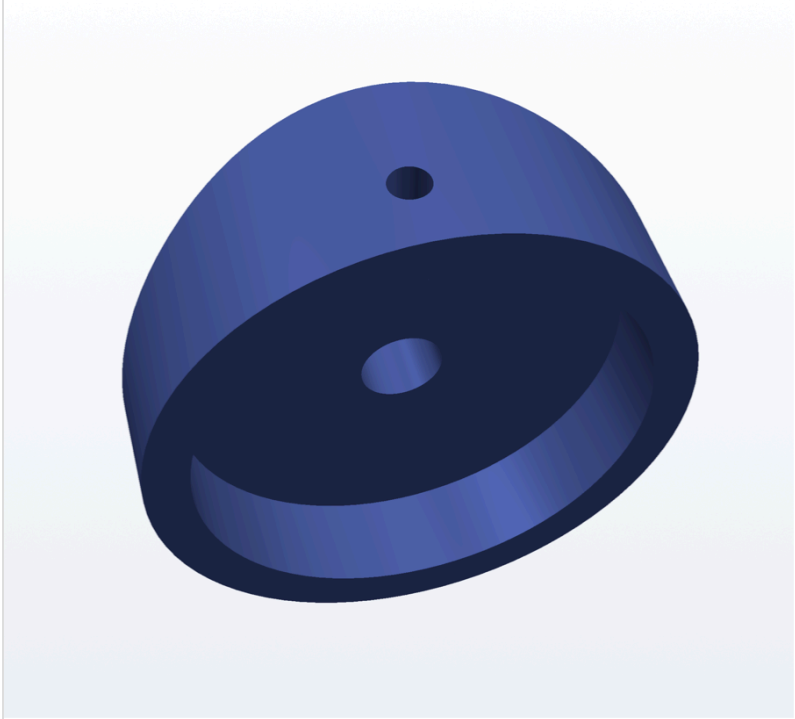
SUB-ASSEMBLY
pods

ENVELOPE W X D X H
18.0 x 46.0 x 46.0 mm (ESTIMATE, CAD)

SOLID VOLUME
8.1 cm3 (CAD, before infill)

DESIGN TARGET — not printed or test-fitted · CONCEPT DESIGN / NOT BUILT HARDWARE
System logic: Cory Davis (NavisWORLD) · CAD geometry: AI-generated (CadQuery) · PX = third-party module proxy

ZS-202
Ear pod, right (light)



TYPE
PRINTED / MADE PART — AI-generated CAD geometry

FILE
ZS-202_ear_pod_R_light

MATERIAL / COLOUR
PETG matte navy #161C33

PRINT ORIENTATION / SETTINGS
Outer face up; 0.2 mm, 4 walls

QTY (THIS FILE)
1

FASTENERS / FIXING
Flanged peg captured; 1x M3x10 PT into ear

SUB-ASSEMBLY
pods

ENVELOPE W X D X H
18.0 x 46.0 x 46.0 mm (ESTIMATE, CAD)

SOLID VOLUME
8.1 cm3 (CAD, before infill)

DESIGN TARGET — not printed or test-fitted · CONCEPT DESIGN / NOT BUILT HARDWARE
System logic: Cory Davis (NavisWORLD) · CAD geometry: AI-generated (CadQuery) · PX = third-party module proxy

ZS-203
Pod rim diffuser, left

TYPE
PRINTED / MADE PART — AI-generated CAD geometry

FILE
ZS-203_pod_rim_diffuser_L

MATERIAL / COLOUR
Translucent white PETG (violet from LEDs)

PRINT ORIENTATION / SETTINGS
Flat; 0.2 mm, 100%

QTY (THIS FILE)
1

FASTENERS / FIXING
Press / glue into pod

SUB-ASSEMBLY
pods

ENVELOPE W X D X H
4.0 x 46.0 x 46.0 mm (ESTIMATE, CAD)

SOLID VOLUME
2.3 cm3 (CAD, before infill)

DESIGN TARGET — not printed or test-fitted · CONCEPT DESIGN / NOT BUILT HARDWARE
System logic: Cory Davis (NavisWORLD) · CAD geometry: AI-generated (CadQuery) · PX = third-party module proxy

ZS-203
Pod rim diffuser, right

TYPE
PRINTED / MADE PART — AI-generated CAD geometry

FILE
ZS-203_pod_rim_diffuser_R

MATERIAL / COLOUR
Translucent white PETG

PRINT ORIENTATION / SETTINGS
Flat; 0.2 mm, 100%

QTY (THIS FILE)
1

FASTENERS / FIXING
Press / glue into pod

SUB-ASSEMBLY
pods

ENVELOPE W X D X H
4.0 x 46.0 x 46.0 mm (ESTIMATE, CAD)

SOLID VOLUME
2.3 cm3 (CAD, before infill)

DESIGN TARGET — not printed or test-fitted · CONCEPT DESIGN / NOT BUILT HARDWARE
System logic: Cory Davis (NavisWORLD) · CAD geometry: AI-generated (CadQuery) · PX = third-party module proxy

ZS-204
Pod cap, left (mic holes)

TYPE
PRINTED / MADE PART — AI-generated CAD geometry

FILE
ZS-204_pod_cap_L_mic

MATERIAL / COLOUR
PETG gloss black

PRINT ORIENTATION / SETTINGS
Outer face up; 0.12 mm

QTY (THIS FILE)
1

FASTENERS / FIXING
Snap / glue into rim

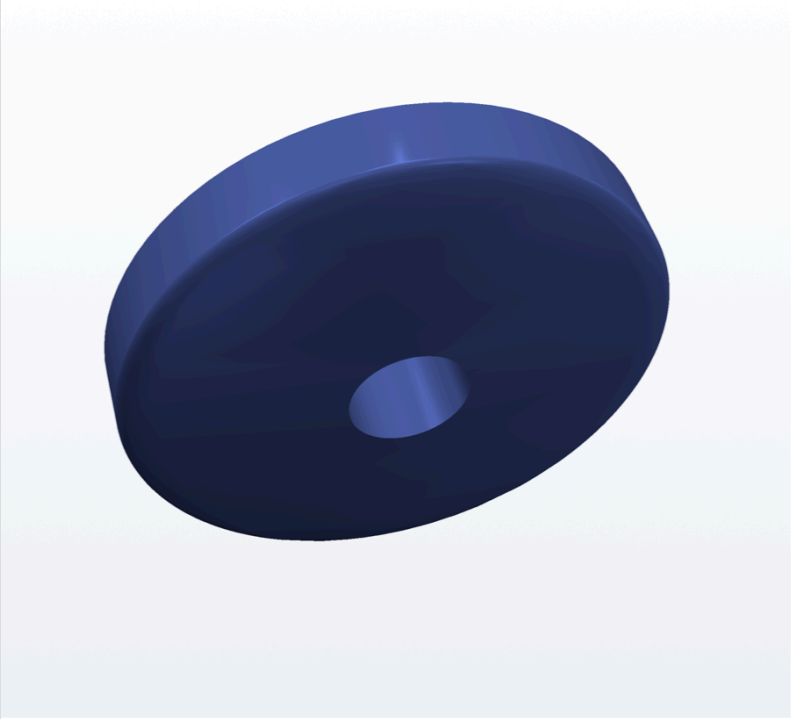
SUB-ASSEMBLY
pods

ENVELOPE W X D X H
7.0 x 36.6 x 36.6 mm (ESTIMATE, CAD)

SOLID VOLUME
6.5 cm3 (CAD, before infill)

DESIGN TARGET — not printed or test-fitted · CONCEPT DESIGN / NOT BUILT HARDWARE
System logic: Cory Davis (NavisWORLD) · CAD geometry: AI-generated (CadQuery) · PX = third-party module proxy

ZS-205
Pod cap, right (light window)



TYPE
PRINTED / MADE PART — AI-generated CAD geometry

FILE
ZS-205_pod_cap_R_light

MATERIAL / COLOUR
PETG gloss black + clear window

PRINT ORIENTATION / SETTINGS
Outer face up; 0.12 mm

QTY (THIS FILE)
1

FASTENERS / FIXING
Snap / glue into rim

SUB-ASSEMBLY
pods

ENVELOPE W X D X H
6.9 x 36.6 x 36.6 mm (ESTIMATE, CAD)

SOLID VOLUME
6.2 cm3 (CAD, before infill)

DESIGN TARGET — not printed or test-fitted · CONCEPT DESIGN / NOT BUILT HARDWARE
System logic: Cory Davis (NavisWORLD) · CAD geometry: AI-generated (CadQuery) · PX = third-party module proxy

ZS-206
Rabbit ear, left

TYPE
PRINTED / MADE PART — AI-generated CAD geometry

FILE
ZS-206_ear_L

MATERIAL / COLOUR
PETG matte navy #161C33

PRINT ORIENTATION / SETTINGS
Lying on back face, slots up; 0.16 mm, 4 walls

QTY (THIS FILE)
1

FASTENERS / FIXING
1x M3x10 PT (ZS-F02) from inside pod

SUB-ASSEMBLY
pods

ENVELOPE W X D X H
35.2 x 14.0 x 84.0 mm (ESTIMATE, CAD)

SOLID VOLUME
25.9 cm3 (CAD, before infill)

DESIGN TARGET — not printed or test-fitted · CONCEPT DESIGN / NOT BUILT HARDWARE
System logic: Cory Davis (NavisWORLD) · CAD geometry: AI-generated (CadQuery) · PX = third-party module proxy

ZS-206
Rabbit ear, right

TYPE
PRINTED / MADE PART — AI-generated CAD geometry

FILE
ZS-206_ear_R

MATERIAL / COLOUR
PETG matte navy #161C33

PRINT ORIENTATION / SETTINGS
Lying on back face, slots up; 0.16 mm

QTY (THIS FILE)
1

FASTENERS / FIXING
1x M3x10 PT (ZS-F02)

SUB-ASSEMBLY
pods

ENVELOPE W X D X H
35.2 x 14.0 x 84.0 mm (ESTIMATE, CAD)

SOLID VOLUME
25.9 cm3 (CAD, before infill)

DESIGN TARGET — not printed or test-fitted · CONCEPT DESIGN / NOT BUILT HARDWARE
System logic: Cory Davis (NavisWORLD) · CAD geometry: AI-generated (CadQuery) · PX = third-party module proxy

ZS-301
Torso shell, front

TYPE
PRINTED / MADE PART — AI-generated CAD geometry

FILE
ZS-301_torso_shell_front

MATERIAL / COLOUR
PETG matte navy #161C33

PRINT ORIENTATION / SETTINGS
Split face down; 0.2 mm, 4 walls, supports

QTY (THIS FILE)
1

FASTENERS / FIXING
4x M3 heat-set inserts; receives 4x RB-008

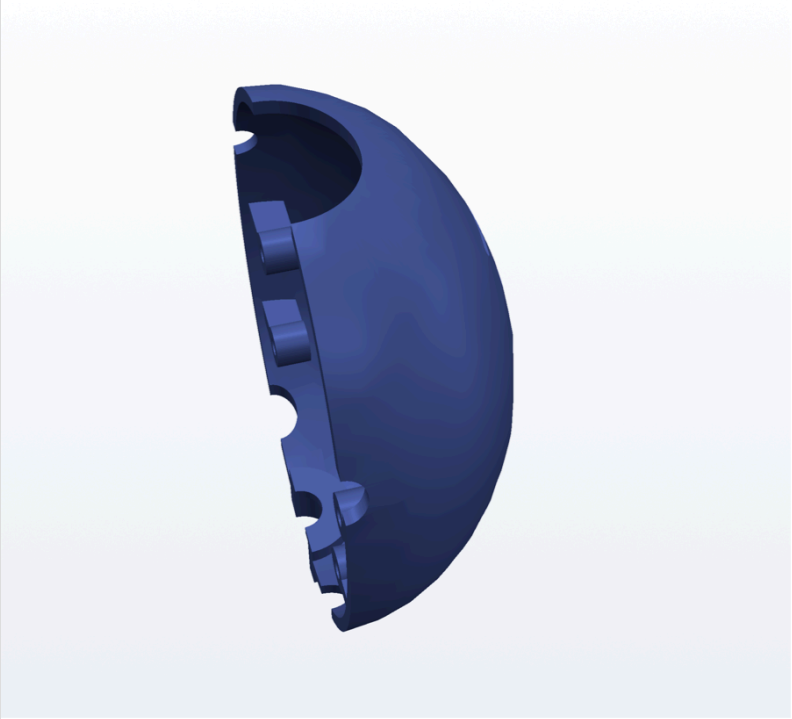
SUB-ASSEMBLY
torso

ENVELOPE W X D X H
88.0 x 37.2 x 102.2 mm (ESTIMATE, CAD)

SOLID VOLUME
25.1 cm3 (CAD, before infill)

DESIGN TARGET — not printed or test-fitted · CONCEPT DESIGN / NOT BUILT HARDWARE
System logic: Cory Davis (NavisWORLD) · CAD geometry: AI-generated (CadQuery) · PX = third-party module proxy

ZS-302
Torso shell, back



TYPE
PRINTED / MADE PART — AI-generated CAD geometry

FILE
ZS-302_torso_shell_back

MATERIAL / COLOUR
PETG matte navy #161C33

PRINT ORIENTATION / SETTINGS
Split face down; 0.2 mm, 4 walls

QTY (THIS FILE)
1

FASTENERS / FIXING
4x RB-008 M3x8

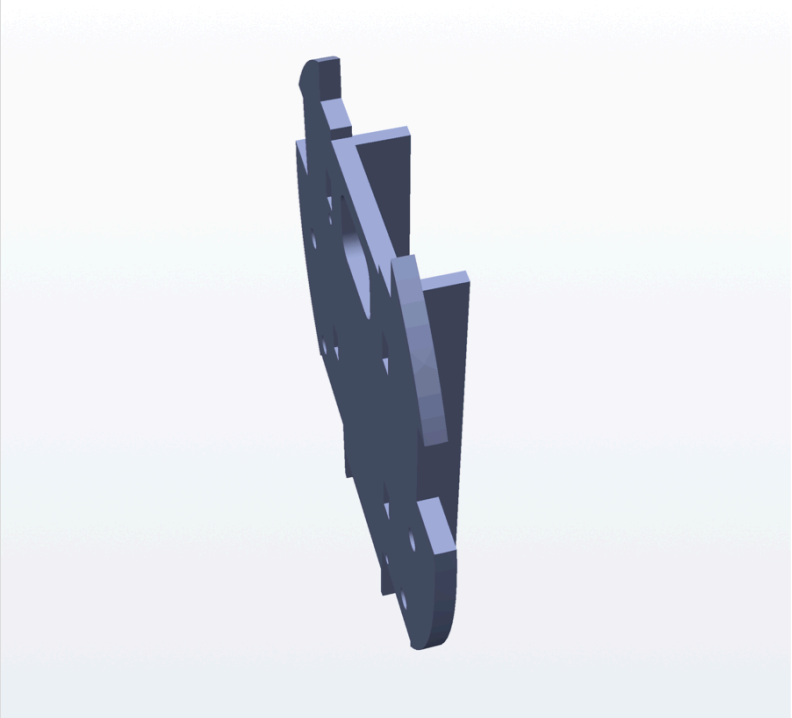
SUB-ASSEMBLY
torso

ENVELOPE W X D X H
88.0 x 38.0 x 102.2 mm (ESTIMATE, CAD)

SOLID VOLUME
28.5 cm3 (CAD, before infill)

DESIGN TARGET — not printed or test-fitted · CONCEPT DESIGN / NOT BUILT HARDWARE
System logic: Cory Davis (NavisWORLD) · CAD geometry: AI-generated (CadQuery) · PX = third-party module proxy

ZS-303
Torso bulkhead / ESP32 cradle



TYPE
PRINTED / MADE PART — AI-generated CAD geometry

FILE
ZS-303_torso_bulkhead_esp32_cradle

MATERIAL / COLOUR
PETG grey

PRINT ORIENTATION / SETTINGS
Flat, rails up; 0.2 mm, 30%

QTY (THIS FILE)
1

FASTENERS / FIXING
Clamped by 4x RB-008; 2x zip ties (ZS-F05) hold DevKit

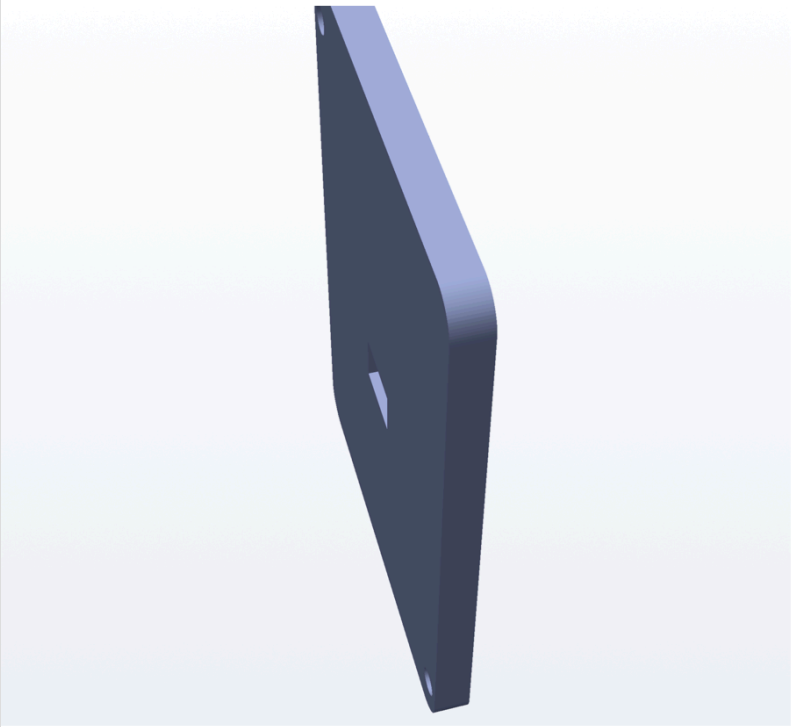
SUB-ASSEMBLY
torso

ENVELOPE W X D X H
82.6 x 11.5 x 87.3 mm (ESTIMATE, CAD)

SOLID VOLUME
16.1 cm3 (CAD, before infill)

DESIGN TARGET — not printed or test-fitted · CONCEPT DESIGN / NOT BUILT HARDWARE
System logic: Cory Davis (NavisWORLD) · CAD geometry: AI-generated (CadQuery) · PX = third-party module proxy

ZS-304
Sensor sled (IMU + BME280)



TYPE
PRINTED / MADE PART — AI-generated CAD geometry

FILE
ZS-304_sensor_sled

MATERIAL / COLOUR
PETG grey

PRINT ORIENTATION / SETTINGS
Flat; 0.2 mm

QTY (THIS FILE)
1

FASTENERS / FIXING
2x M2x6 PT (ZS-F03) into ZS-303

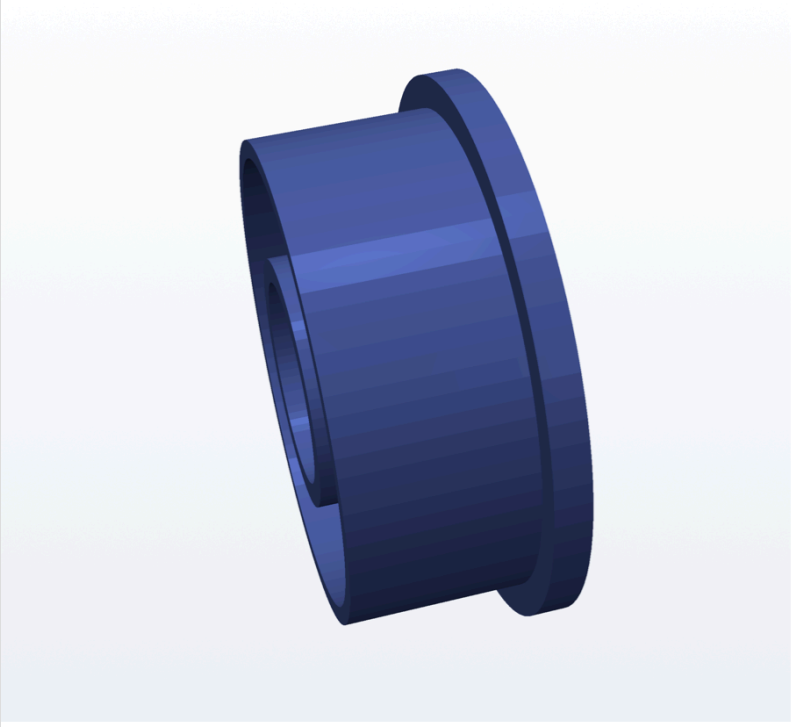
SUB-ASSEMBLY
torso

ENVELOPE W X D X H
36.0 x 3.0 x 52.0 mm (ESTIMATE, CAD)

SOLID VOLUME
5.5 cm3 (CAD, before infill)

DESIGN TARGET — not printed or test-fitted · CONCEPT DESIGN / NOT BUILT HARDWARE
System logic: Cory Davis (NavisWORLD) · CAD geometry: AI-generated (CadQuery) · PX = third-party module proxy

ZS-305
Chest core housing



TYPE
PRINTED / MADE PART — AI-generated CAD geometry

FILE
ZS-305_chest_core_housing

MATERIAL / COLOUR
PETG black

PRINT ORIENTATION / SETTINGS
Flange down; 0.16 mm, 100%

QTY (THIS FILE)
1

FASTENERS / FIXING
Flange captured behind torso front shell

SUB-ASSEMBLY
torso

ENVELOPE W X D X H
46.0 x 18.0 x 46.0 mm (ESTIMATE, CAD)

SOLID VOLUME
8.1 cm3 (CAD, before infill)

DESIGN TARGET — not printed or test-fitted · CONCEPT DESIGN / NOT BUILT HARDWARE
System logic: Cory Davis (NavisWORLD) · CAD geometry: AI-generated (CadQuery) · PX = third-party module proxy

ZS-306
Chest ring diffuser

TYPE
PRINTED / MADE PART — AI-generated CAD geometry

FILE
ZS-306_chest_ring_diffuser

MATERIAL / COLOUR
Translucent white PETG (cyan from LEDs)

PRINT ORIENTATION / SETTINGS
Flat; 100%

QTY (THIS FILE)
1

FASTENERS / FIXING
Press / glue

SUB-ASSEMBLY
torso

ENVELOPE W X D X H
37.2 x 12.5 x 37.2 mm (ESTIMATE, CAD)

SOLID VOLUME
9.0 cm3 (CAD, before infill)

DESIGN TARGET — not printed or test-fitted · CONCEPT DESIGN / NOT BUILT HARDWARE
System logic: Cory Davis (NavisWORLD) · CAD geometry: AI-generated (CadQuery) · PX = third-party module proxy

ZS-307
Chest centre lens

TYPE
PRINTED / MADE PART — AI-generated CAD geometry

FILE
ZS-307_chest_center_lens

MATERIAL / COLOUR
Clear (if MAX30102) or violet translucent

PRINT ORIENTATION / SETTINGS
Dome up; resin or PETG 100%

QTY (THIS FILE)
1

FASTENERS / FIXING
Glue into baffle

SUB-ASSEMBLY
torso

ENVELOPE W X D X H
16.6 x 16.0 x 16.6 mm (ESTIMATE, CAD)

SOLID VOLUME
3.4 cm3 (CAD, before infill)

DESIGN TARGET — not printed or test-fitted · CONCEPT DESIGN / NOT BUILT HARDWARE
System logic: Cory Davis (NavisWORLD) · CAD geometry: AI-generated (CadQuery) · PX = third-party module proxy

ZS-401
Arm + mitten hand, left

TYPE
PRINTED / MADE PART — AI-generated CAD geometry

FILE
ZS-401_arm_L

MATERIAL / COLOUR
PETG matte navy #161C33

PRINT ORIENTATION / SETTINGS
Peg up; 0.16 mm, supports under hand

QTY (THIS FILE)
1

FASTENERS / FIXING
Flanged peg captured by torso split

SUB-ASSEMBLY
limbs

ENVELOPE W X D X H
29.5 x 26.0 x 63.0 mm (ESTIMATE, CAD)

SOLID VOLUME
15.4 cm3 (CAD, before infill)

DESIGN TARGET — not printed or test-fitted · CONCEPT DESIGN / NOT BUILT HARDWARE
System logic: Cory Davis (NavisWORLD) · CAD geometry: AI-generated (CadQuery) · PX = third-party module proxy

ZS-401
Arm + mitten hand, right

TYPE
PRINTED / MADE PART — AI-generated CAD geometry

FILE
ZS-401_arm_R

MATERIAL / COLOUR
PETG matte navy #161C33

PRINT ORIENTATION / SETTINGS
Peg up; 0.16 mm

QTY (THIS FILE)
1

FASTENERS / FIXING
Flanged peg captured

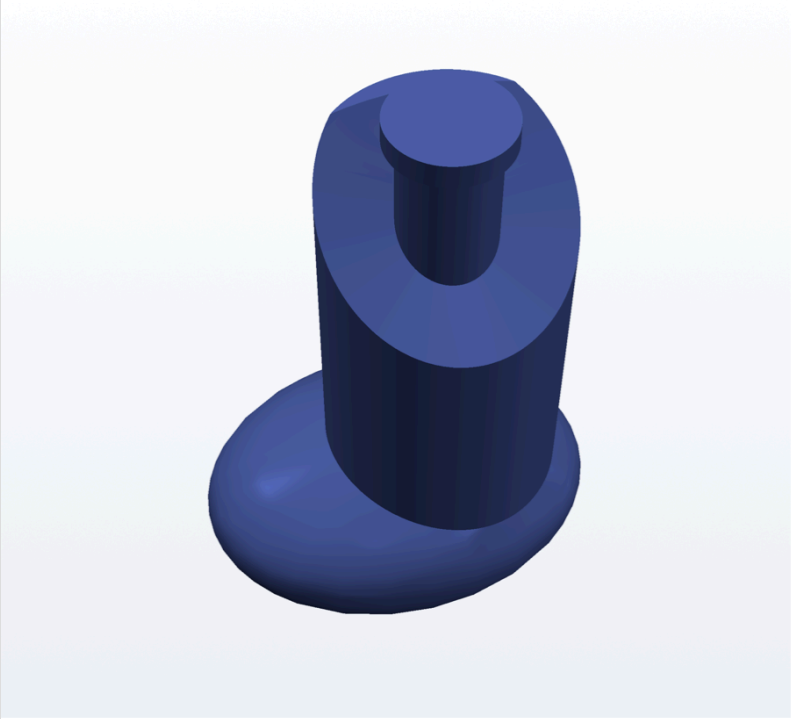
SUB-ASSEMBLY
limbs

ENVELOPE W X D X H
29.5 x 26.0 x 63.0 mm (ESTIMATE, CAD)

SOLID VOLUME
15.4 cm3 (CAD, before infill)

DESIGN TARGET — not printed or test-fitted · CONCEPT DESIGN / NOT BUILT HARDWARE
System logic: Cory Davis (NavisWORLD) · CAD geometry: AI-generated (CadQuery) · PX = third-party module proxy

ZS-402
Leg + foot, left



TYPE
PRINTED / MADE PART — AI-generated CAD geometry

FILE
ZS-402_leg_L

MATERIAL / COLOUR
PETG matte navy #161C33

PRINT ORIENTATION / SETTINGS
Peg up; 0.16 mm, supports under foot

QTY (THIS FILE)
1

FASTENERS / FIXING
Flanged peg captured

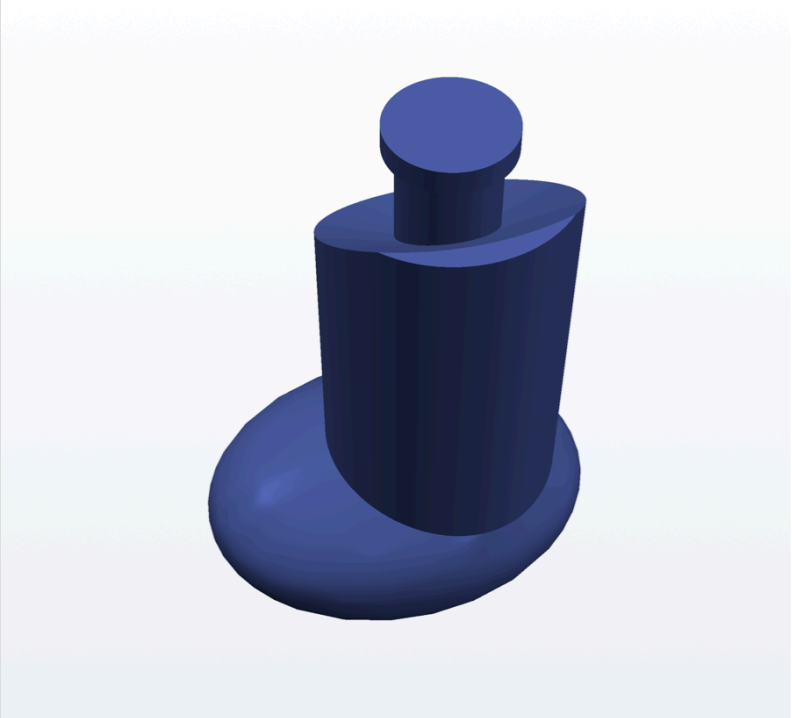
SUB-ASSEMBLY
limbs

ENVELOPE W X D X H
26.0 x 34.0 x 50.0 mm (ESTIMATE, CAD)

SOLID VOLUME
16.0 cm3 (CAD, before infill)

DESIGN TARGET — not printed or test-fitted · CONCEPT DESIGN / NOT BUILT HARDWARE
System logic: Cory Davis (NavisWORLD) · CAD geometry: AI-generated (CadQuery) · PX = third-party module proxy

ZS-402
Leg + foot, right



TYPE
PRINTED / MADE PART — AI-generated CAD geometry

FILE
ZS-402_leg_R

MATERIAL / COLOUR
PETG matte navy #161C33

PRINT ORIENTATION / SETTINGS
Peg up; 0.16 mm

QTY (THIS FILE)
1

FASTENERS / FIXING
Flanged peg captured

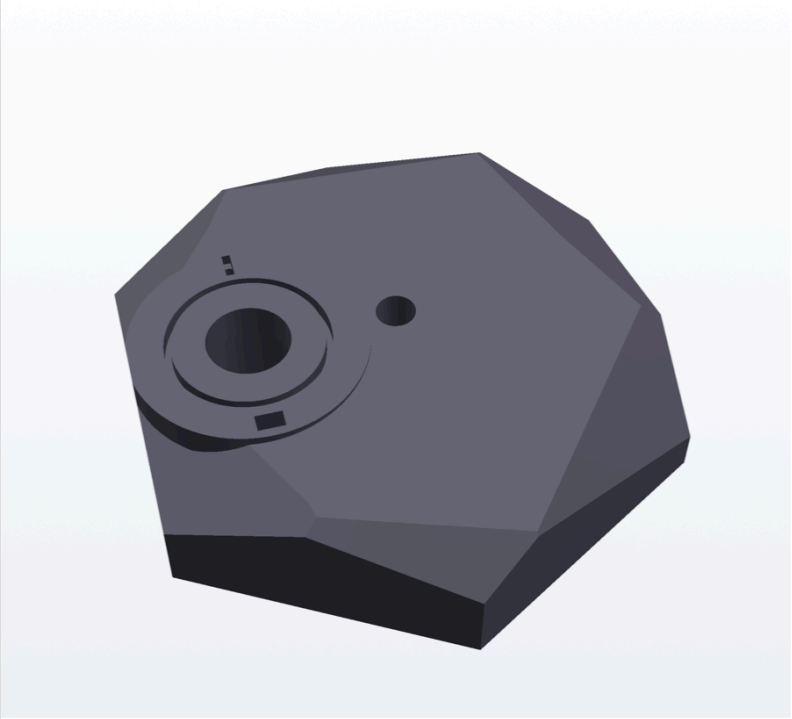
SUB-ASSEMBLY
limbs

ENVELOPE W X D X H
26.0 x 34.0 x 50.0 mm (ESTIMATE, CAD)

SOLID VOLUME
16.0 cm3 (CAD, before infill)

DESIGN TARGET — not printed or test-fitted · CONCEPT DESIGN / NOT BUILT HARDWARE
System logic: Cory Davis (NavisWORLD) · CAD geometry: AI-generated (CadQuery) · PX = third-party module proxy

ZS-501
Pedestal (rock hex) + station body



TYPE
PRINTED / MADE PART — AI-generated CAD geometry

FILE
ZS-501_pedestal

MATERIAL / COLOUR
PLA dark grey/black; 15% infill, station walls 100%

PRINT ORIENTATION / SETTINGS
Bottom down; 0.28 mm, 3 walls

QTY (THIS FILE)
1

FASTENERS / FIXING
3x felt pads; hosts ZS-601/604 screws

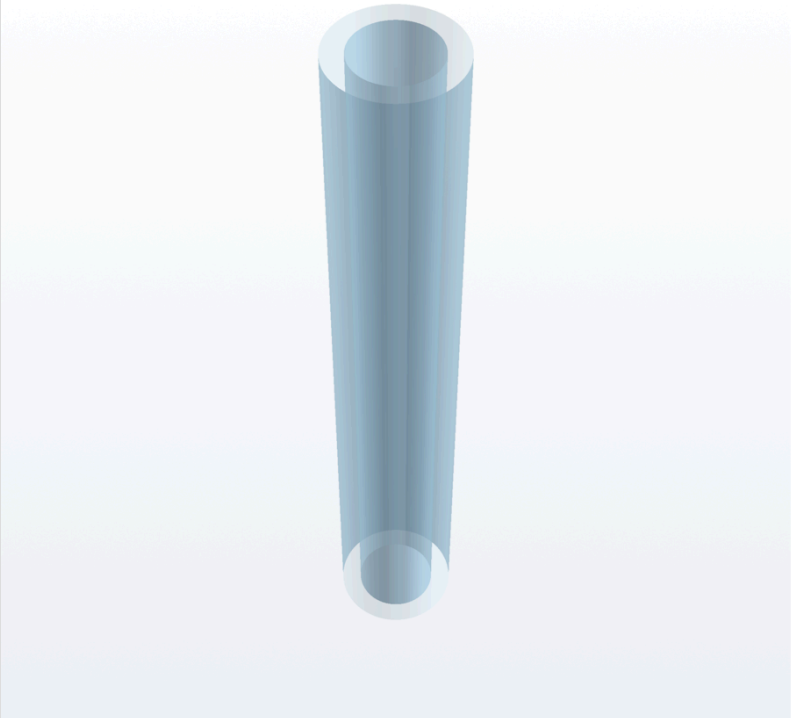
SUB-ASSEMBLY
pedestal

ENVELOPE W X D X H
190.0 x 164.5 x 45.0 mm (ESTIMATE, CAD)

SOLID VOLUME
884.2 cm3 (CAD, before infill)

DESIGN TARGET — not printed or test-fitted · CONCEPT DESIGN / NOT BUILT HARDWARE
System logic: Cory Davis (NavisWORLD) · CAD geometry: AI-generated (CadQuery) · PX = third-party module proxy

ZS-502
Support tube (bought, cut)



TYPE
BOUGHT STOCK (cut to length) — CAD geometry AI-generated

FILE
ZS-502_support_tube_acrylic

MATERIAL / COLOUR
Clear acrylic tube 12 OD x 8 ID x 75

PRINT ORIENTATION / SETTINGS
Cut + deburr (not printed)

QTY (THIS FILE)
1

FASTENERS / FIXING
Glued into pedestal; slip fit into torso

SUB-ASSEMBLY
pedestal

ENVELOPE W X D X H
12.0 x 12.0 x 75.0 mm (ESTIMATE, CAD)

SOLID VOLUME
4.7 cm3 (CAD, before infill)

DESIGN TARGET — not printed or test-fitted · CONCEPT DESIGN / NOT BUILT HARDWARE
System logic: Cory Davis (NavisWORLD) · CAD geometry: AI-generated (CadQuery) · PX = third-party module proxy

ZS-601
Station LED plug

TYPE
PRINTED / MADE PART — AI-generated CAD geometry

FILE
ZS-601_station_led_plug

MATERIAL / COLOUR
White PETG; cavity PTFE-tape lined

PRINT ORIENTATION / SETTINGS
Flange down; 0.2 mm, 100%

QTY (THIS FILE)
1

FASTENERS / FIXING
2x M3x8 PT into pedestal

SUB-ASSEMBLY
station

ENVELOPE W X D X H
44.0 x 44.0 x 31.0 mm (ESTIMATE, CAD)

SOLID VOLUME
22.7 cm3 (CAD, before infill)

DESIGN TARGET — not printed or test-fitted · CONCEPT DESIGN / NOT BUILT HARDWARE
System logic: Cory Davis (NavisWORLD) · CAD geometry: AI-generated (CadQuery) · PX = third-party module proxy

ZS-602
Light shield (removable)

TYPE
PRINTED / MADE PART — AI-generated CAD geometry

FILE
ZS-602_light_shield

MATERIAL / COLOUR
Opaque black PETG/PLA; inside matte black

PRINT ORIENTATION / SETTINGS
Top down; 4 walls, 100% top

QTY (THIS FILE)
1

FASTENERS / FIXING
Magnetic pogo + gravity (no screws)

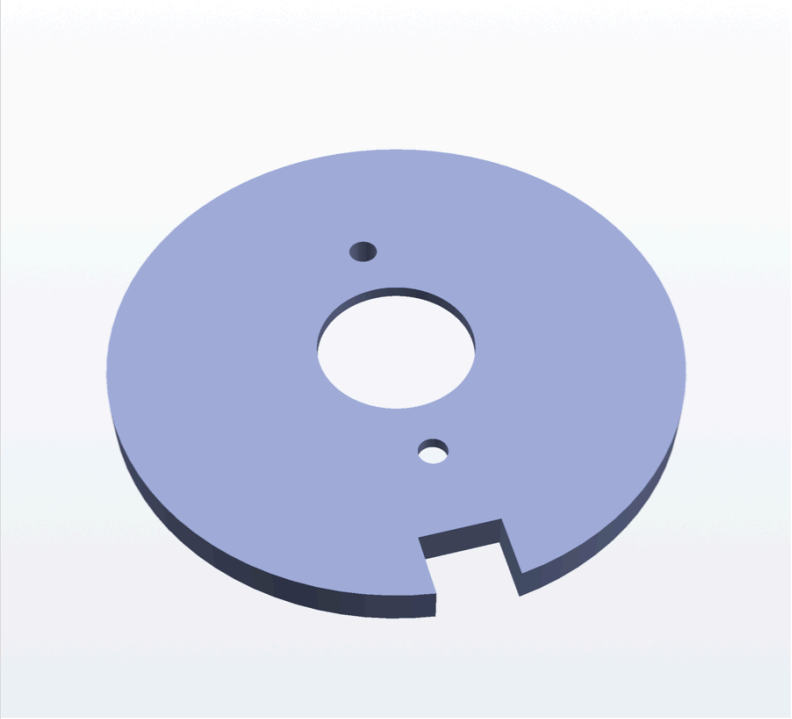
SUB-ASSEMBLY
station

ENVELOPE W X D X H
72.0 x 72.0 x 47.6 mm (ESTIMATE, CAD)

SOLID VOLUME
32.4 cm3 (CAD, before infill)

DESIGN TARGET — not printed or test-fitted · CONCEPT DESIGN / NOT BUILT HARDWARE
System logic: Cory Davis (NavisWORLD) · CAD geometry: AI-generated (CadQuery) · PX = third-party module proxy

ZS-603
Sensor mount disc (AS7341)



TYPE
PRINTED / MADE PART — AI-generated CAD geometry

FILE
ZS-603_sensor_mount

MATERIAL / COLOUR
Matte black PETG

PRINT ORIENTATION / SETTINGS
Flat; 100%

QTY (THIS FILE)
1

FASTENERS / FIXING
2x M2x5 to AS7341 board; glued into shield

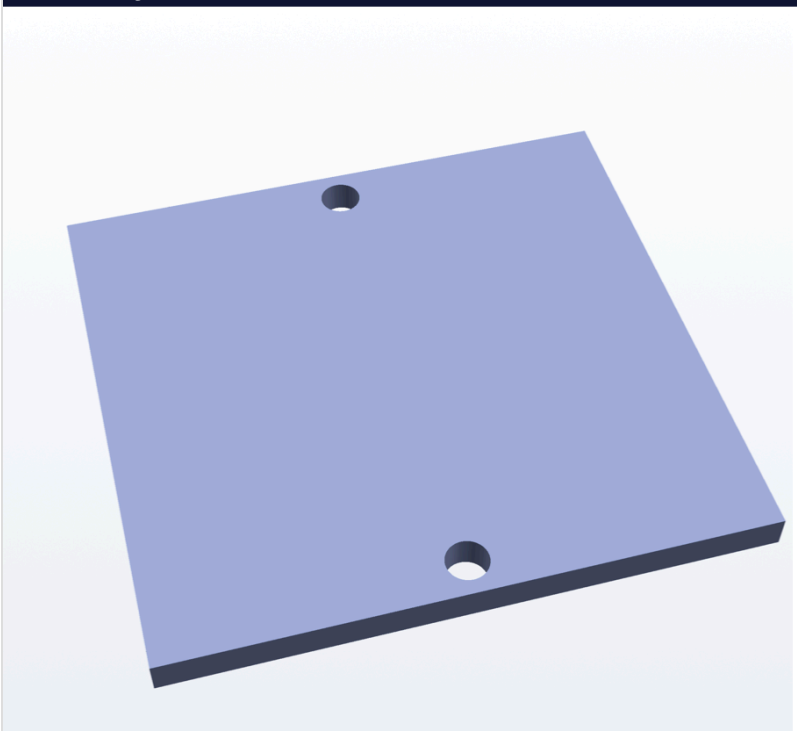
SUB-ASSEMBLY
station

ENVELOPE W X D X H
43.4 x 43.6 x 3.0 mm (ESTIMATE, CAD)

SOLID VOLUME
3.6 cm3 (CAD, before infill)

DESIGN TARGET — not printed or test-fitted · CONCEPT DESIGN / NOT BUILT HARDWARE
System logic: Cory Davis (NavisWORLD) · CAD geometry: AI-generated (CadQuery) · PX = third-party module proxy

ZS-604
Driver bay cover



TYPE
PRINTED / MADE PART — AI-generated CAD geometry

FILE
ZS-604_driver_bay_cover

MATERIAL / COLOUR
PLA black

PRINT ORIENTATION / SETTINGS
Flat; 0.2 mm

QTY (THIS FILE)
1

FASTENERS / FIXING
2x M2x6 PT into pedestal

SUB-ASSEMBLY
station

ENVELOPE W X D X H
32.0 x 34.0 x 2.0 mm (ESTIMATE, CAD)

SOLID VOLUME
2.2 cm3 (CAD, before infill)

DESIGN TARGET — not printed or test-fitted · CONCEPT DESIGN / NOT BUILT HARDWARE
System logic: Cory Davis (NavisWORLD) · CAD geometry: AI-generated (CadQuery) · PX = third-party module proxy

ZS-605
Dark-reference plug

TYPE
PRINTED / MADE PART — AI-generated CAD geometry

FILE
ZS-605_dark_reference_plug

MATERIAL / COLOUR
Opaque black PLA 100%

PRINT ORIENTATION / SETTINGS
Knob up; 100%

QTY (THIS FILE)
1

FASTENERS / FIXING
None (drop into cup)

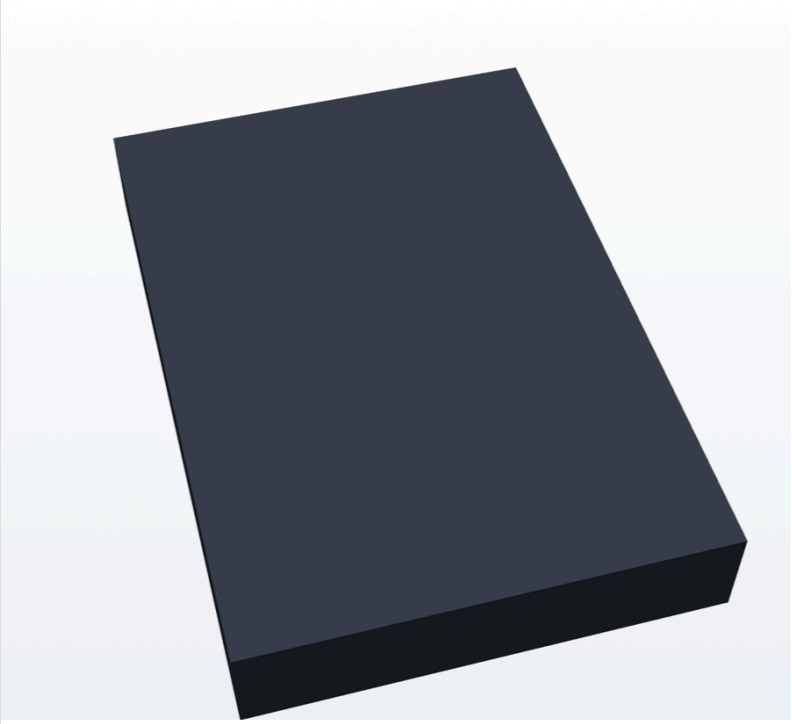
SUB-ASSEMBLY
station

ENVELOPE W X D X H
25.4 x 25.4 x 20.0 mm (ESTIMATE, CAD)

SOLID VOLUME
7.0 cm3 (CAD, before infill)

DESIGN TARGET — not printed or test-fitted · CONCEPT DESIGN / NOT BUILT HARDWARE
System logic: Cory Davis (NavisWORLD) · CAD geometry: AI-generated (CadQuery) · PX = third-party module proxy

PX (third-party proxy)
AS7341 11-ch spectral breakout (ZS-S01)



TYPE
THIRD-PARTY MODULE — simplified proxy, not the real part geometry

FILE
PX_AS7341_breakout

MATERIAL / COLOUR
Third-party breakout (Adafruit 4698 or equiv.)

PRINT ORIENTATION / SETTINGS
Bought

QTY (THIS FILE)
1

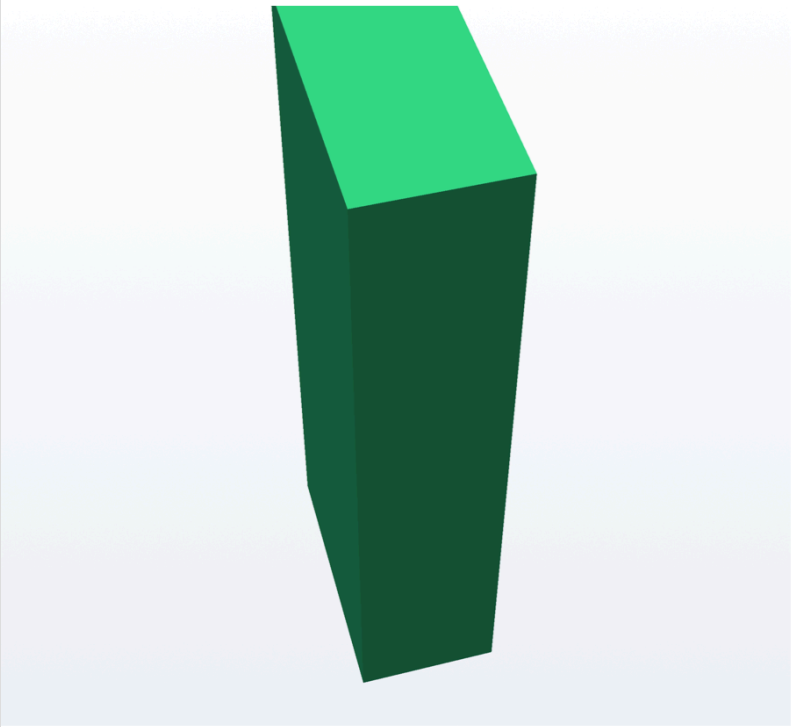
FASTENERS / FIXING
2x M2x5 to ZS-603

SUB-ASSEMBLY
station

ENVELOPE W X D X H
25.6 x 17.8 x 5.0 mm (ESTIMATE, CAD)

DESIGN TARGET — not printed or test-fitted · CONCEPT DESIGN / NOT BUILT HARDWARE
System logic: Cory Davis (NavisWORLD) · CAD geometry: AI-generated (CadQuery) · PX = third-party module proxy

PX (third-party proxy)
ESP32-S3 DevKitC-1 (RB-001)



TYPE
THIRD-PARTY MODULE — simplified proxy, not the real part geometry

FILE
PX_RB-001_esp32s3_devkitc1

MATERIAL / COLOUR
Espressif module — third party

PRINT ORIENTATION / SETTINGS
Bought

QTY (THIS FILE)
1

FASTENERS / FIXING
Rails + 2 zip ties on ZS-303

SUB-ASSEMBLY
torso

ENVELOPE W X D X H
26.0 x 14.0 x 69.0 mm (ESTIMATE, CAD)

DESIGN TARGET — not printed or test-fitted · CONCEPT DESIGN / NOT BUILT HARDWARE
System logic: Cory Davis (NavisWORLD) · CAD geometry: AI-generated (CadQuery) · PX = third-party module proxy

PX (third-party proxy)
ICM-42688-P IMU breakout (RB-002)

TYPE
THIRD-PARTY MODULE — simplified proxy, not the real part geometry

FILE
PX_RB-002_icm42688_imu

MATERIAL / COLOUR
Third-party breakout

PRINT ORIENTATION / SETTINGS
Bought

QTY (THIS FILE)
1

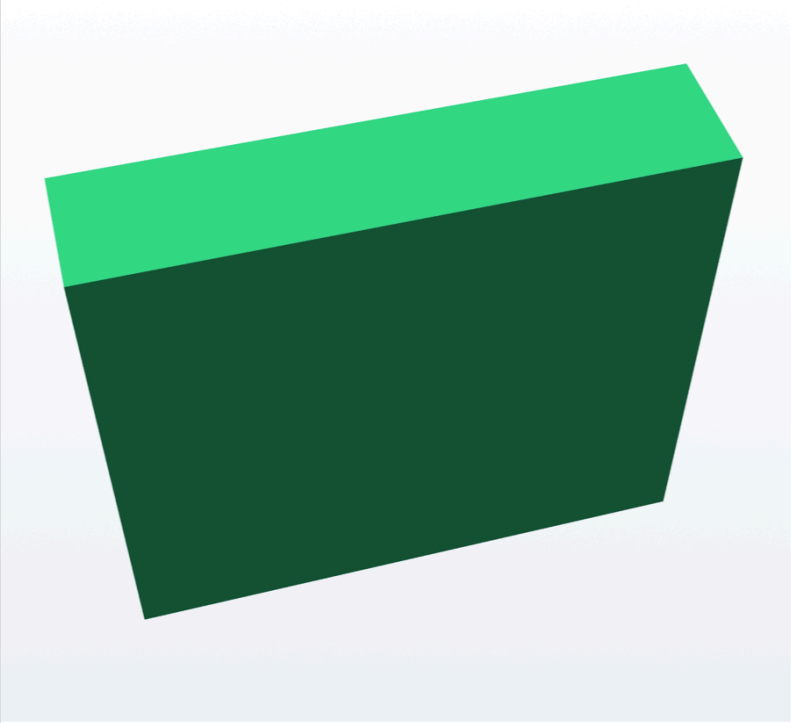
FASTENERS / FIXING
On ZS-304 (tape / M2)

SUB-ASSEMBLY
torso

ENVELOPE W X D X H
23.0 x 4.0 x 18.0 mm (ESTIMATE, CAD)

DESIGN TARGET — not printed or test-fitted · CONCEPT DESIGN / NOT BUILT HARDWARE
System logic: Cory Davis (NavisWORLD) · CAD geometry: AI-generated (CadQuery) · PX = third-party module proxy

PX (third-party proxy)
BH1750 light breakout (RB-003)



TYPE
THIRD-PARTY MODULE — simplified proxy, not the real part geometry

FILE
PX_RB-003_bh1750_podR

MATERIAL / COLOUR
Third-party breakout

PRINT ORIENTATION / SETTINGS
Bought

QTY (THIS FILE)
1

FASTENERS / FIXING
Glued in right pod

SUB-ASSEMBLY
pods

ENVELOPE W X D X H
4.0 x 18.0 x 16.0 mm (ESTIMATE, CAD)

DESIGN TARGET — not printed or test-fitted · CONCEPT DESIGN / NOT BUILT HARDWARE
System logic: Cory Davis (NavisWORLD) · CAD geometry: AI-generated (CadQuery) · PX = third-party module proxy

PX (third-party proxy)
BME280 breakout (RB-004)

TYPE
THIRD-PARTY MODULE — simplified proxy, not the real part geometry

FILE
PX_RB-004_bme280

MATERIAL / COLOUR
Third-party breakout

PRINT ORIENTATION / SETTINGS
Bought

QTY (THIS FILE)
1

FASTENERS / FIXING
On ZS-304

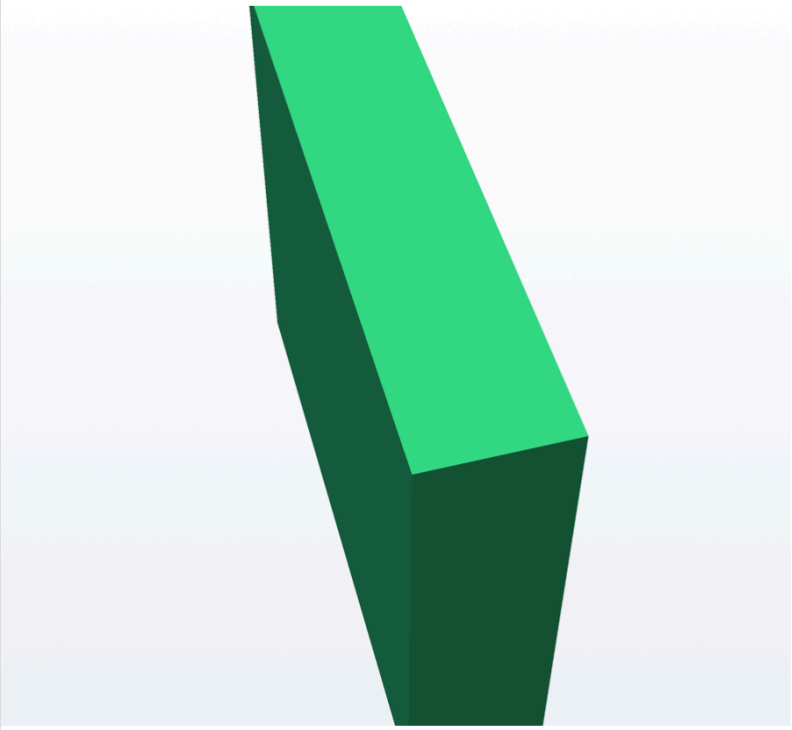
SUB-ASSEMBLY
torso

ENVELOPE W X D X H
18.0 x 4.0 x 16.0 mm (ESTIMATE, CAD)

DESIGN TARGET — not printed or test-fitted · CONCEPT DESIGN / NOT BUILT HARDWARE
System logic: Cory Davis (NavisWORLD) · CAD geometry: AI-generated (CadQuery) · PX = third-party module proxy

PX (third-party proxy)

MAX30102 breakout (RB-005, optional, non-medical)



TYPE
THIRD-PARTY MODULE — simplified proxy, not the real part geometry

FILE
PX_RB-005_max30102

MATERIAL / COLOUR
Third-party breakout

PRINT ORIENTATION / SETTINGS
Bought

QTY (THIS FILE)
1

FASTENERS / FIXING
Glued behind ZS-305

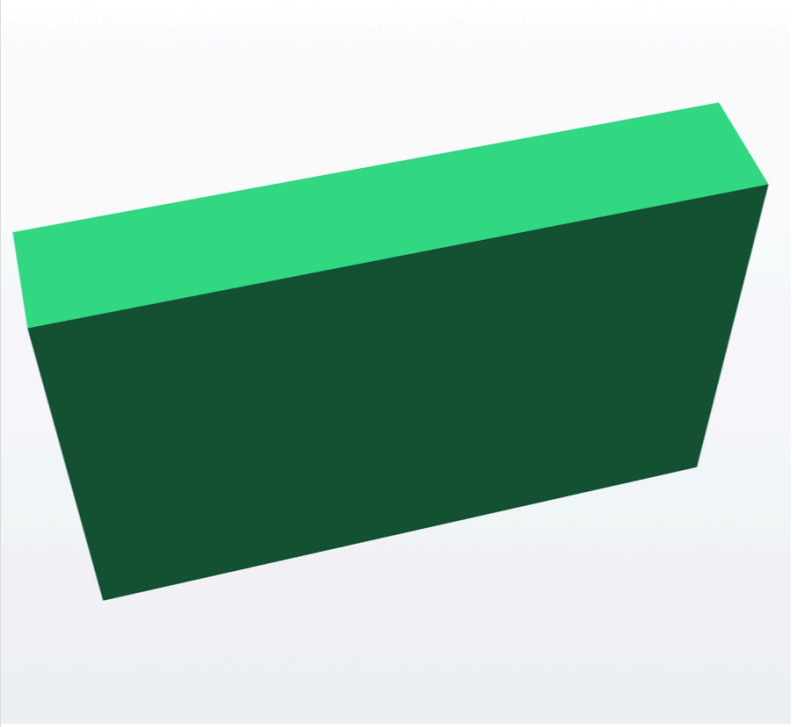
SUB-ASSEMBLY
torso

ENVELOPE W X D X H
22.0 x 5.0 x 18.0 mm (ESTIMATE, CAD)

DESIGN TARGET — not printed or test-fitted · CONCEPT DESIGN / NOT BUILT HARDWARE
System logic: Cory Davis (NavisWORLD) · CAD geometry: AI-generated (CadQuery) · PX = third-party module proxy

PX (third-party proxy)

INMP441 I2S mic breakout (RB-006)



TYPE
THIRD-PARTY MODULE — simplified proxy, not the real part geometry

FILE
PX_RB-006_inmp441_podL

MATERIAL / COLOUR
Third-party breakout

PRINT ORIENTATION / SETTINGS
Bought

QTY (THIS FILE)
1

FASTENERS / FIXING
Glued in left pod

SUB-ASSEMBLY
pods

ENVELOPE W X D X H
4.0 x 23.0 x 15.0 mm (ESTIMATE, CAD)

DESIGN TARGET — not printed or test-fitted · CONCEPT DESIGN / NOT BUILT HARDWARE
System logic: Cory Davis (NavisWORLD) · CAD geometry: AI-generated (CadQuery) · PX = third-party module proxy

PX (third-party proxy)
NEO-M9N-class GNSS (RB-007, optional)

TYPE
THIRD-PARTY MODULE — simplified proxy, not the real part geometry

FILE
PX_RB-007_gnss_head

MATERIAL / COLOUR
Third-party breakout

PRINT ORIENTATION / SETTINGS
Bought

QTY (THIS FILE)
1

FASTENERS / FIXING
Tape on ZS-103 crown

SUB-ASSEMBLY
head

ENVELOPE W X D X H
30.0 x 6.0 x 25.0 mm (ESTIMATE, CAD)

DESIGN TARGET — not printed or test-fitted · CONCEPT DESIGN / NOT BUILT HARDWARE
System logic: Cory Davis (NavisWORLD) · CAD geometry: AI-generated (CadQuery) · PX = third-party module proxy

PX (third-party proxy)
Crystal sample (example only)

TYPE
THIRD-PARTY MODULE — simplified proxy, not the real part geometry

FILE
PX_crystal_sample

MATERIAL / COLOUR
User's crystal

PRINT ORIENTATION / SETTINGS
—

QTY (THIS FILE)
1

FASTENERS / FIXING
Sits in cup

SUB-ASSEMBLY
station

ENVELOPE W X D X H
17.4 x 17.4 x 33.0 mm (ESTIMATE, CAD)

DESIGN TARGET — not printed or test-fitted · CONCEPT DESIGN / NOT BUILT HARDWARE
System logic: Cory Davis (NavisWORLD) · CAD geometry: AI-generated (CadQuery) · PX = third-party module proxy

PX (third-party proxy)
Window disc 29.6 x 2 (ZS-S02)

TYPE
THIRD-PARTY MODULE — simplified proxy, not the real part geometry

FILE
PX_crystal_window_borosilicate

MATERIAL / COLOUR
Borosilicate / fused silica

PRINT ORIENTATION / SETTINGS
Bought

QTY (THIS FILE)
1

FASTENERS / FIXING
Held by ZS-601

SUB-ASSEMBLY
station

ENVELOPE W X D X H
29.6 x 29.6 x 2.0 mm (ESTIMATE, CAD)

DESIGN TARGET — not printed or test-fitted · CONCEPT DESIGN / NOT BUILT HARDWARE
System logic: Cory Davis (NavisWORLD) · CAD geometry: AI-generated (CadQuery) · PX = third-party module proxy

PX (third-party proxy)
Crown accent LED board, 2x WS2812B (ZS-E03)

TYPE
THIRD-PARTY MODULE — simplified proxy, not the real part geometry

FILE
PX_led_accent_pcb

MATERIAL / COLOUR
Third-party / custom

PRINT ORIENTATION / SETTINGS
Bought

QTY (THIS FILE)
1

FASTENERS / FIXING
Glued under ZS-107

SUB-ASSEMBLY
head

ENVELOPE W X D X H
40.0 x 8.0 x 3.0 mm (ESTIMATE, CAD)

DESIGN TARGET — not printed or test-fitted · CONCEPT DESIGN / NOT BUILT HARDWARE
System logic: Cory Davis (NavisWORLD) · CAD geometry: AI-generated (CadQuery) · PX = third-party module proxy

PX (third-party proxy)
Eye LED strip L, 5x WS2812B (ZS-E02)

TYPE
THIRD-PARTY MODULE — simplified proxy, not the real part geometry

FILE
PX_led_eyeL

MATERIAL / COLOUR
Third-party strip

PRINT ORIENTATION / SETTINGS
Bought

QTY (THIS FILE)
1

FASTENERS / FIXING
Adhesive on ZS-105

SUB-ASSEMBLY
head

ENVELOPE W X D X H
28.0 x 2.0 x 15.0 mm (ESTIMATE, CAD)

DESIGN TARGET — not printed or test-fitted · CONCEPT DESIGN / NOT BUILT HARDWARE
System logic: Cory Davis (NavisWORLD) · CAD geometry: AI-generated (CadQuery) · PX = third-party module proxy

PX (third-party proxy)
Eye LED strip R, 5x WS2812B (ZS-E02)

TYPE
THIRD-PARTY MODULE — simplified proxy, not the real part geometry

FILE
PX_led_eyeR

MATERIAL / COLOUR
Third-party strip

PRINT ORIENTATION / SETTINGS
Bought

QTY (THIS FILE)
1

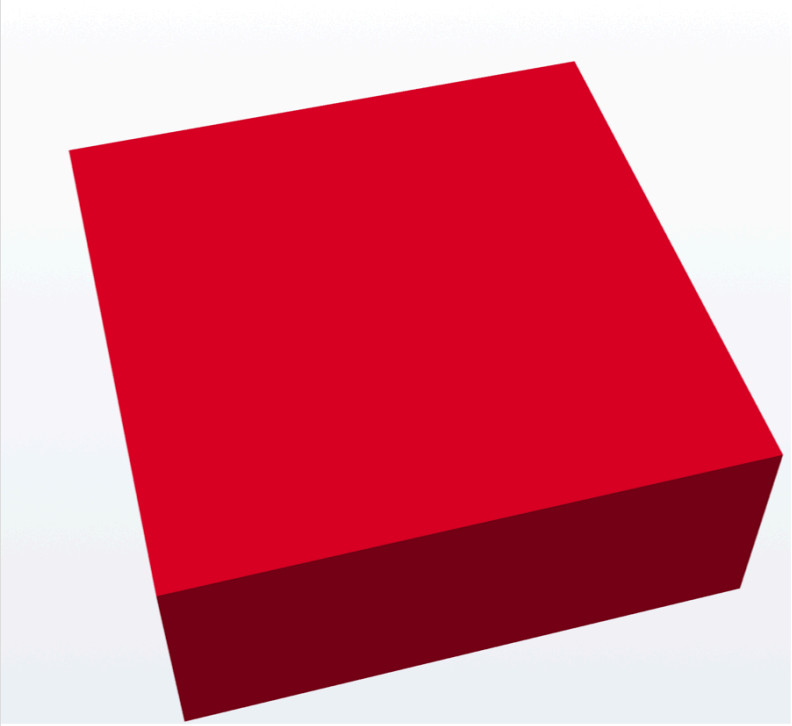
FASTENERS / FIXING
Adhesive on ZS-105

SUB-ASSEMBLY
head

ENVELOPE W X D X H
28.0 x 2.0 x 15.0 mm (ESTIMATE, CAD)

DESIGN TARGET — not printed or test-fitted · CONCEPT DESIGN / NOT BUILT HARDWARE
System logic: Cory Davis (NavisWORLD) · CAD geometry: AI-generated (CadQuery) · PX = third-party module proxy

PX (third-party proxy)
940 nm IR LED (ZS-S06, optional)



TYPE
THIRD-PARTY MODULE — simplified proxy, not the real part geometry

FILE
PX_led_ir940

MATERIAL / COLOUR
Third-party LED

PRINT ORIENTATION / SETTINGS
Bought

QTY (THIS FILE)
1

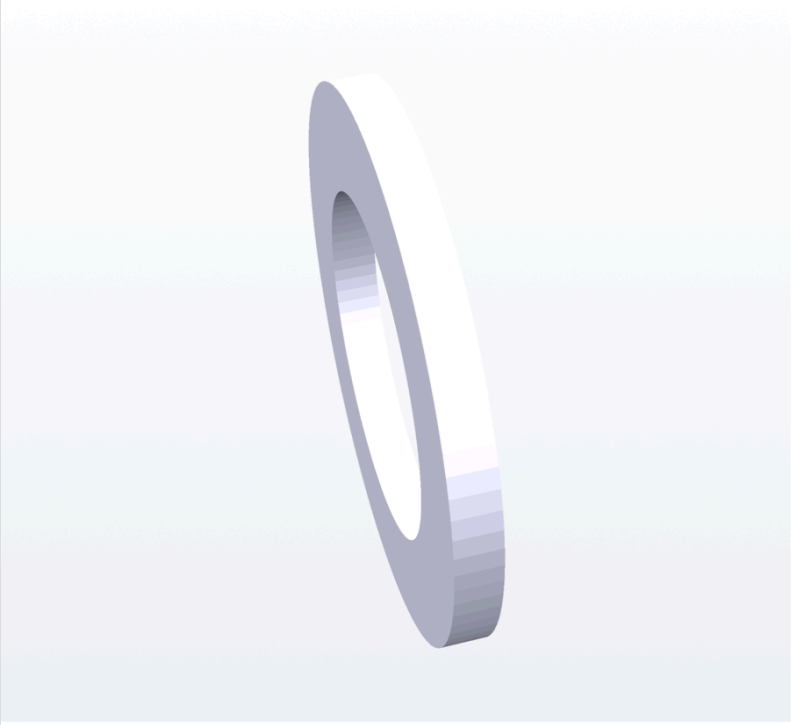
FASTENERS / FIXING
Soldered to ZS-S07

SUB-ASSEMBLY
station

ENVELOPE W X D X H
3.5 x 3.5 x 1.4 mm (ESTIMATE, CAD)

DESIGN TARGET — not printed or test-fitted · CONCEPT DESIGN / NOT BUILT HARDWARE
System logic: Cory Davis (NavisWORLD) · CAD geometry: AI-generated (CadQuery) · PX = third-party module proxy

PX (third-party proxy)
WS2812B 12-LED ring, chest (ZS-E01)



TYPE
THIRD-PARTY MODULE — simplified proxy, not the real part geometry

FILE
PX_led_ring12_chest

MATERIAL / COLOUR
Third-party ring

PRINT ORIENTATION / SETTINGS
Bought

QTY (THIS FILE)
1

FASTENERS / FIXING
Over ZS-305 baffle

SUB-ASSEMBLY
torso

ENVELOPE W X D X H
37.0 x 3.0 x 37.0 mm (ESTIMATE, CAD)

DESIGN TARGET — not printed or test-fitted · CONCEPT DESIGN / NOT BUILT HARDWARE
System logic: Cory Davis (NavisWORLD) · CAD geometry: AI-generated (CadQuery) · PX = third-party module proxy

PX (third-party proxy)
WS2812B 12-LED ring, left pod (ZS-E01)

TYPE
THIRD-PARTY MODULE — simplified proxy, not the real part geometry

FILE
PX_led_ring12_podL

MATERIAL / COLOUR
Third-party ring

PRINT ORIENTATION / SETTINGS
Bought

QTY (THIS FILE)
1

FASTENERS / FIXING
In pod cavity

SUB-ASSEMBLY
pods

ENVELOPE W X D X H
3.0 x 37.0 x 37.0 mm (ESTIMATE, CAD)

DESIGN TARGET — not printed or test-fitted · CONCEPT DESIGN / NOT BUILT HARDWARE
System logic: Cory Davis (NavisWORLD) · CAD geometry: AI-generated (CadQuery) · PX = third-party module proxy

PX (third-party proxy)
WS2812B 12-LED ring, right pod (ZS-E01)

TYPE
THIRD-PARTY MODULE — simplified proxy, not the real part geometry

FILE
PX_led_ring12_podR

MATERIAL / COLOUR
Third-party ring

PRINT ORIENTATION / SETTINGS
Bought

QTY (THIS FILE)
1

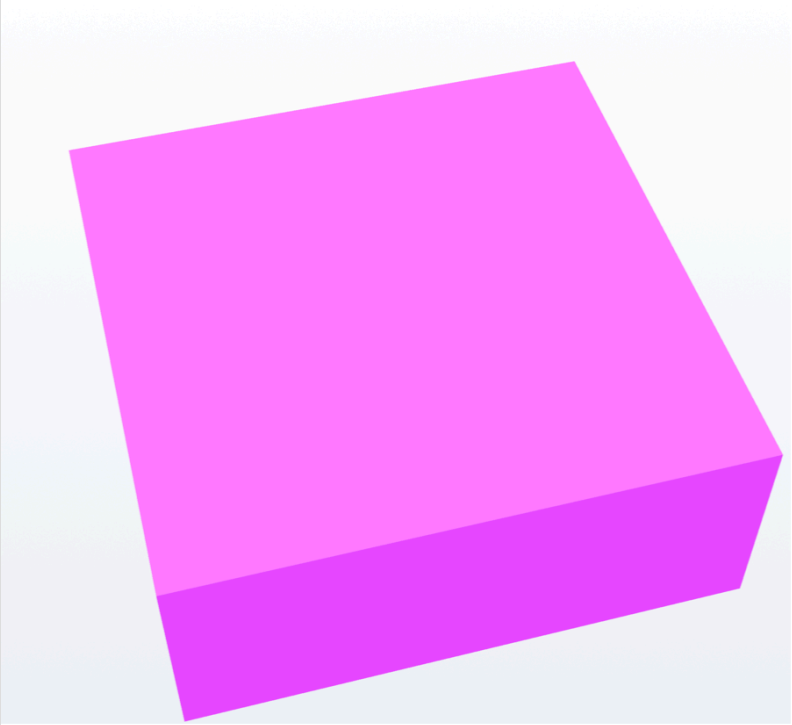
FASTENERS / FIXING
In pod cavity

SUB-ASSEMBLY
pods

ENVELOPE W X D X H
3.0 x 37.0 x 37.0 mm (ESTIMATE, CAD)

DESIGN TARGET — not printed or test-fitted · CONCEPT DESIGN / NOT BUILT HARDWARE
System logic: Cory Davis (NavisWORLD) · CAD geometry: AI-generated (CadQuery) · PX = third-party module proxy

PX (third-party proxy)
365 nm UV-A LED (ZS-S05) — EYE HAZARD



TYPE
THIRD-PARTY MODULE — simplified proxy, not the real part geometry

FILE
PX_led_uv365

MATERIAL / COLOUR
Third-party LED

PRINT ORIENTATION / SETTINGS
Bought

QTY (THIS FILE)
1

FASTENERS / FIXING
Soldered to ZS-S07

SUB-ASSEMBLY
station

ENVELOPE W X D X H
3.5 x 3.5 x 1.4 mm (ESTIMATE, CAD)

DESIGN TARGET — not printed or test-fitted · CONCEPT DESIGN / NOT BUILT HARDWARE
System logic: Cory Davis (NavisWORLD) · CAD geometry: AI-generated (CadQuery) · PX = third-party module proxy

PX (third-party proxy)
White LED (ZS-S04)

TYPE
THIRD-PARTY MODULE — simplified proxy, not the real part geometry

FILE
PX_led_white

MATERIAL / COLOUR
Third-party LED

PRINT ORIENTATION / SETTINGS
Bought

QTY (THIS FILE)
1

FASTENERS / FIXING
Soldered to ZS-S07

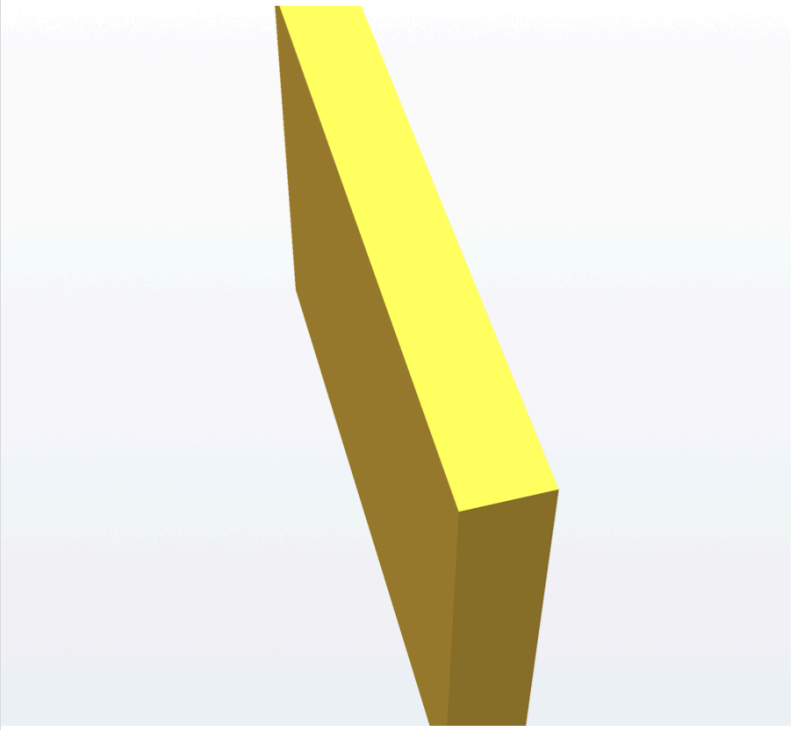
SUB-ASSEMBLY
station

ENVELOPE W X D X H
3.5 x 3.5 x 1.4 mm (ESTIMATE, CAD)

DESIGN TARGET — not printed or test-fitted · CONCEPT DESIGN / NOT BUILT HARDWARE
System logic: Cory Davis (NavisWORLD) · CAD geometry: AI-generated (CadQuery) · PX = third-party module proxy

PX (third-party proxy)

1S LiPo 503450 ~1000 mAh, protected (ZS-B01)



TYPE
THIRD-PARTY MODULE — simplified proxy, not the real part geometry

FILE
PX_lipo_503450_protected

MATERIAL / COLOUR
Third-party cell WITH PCM

PRINT ORIENTATION / SETTINGS
Bought

QTY (THIS FILE)
1

FASTENERS / FIXING
Slides into ZS-109

SUB-ASSEMBLY
power

ENVELOPE W X D X H
50.0 x 6.0 x 34.0 mm (ESTIMATE, CAD)

DESIGN TARGET — not printed or test-fitted · CONCEPT DESIGN / NOT BUILT HARDWARE
System logic: Cory Davis (NavisWORLD) · CAD geometry: AI-generated (CadQuery) · PX = third-party module proxy

PX (third-party proxy)

Magnet 6 x 2 mm (ZS-S10)

TYPE
THIRD-PARTY MODULE — simplified proxy, not the real part geometry

FILE
PX_magnet_6x2

MATERIAL / COLOUR
NdFeB

PRINT ORIENTATION / SETTINGS
Bought

QTY (THIS FILE)
1

FASTENERS / FIXING
Glued in ZS-602 flange

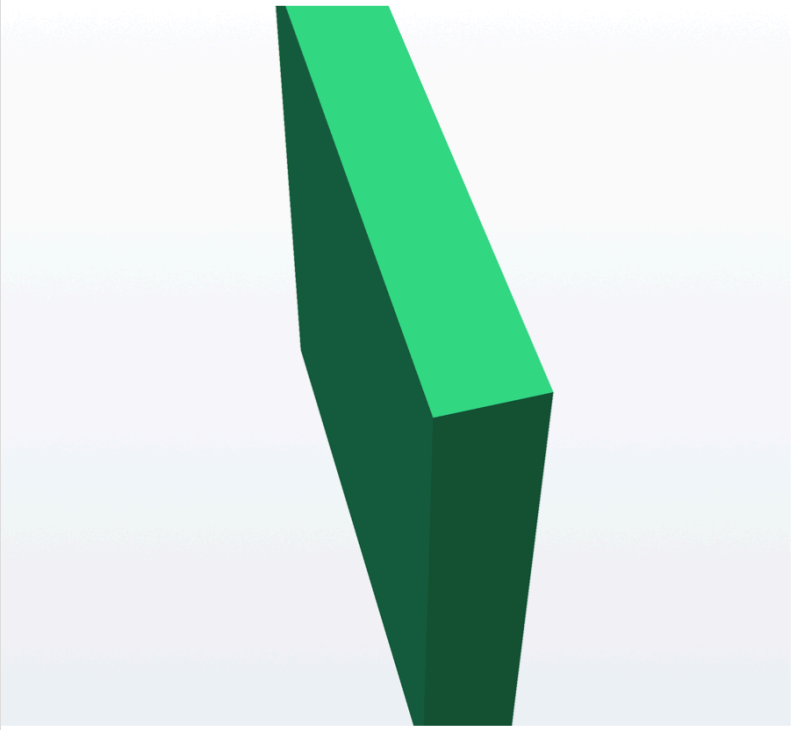
SUB-ASSEMBLY
station

ENVELOPE W X D X H
6.0 x 6.0 x 2.0 mm (ESTIMATE, CAD)

DESIGN TARGET — not printed or test-fitted · CONCEPT DESIGN / NOT BUILT HARDWARE
System logic: Cory Davis (NavisWORLD) · CAD geometry: AI-generated (CadQuery) · PX = third-party module proxy

PX (third-party proxy)

MAX98357A I2S amp breakout (ZS-E05)



TYPE
THIRD-PARTY MODULE — simplified proxy, not the real part geometry

FILE
PX_max98357a_amp

MATERIAL / COLOUR
Third-party breakout

PRINT ORIENTATION / SETTINGS
Bought

QTY (THIS FILE)
1

FASTENERS / FIXING
Tape / M2 on ZS-103

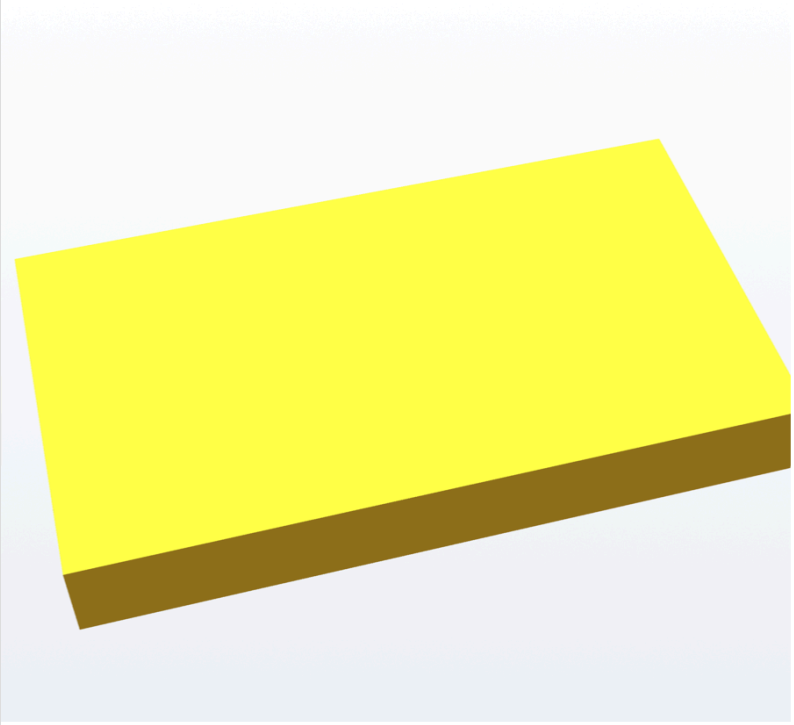
SUB-ASSEMBLY
head

ENVELOPE W X D X H
18.0 x 3.0 x 18.0 mm (ESTIMATE, CAD)

DESIGN TARGET — not printed or test-fitted · CONCEPT DESIGN / NOT BUILT HARDWARE
System logic: Cory Davis (NavisWORLD) · CAD geometry: AI-generated (CadQuery) · PX = third-party module proxy

PX (third-party proxy)

Pogo pads 4-pin (ZS-S11)



TYPE
THIRD-PARTY MODULE — simplified proxy, not the real part geometry

FILE
PX_pogo_pads_4p

MATERIAL / COLOUR
Third-party magnetic pogo

PRINT ORIENTATION / SETTINGS
Bought

QTY (THIS FILE)
1

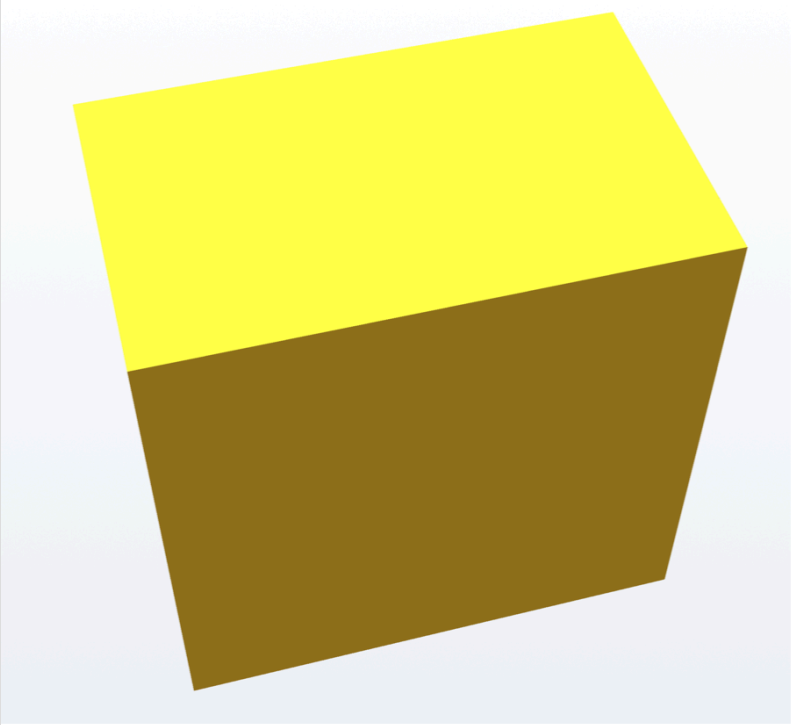
FASTENERS / FIXING
Glued in ZS-602 flange pocket

SUB-ASSEMBLY
station

ENVELOPE W X D X H
5.0 x 9.0 x 1.2 mm (ESTIMATE, CAD)

DESIGN TARGET — not printed or test-fitted · CONCEPT DESIGN / NOT BUILT HARDWARE
System logic: Cory Davis (NavisWORLD) · CAD geometry: AI-generated (CadQuery) · PX = third-party module proxy

PX (third-party proxy)
Pogo receptacle 4-pin (ZS-S11)



TYPE
THIRD-PARTY MODULE — simplified proxy, not the real part geometry

FILE
PX_pogo_receptacle_4p

MATERIAL / COLOUR
Third-party magnetic pogo

PRINT ORIENTATION / SETTINGS
Bought

QTY (THIS FILE)
1

FASTENERS / FIXING
Glued in pedestal pocket

SUB-ASSEMBLY
station

ENVELOPE W X D X H
5.0 x 8.0 x 8.0 mm (ESTIMATE, CAD)

DESIGN TARGET — not printed or test-fitted · CONCEPT DESIGN / NOT BUILT HARDWARE
System logic: Cory Davis (NavisWORLD) · CAD geometry: AI-generated (CadQuery) · PX = third-party module proxy

PX (third-party proxy)
Adafruit PowerBoost 1000C (ZS-B02)

TYPE
THIRD-PARTY MODULE — simplified proxy, not the real part geometry

FILE
PX_powerboost_1000c

MATERIAL / COLOUR
Third-party module

PRINT ORIENTATION / SETTINGS
Bought

QTY (THIS FILE)
1

FASTENERS / FIXING
2x M2 to ZS-103

SUB-ASSEMBLY
power

ENVELOPE W X D X H
36.0 x 6.0 x 23.0 mm (ESTIMATE, CAD)

DESIGN TARGET — not printed or test-fitted · CONCEPT DESIGN / NOT BUILT HARDWARE
System logic: Cory Davis (NavisWORLD) · CAD geometry: AI-generated (CadQuery) · PX = third-party module proxy

PX (third-party proxy)
PTFE white reference 25.4 x 3 (ZS-S12)

TYPE
THIRD-PARTY MODULE — simplified proxy, not the real part geometry

FILE
PX_ptfe_white_reference_25x3

MATERIAL / COLOUR
Virgin PTFE

PRINT ORIENTATION / SETTINGS
Bought

QTY (THIS FILE)
1

FASTENERS / FIXING
Drop into cup (calibration)

SUB-ASSEMBLY
station

ENVELOPE W X D X H
25.4 x 25.4 x 3.0 mm (ESTIMATE, CAD)

DESIGN TARGET — not printed or test-fitted · CONCEPT DESIGN / NOT BUILT HARDWARE
System logic: Cory Davis (NavisWORLD) · CAD geometry: AI-generated (CadQuery) · PX = third-party module proxy

PX (third-party proxy)
Reed switch (ZS-S09) — UV interlock

TYPE
THIRD-PARTY MODULE — simplified proxy, not the real part geometry

FILE
PX_reed_switch

MATERIAL / COLOUR
Glass reed, NO

PRINT ORIENTATION / SETTINGS
Bought

QTY (THIS FILE)
1

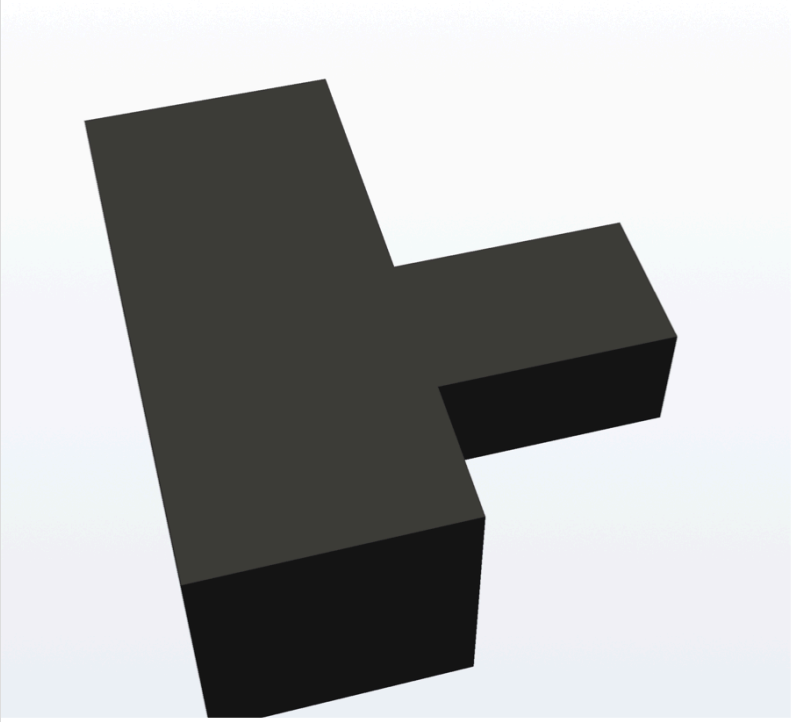
FASTENERS / FIXING
Glued in pedestal slot

SUB-ASSEMBLY
station

ENVELOPE W X D X H
8.0 x 2.2 x 2.2 mm (ESTIMATE, CAD)

DESIGN TARGET — not printed or test-fitted · CONCEPT DESIGN / NOT BUILT HARDWARE
System logic: Cory Davis (NavisWORLD) · CAD geometry: AI-generated (CadQuery) · PX = third-party module proxy

PX (third-party proxy)
Slide switch EN (ZS-B03)



TYPE
THIRD-PARTY MODULE — simplified proxy, not the real part geometry

FILE
PX_slide_switch_EN

MATERIAL / COLOUR
Third-party switch

PRINT ORIENTATION / SETTINGS
Bought

QTY (THIS FILE)
1

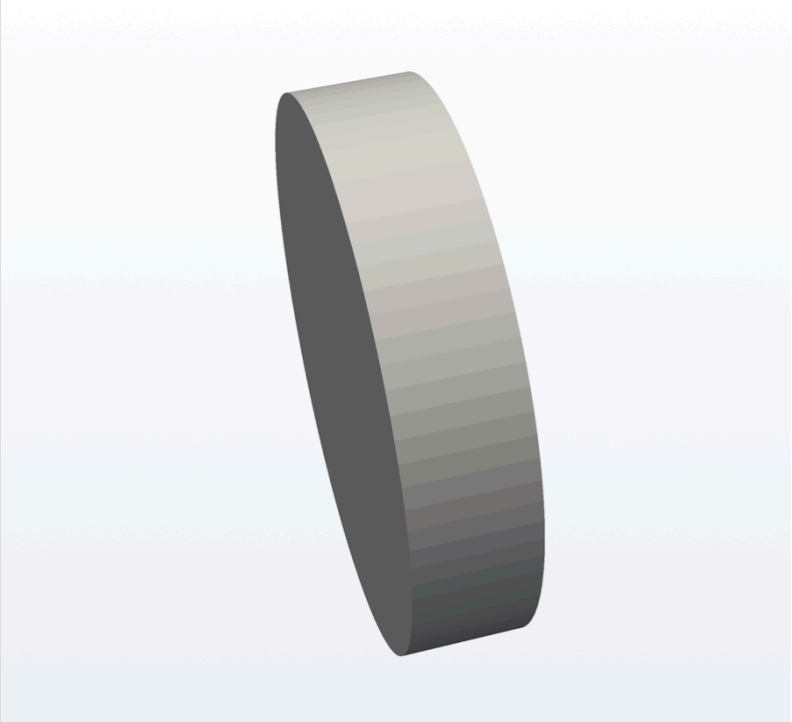
FASTENERS / FIXING
Glued in ZS-102 slot

SUB-ASSEMBLY
power

ENVELOPE W X D X H
9.0 x 8.0 x 4.0 mm (ESTIMATE, CAD)

DESIGN TARGET — not printed or test-fitted · CONCEPT DESIGN / NOT BUILT HARDWARE
System logic: Cory Davis (NavisWORLD) · CAD geometry: AI-generated (CadQuery) · PX = third-party module proxy

PX (third-party proxy)
28 mm 8 ohm speaker (ZS-E06)



TYPE
THIRD-PARTY MODULE — simplified proxy, not the real part geometry

FILE
PX_speaker28

MATERIAL / COLOUR
Third-party

PRINT ORIENTATION / SETTINGS
Bought

QTY (THIS FILE)
1

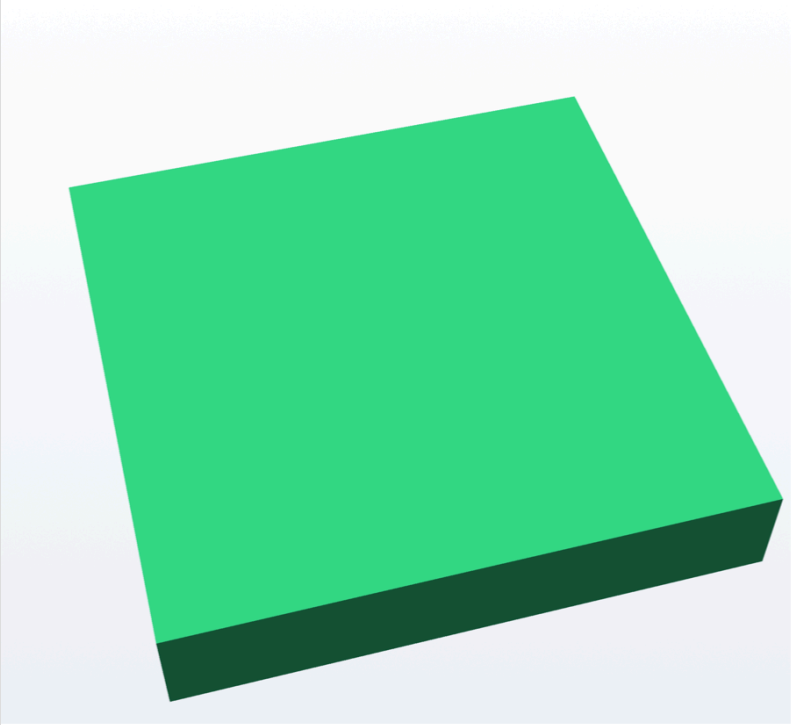
FASTENERS / FIXING
Glued in ZS-102 ring

SUB-ASSEMBLY
head

ENVELOPE W X D X H
28.0 x 6.0 x 28.0 mm (ESTIMATE, CAD)

DESIGN TARGET — not printed or test-fitted · CONCEPT DESIGN / NOT BUILT HARDWARE
System logic: Cory Davis (NavisWORLD) · CAD geometry: AI-generated (CadQuery) · PX = third-party module proxy

PX (third-party proxy)
Station driver board, Q1-Q3 (ZS-S08)



TYPE
THIRD-PARTY MODULE — simplified proxy, not the real part geometry

FILE
PX_station_driver_board

MATERIAL / COLOUR
Custom perfboard

PRINT ORIENTATION / SETTINGS
Build

QTY (THIS FILE)
1

FASTENERS / FIXING
Sits in bay under ZS-604

SUB-ASSEMBLY
station

ENVELOPE W X D X H
22.0 x 22.0 x 4.0 mm (ESTIMATE, CAD)

DESIGN TARGET — not printed or test-fitted · CONCEPT DESIGN / NOT BUILT HARDWARE
System logic: Cory Davis (NavisWORLD) · CAD geometry: AI-generated (CadQuery) · PX = third-party module proxy

PX (third-party proxy)
Station LED board 20x20 (ZS-S07)

TYPE
THIRD-PARTY MODULE — simplified proxy, not the real part geometry

FILE
PX_station_led_board

MATERIAL / COLOUR
Custom perfboard

PRINT ORIENTATION / SETTINGS
Build

QTY (THIS FILE)
1

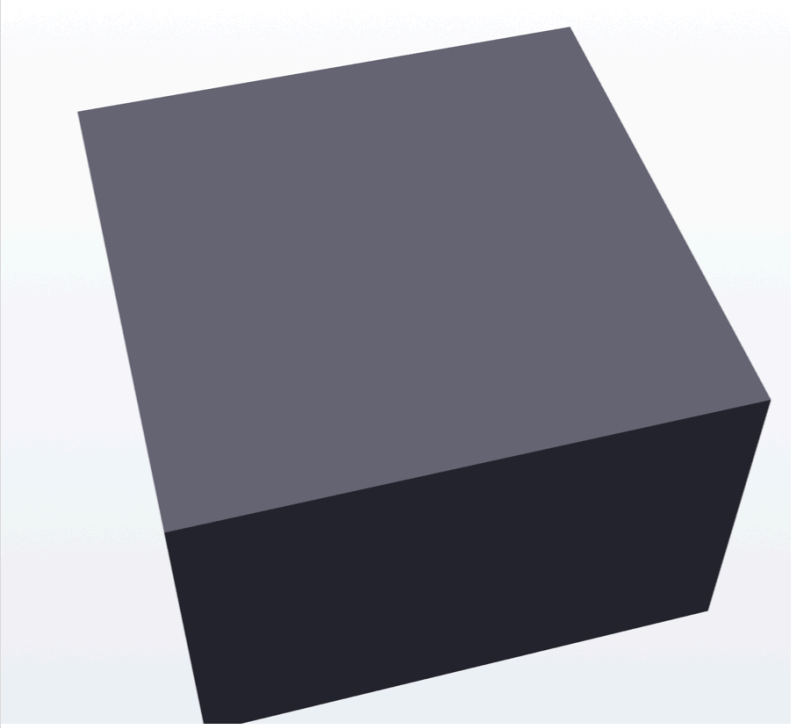
FASTENERS / FIXING
In ZS-601 pocket

SUB-ASSEMBLY
station

ENVELOPE W X D X H
20.0 x 20.0 x 1.6 mm (ESTIMATE, CAD)

DESIGN TARGET — not printed or test-fitted · CONCEPT DESIGN / NOT BUILT HARDWARE
System logic: Cory Davis (NavisWORLD) · CAD geometry: AI-generated (CadQuery) · PX = third-party module proxy

PX (third-party proxy)
USB-C receptacle breakout (ZS-E07)



TYPE
THIRD-PARTY MODULE — simplified proxy, not the real part geometry

FILE
PX_usbc_breakout_pedestal

MATERIAL / COLOUR
Third-party breakout

PRINT ORIENTATION / SETTINGS
Bought

QTY (THIS FILE)
1

FASTENERS / FIXING
Glued in pedestal back pocket

SUB-ASSEMBLY
pedestal

ENVELOPE W X D X H
12.0 x 12.0 x 8.0 mm (ESTIMATE, CAD)

DESIGN TARGET — not printed or test-fitted · CONCEPT DESIGN / NOT BUILT HARDWARE
System logic: Cory Davis (NavisWORLD) · CAD geometry: AI-generated (CadQuery) · PX = third-party module proxy

PX (third-party proxy)
UV-cut filter 20 x 2 (ZS-S03, optional)

TYPE
THIRD-PARTY MODULE — simplified proxy, not the real part geometry

FILE
PX_uv_cut_filter_optional

MATERIAL / COLOUR
~420 nm longpass glass or UV-block acrylic

PRINT ORIENTATION / SETTINGS
Bought

QTY (THIS FILE)
1

FASTENERS / FIXING
In ZS-603 recess

SUB-ASSEMBLY
station

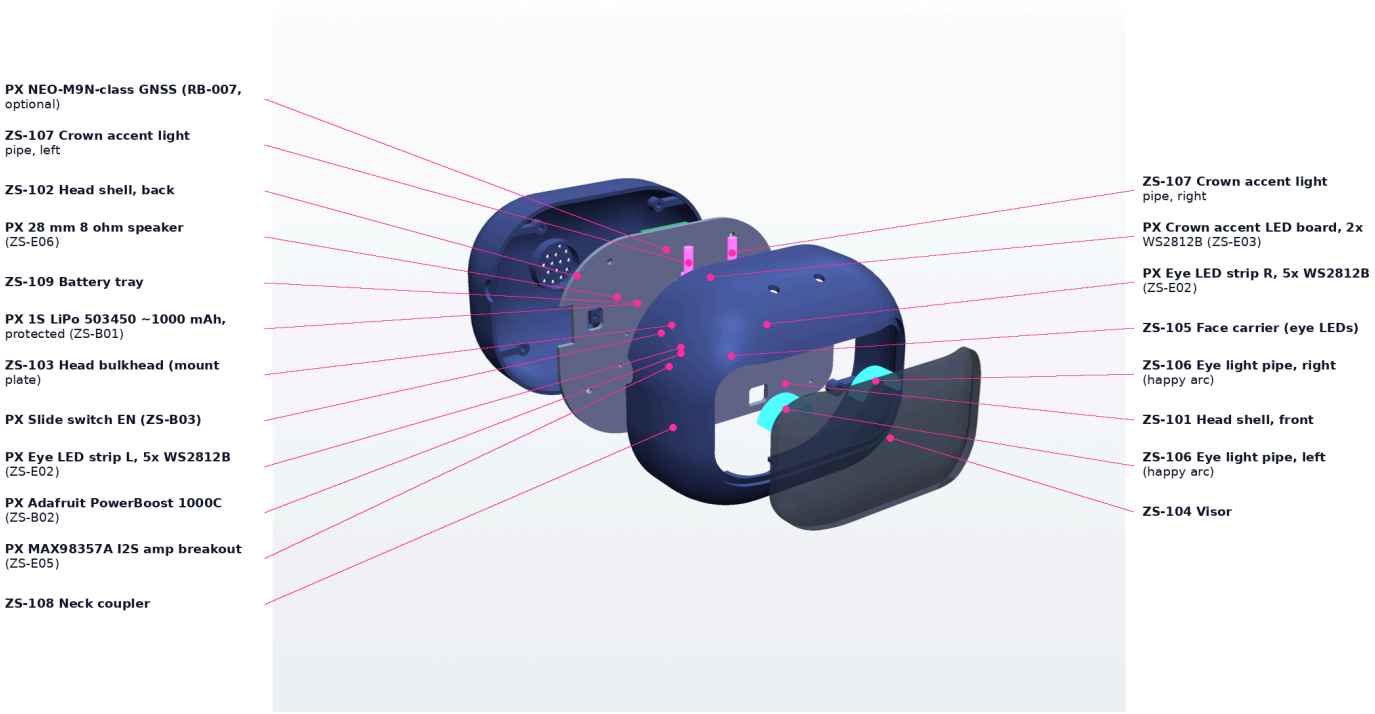
ENVELOPE W X D X H
20.0 x 20.0 x 2.0 mm (ESTIMATE, CAD)

DESIGN TARGET — not printed or test-fitted · CONCEPT DESIGN / NOT BUILT HARDWARE
System logic: Cory Davis (NavisWORLD) · CAD geometry: AI-generated (CadQuery) · PX = third-party module proxy

6. Sub-assemblies

Head (incl. power bay)

Head sub-assembly (incl. power bay)
Labels: part number + name (PX = third-party module proxy)



DESIGN TARGET — not printed or test-fitted · CONCEPT DESIGN / NOT BUILT HARDWARE
System logic: Cory Davis (NavisWORLD) · CAD geometry: AI-generated (CadQuery) · PX = third-party module proxy

Ear pods

Ear pods + ears (left = mic, right = light)

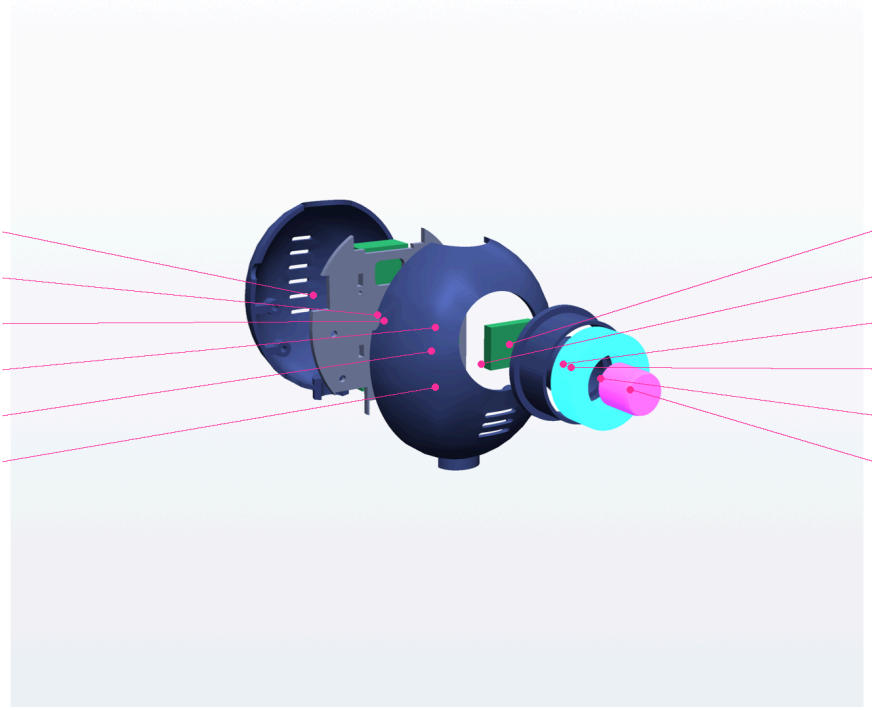
Labels: part number + name (PX = third-party module proxy)

DESIGN TARGET — not printed or test-fitted · CONCEPT DESIGN / NOT BUILT HARDWARE
System logic: Cory Davis (NavisWORLD) · CAD geometry: AI-generated (CadQuery) · PX = third-party module proxy

Torso / chest

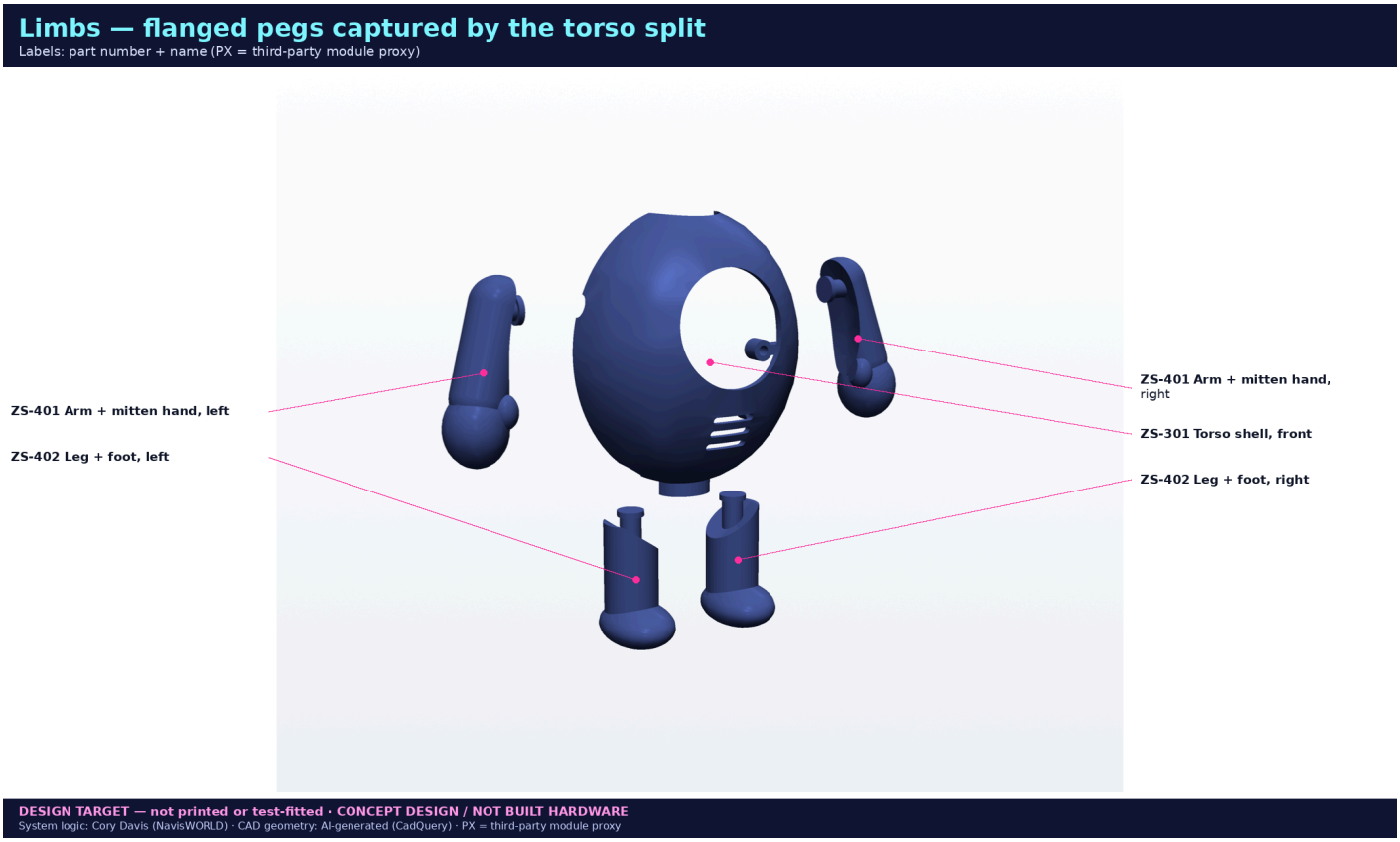
Torso + chest core + sensor sled + ESP32-S3 cradle

Labels: part number + name (PX = third-party module proxy)



DESIGN TARGET — not printed or test-fitted · CONCEPT DESIGN / NOT BUILT HARDWARE
System logic: Cory Davis (NavisWORLD) · CAD geometry: AI-generated (CadQuery) · PX = third-party module proxy

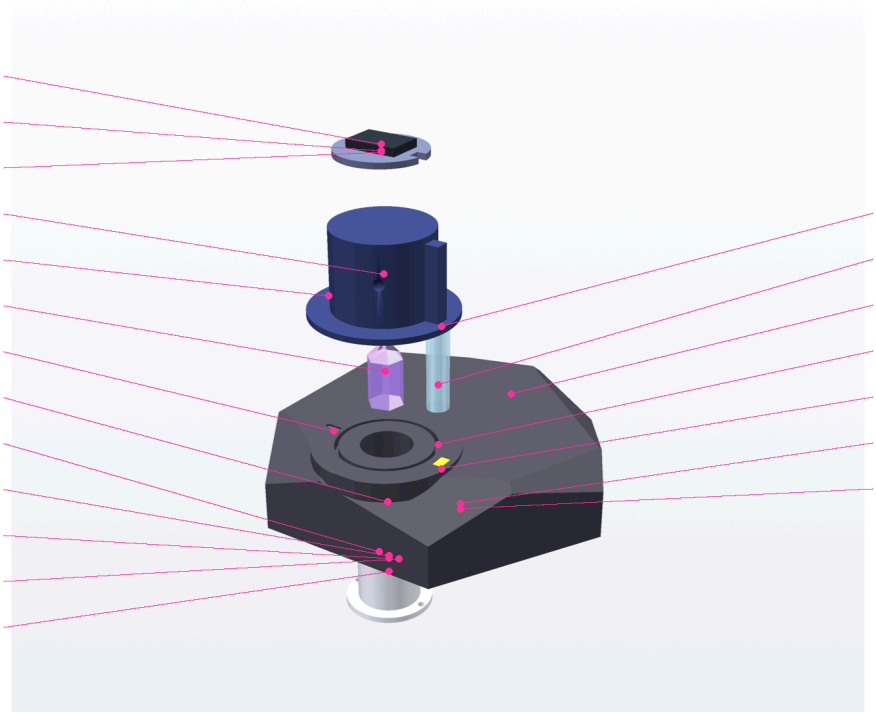
Limbs



Pedestal + crystal station

Pedestal + crystal-reading station

Labels: part number + name (PX = third-party module proxy)

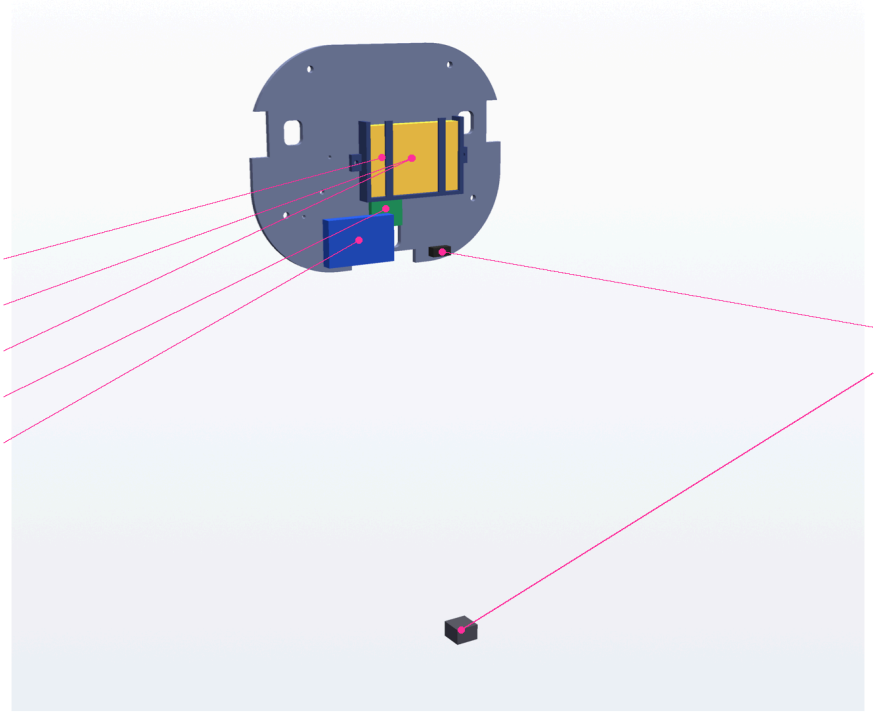


DESIGN TARGET — not printed or test-fitted · CONCEPT DESIGN / NOT BUILT HARDWARE
System logic: Cory Davis (NavisWORLD) · CAD geometry: AI-generated (CadQuery) · PX = third-party module proxy

Power

Power: protected LiPo + PowerBoost 1000C + EN switch (+ pedestal USB-C)

Labels: part number + name (PX = third-party module proxy)

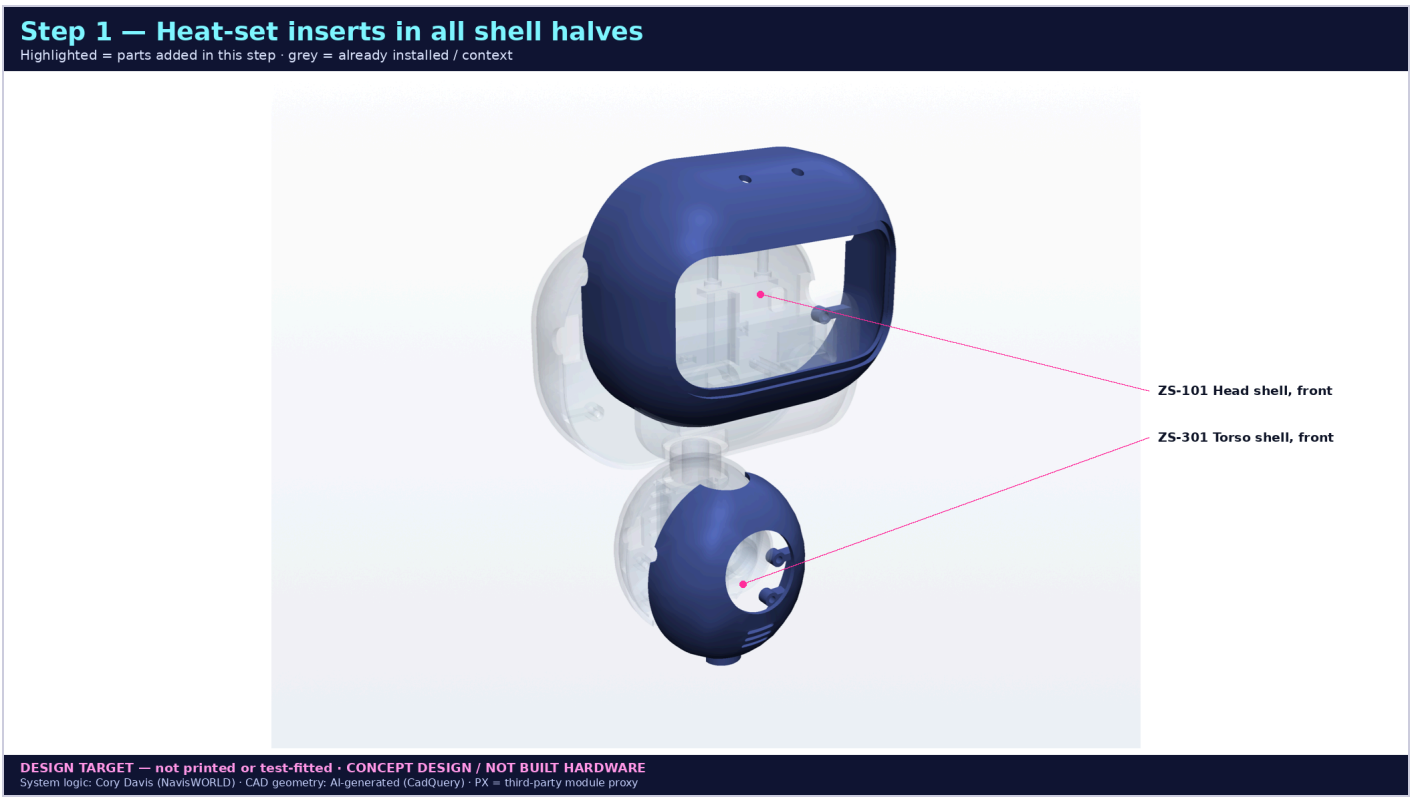


DESIGN TARGET — not printed or test-fitted · CONCEPT DESIGN / NOT BUILT HARDWARE
System logic: Cory Davis (NavisWORLD) · CAD geometry: AI-generated (CadQuery) · PX = third-party module proxy

7. Assembly steps

Each step: image (highlighted = new parts), fasteners, actions, wiring for that step (with its schematic sheet), and a check you must pass before moving on.

Step 1 — Heat-set inserts in all shell halves



Parts added: ZS-101 Head shell, front, ZS-301 Torso shell, front

Fasteners: 8 × ZS-F01 M3 heat-set inserts (short, brass): 4 in the ZS-101 head front bosses, 4 in the ZS-301 torso front bosses. Pilot 4.0 mm (ESTIMATE — match your insert).

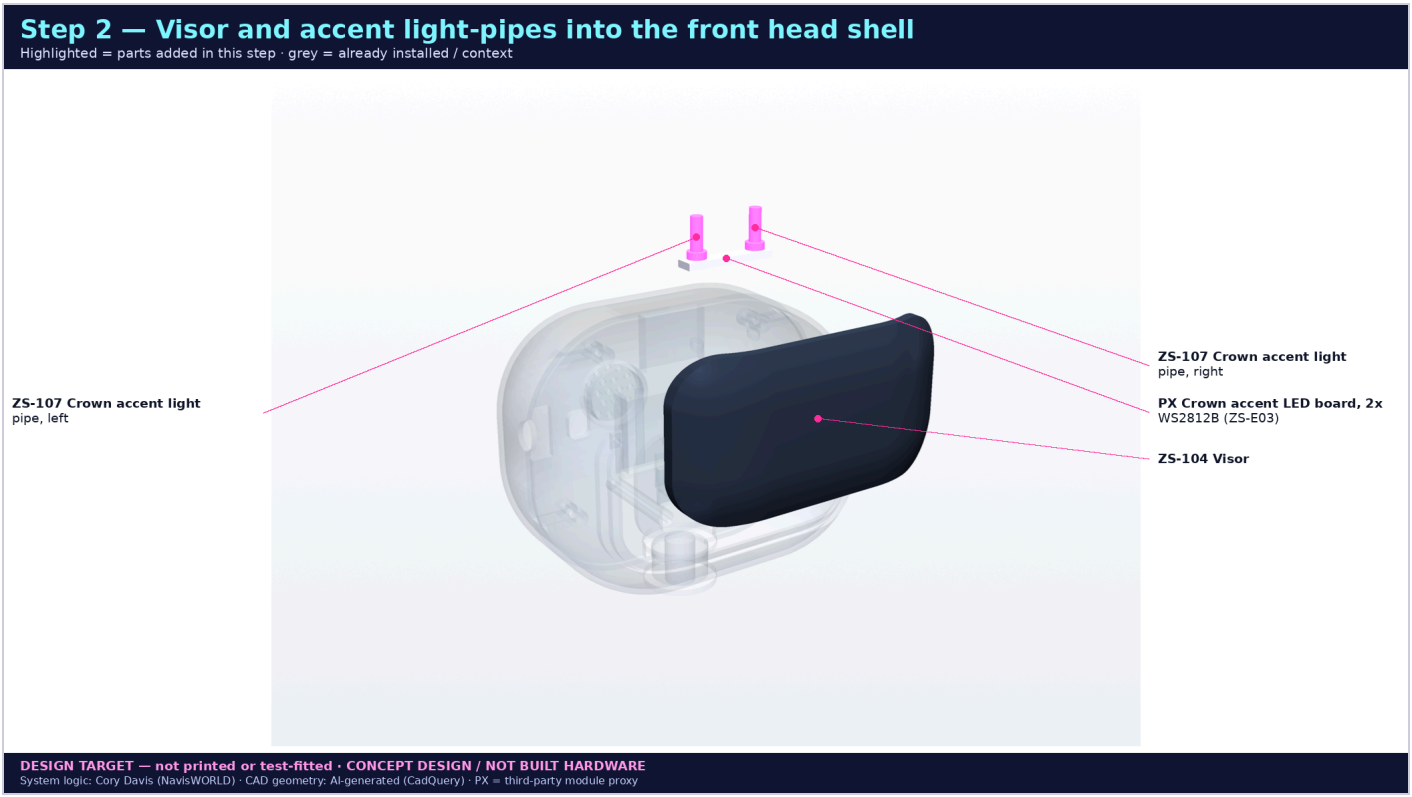
Do:

1. Iron at ~220 °C (PLA) / ~245 °C (PETG). Push each insert straight down with the insert tip; stop 0.2 mm proud and flatten with a flat block while warm.
2. Let cool 1 min before handling. Clear any plastic that squeezed into the thread with an M3 screw.

Wiring: None.

> **CHECK before moving on:** An M3 screw threads fully into every insert by hand; no insert spins or tilts.

Step 2 — Visor and accent light-pipes into the front head shell



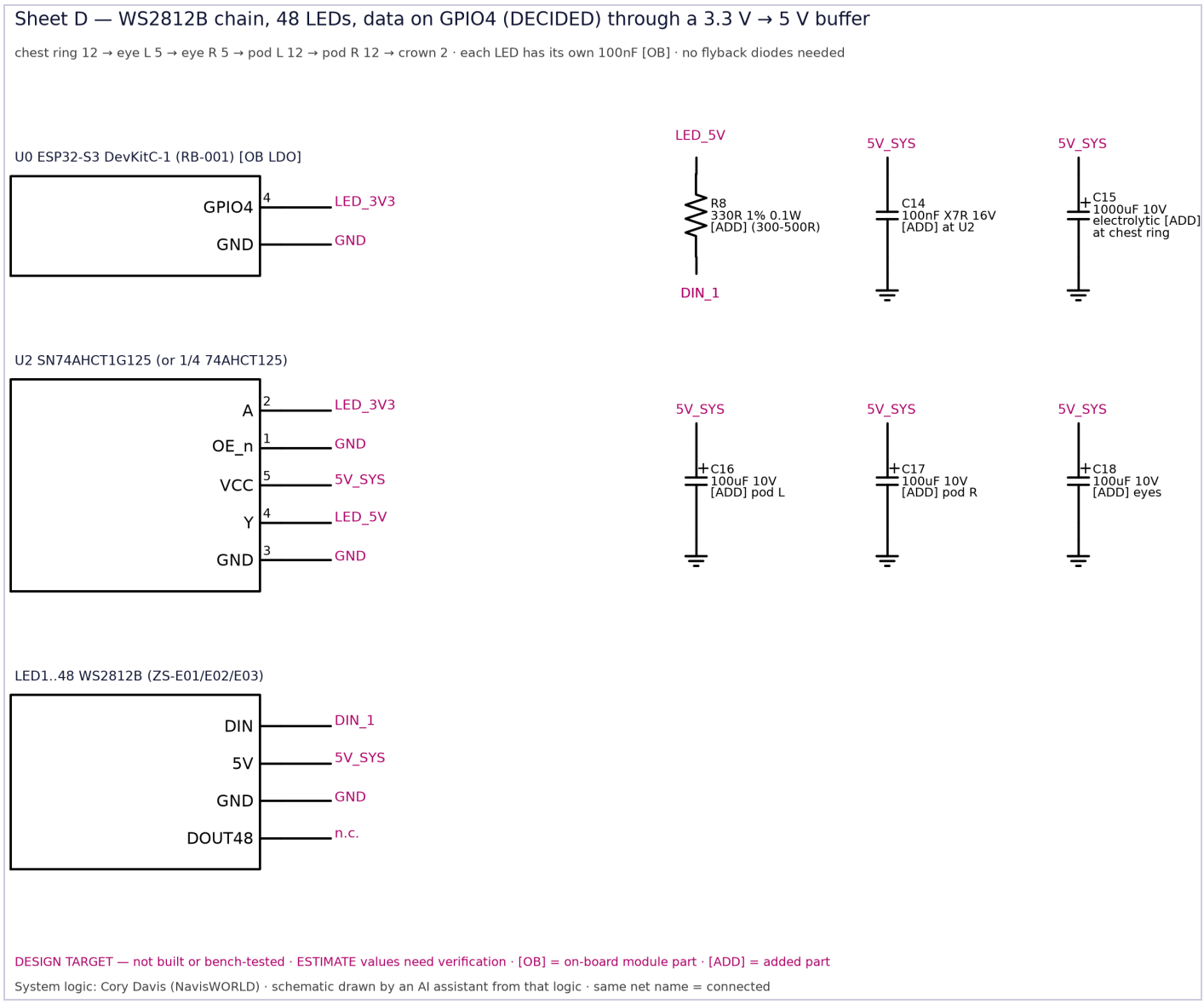
Parts added: ZS-104 Visor, ZS-107 Crown accent light pipe, left, ZS-107 Crown accent light pipe, right, PX Crown accent LED board, 2x WS2812B (ZS-E03)

Fasteners: Visor: clear silicone or CA onto the ZS-101 ledge. Accent pipes: press-fit.

Do:

1. Insert the visor from the inside of ZS-101 so the lip sits on the inner ledge.
2. Press both accent pipes into their slots; the flat faces point inward to the accent LED PCB.
3. Seat the accent PCB (crown pair, last 2 LEDs in the chain) behind the pipes.

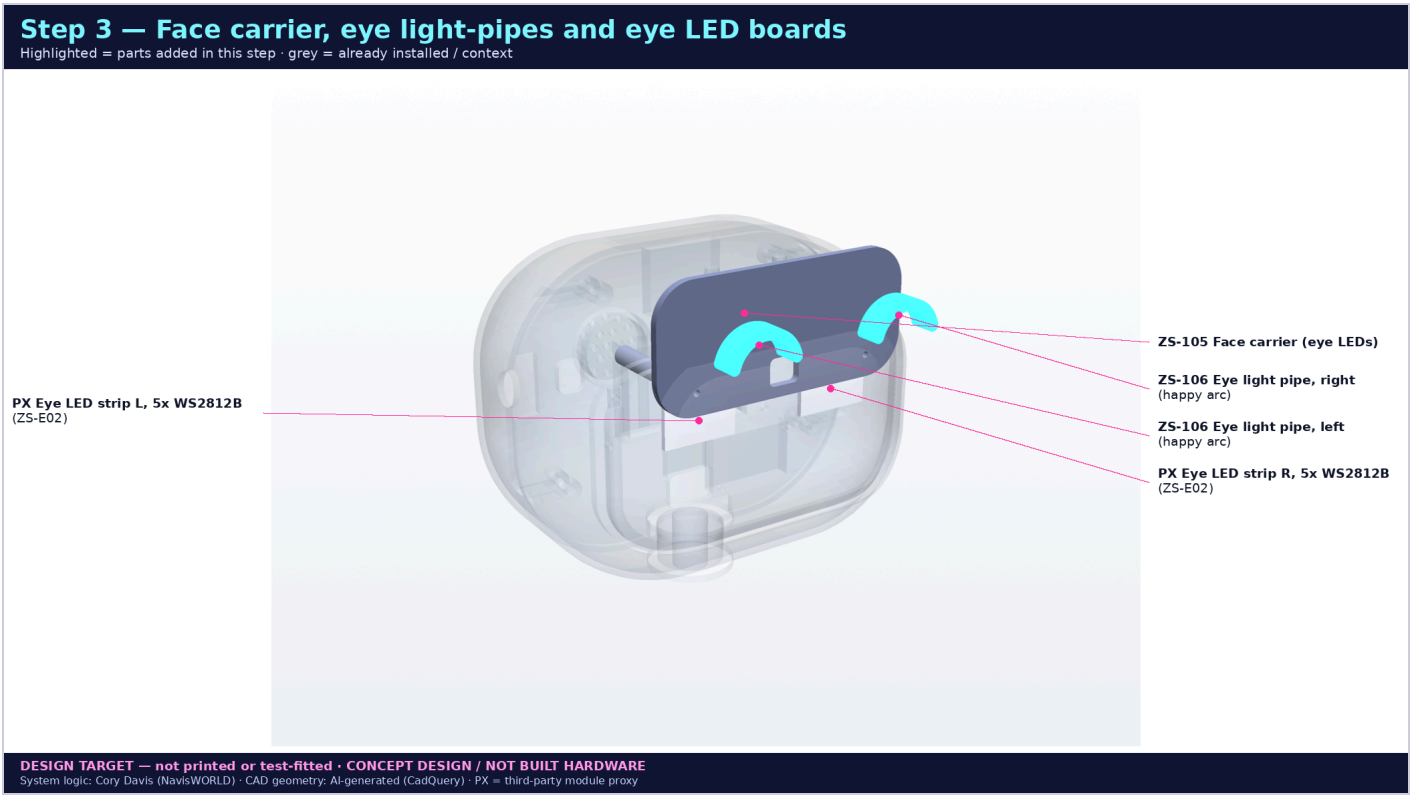
Wiring: Accent PCB: 5V_SYS / GND / DIN from eye R DOUT (26 AWG power, 30 AWG data). See Sheet D.



Wiring for this step: docs/schematics/D_led_chain (pin labels = ESP32-S3 GPIO numbers)

CHECK before moving on: Visor sits flush with no light gap; pipes do not rattle.

Step 3 — Face carrier, eye light-pipes and eye LED boards



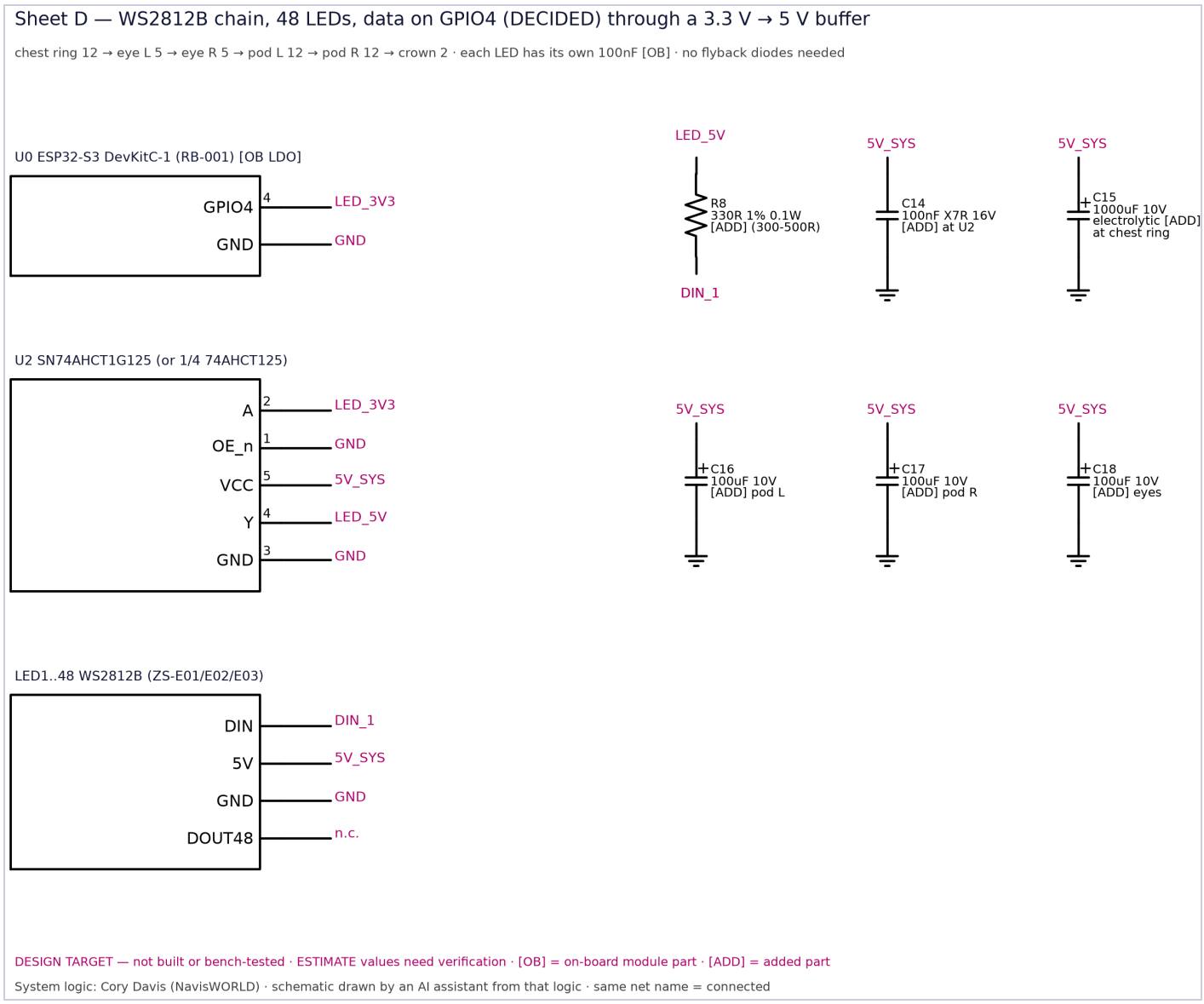
Parts added: ZS-105 Face carrier (eye LEDs), ZS-106 Eye light pipe, left (happy arc), ZS-106 Eye light pipe, right (happy arc), PX Eye LED strip L, 5x WS2812B (ZS-E02), PX Eye LED strip R, 5x WS2812B (ZS-E02)

Fasteners: Face carrier: 2 × 2.5 mm PT × 12 (ZS-F04) into ZS-103 posts. Eye boards: hot glue or M2 × 4 (ESTIMATE).

Do:

1. Press the eye pipes into the face carrier.
2. Mount eye L and eye R LED boards (5 LEDs each) against the pipe ends.
3. Fit C18 (100 µF 10 V) across 5V/GND at the eye boards.

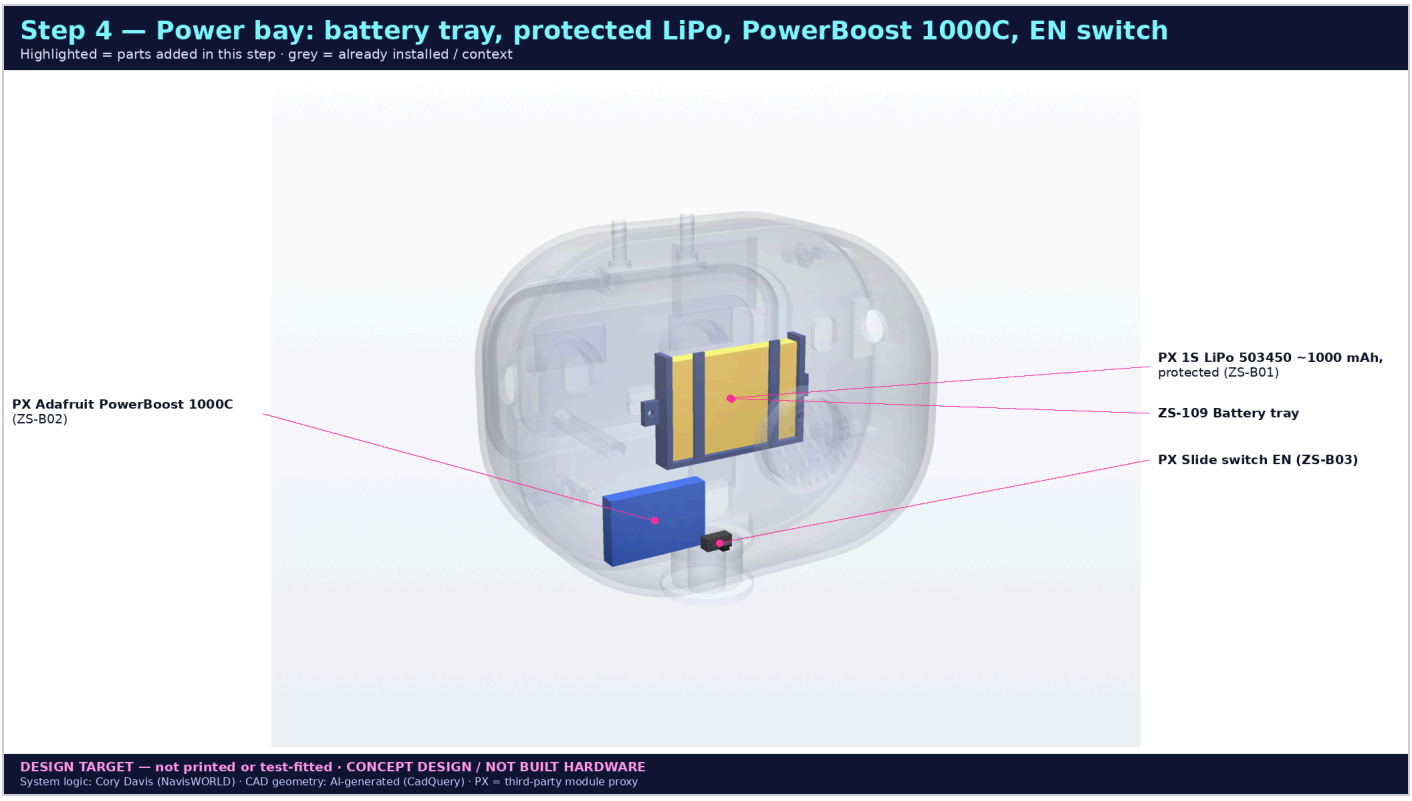
Wiring: Chain: chest ring DOUT → eye L DIN → eye L DOUT → eye R DIN → … (Sheet D). C18 at eyes.



Wiring for this step: docs/schematics/D_led_chain (pin labels = ESP32-S3 GPIO numbers)

CHECK before moving on: Pipes aligned with LEDs; continuity 5V_SYS ↔ every eye LED 5V pin; no short 5V↔GND (> 1 kΩ).

Step 4 — Power bay: battery tray, protected LiPo, PowerBoost 1000C, EN switch



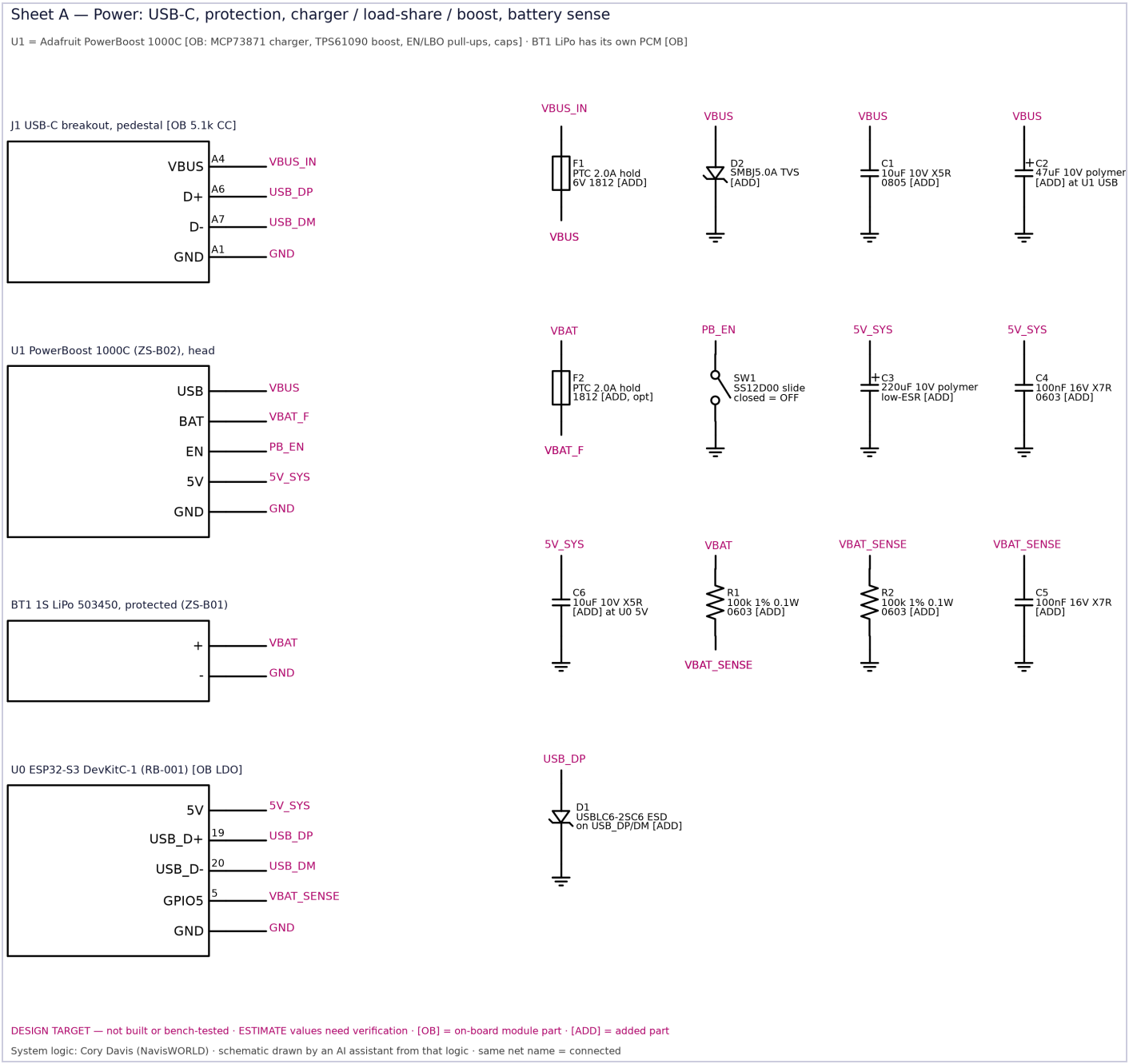
Parts added: ZS-109 Battery tray, PX 1S LiPo 503450 ~1000 mAh, protected (ZS-B01), PX Adafruit PowerBoost 1000C (ZS-B02), PX Slide switch EN (ZS-B03)

Fasteners: Battery tray: 2 × M2 × 6 into ZS-103. PowerBoost: 2 × M2 to ZS-103. Cell: foam tape only — never screws or clamps on the cell.

Do:

1. ADULT step. Check the cell is a PROTECTED cell (PCM board under the yellow tape) and reads 3.6–4.2 V before installing.
2. Fit SW1 so OFF = EN tied to GND. Solder F2 (optional) in the battery + lead, then BT1 to U1 BAT via JST-PH 2.0.
3. Solder C2 (47 µF) at U1 USB input, C3 (220 µF) + C4 (100 nF) at U1 5V output.

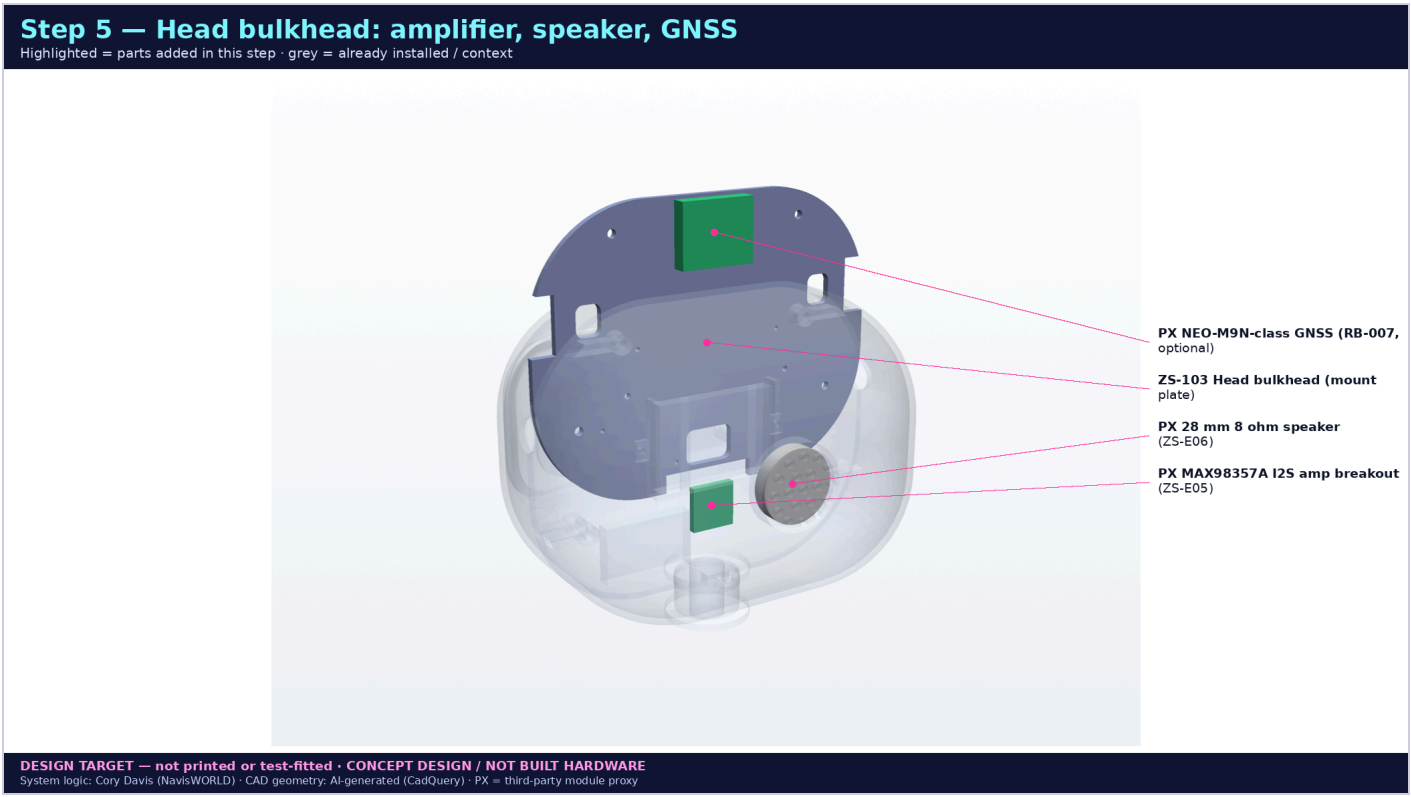
Wiring: Sheet A. BAT: 24 AWG ≤ 75 mm. 5V_SYS and VBUS: 2 × 26 AWG in parallel. EN: 30 AWG.



Wiring for this step: docs/schematics/A_power (pin labels = ESP32-S3 GPIO numbers)

> **CHECK before moving on:** SW1 OFF: U1 5V output < 0.3 V. SW1 ON (no USB): 5.1–5.25 V, blue LED on. Plug USB: charge LED on, 5V stays up.

Step 5 — Head bulkhead: amplifier, speaker, GNSS



Parts added: ZS-103 Head bulkhead (mount plate), PX MAX98357A I2S amp breakout (ZS-E05), PX 28 mm 8 ohm speaker (ZS-E06), PX NEO-M9N-class GNSS (RB-007, optional)

Fasteners: Bulkhead ZS-103 is clamped later by the 4 × RB-008 shell screws. Amp + GNSS: M2 × 4 or VHB tape (ESTIMATE). Speaker: hot glue ring.

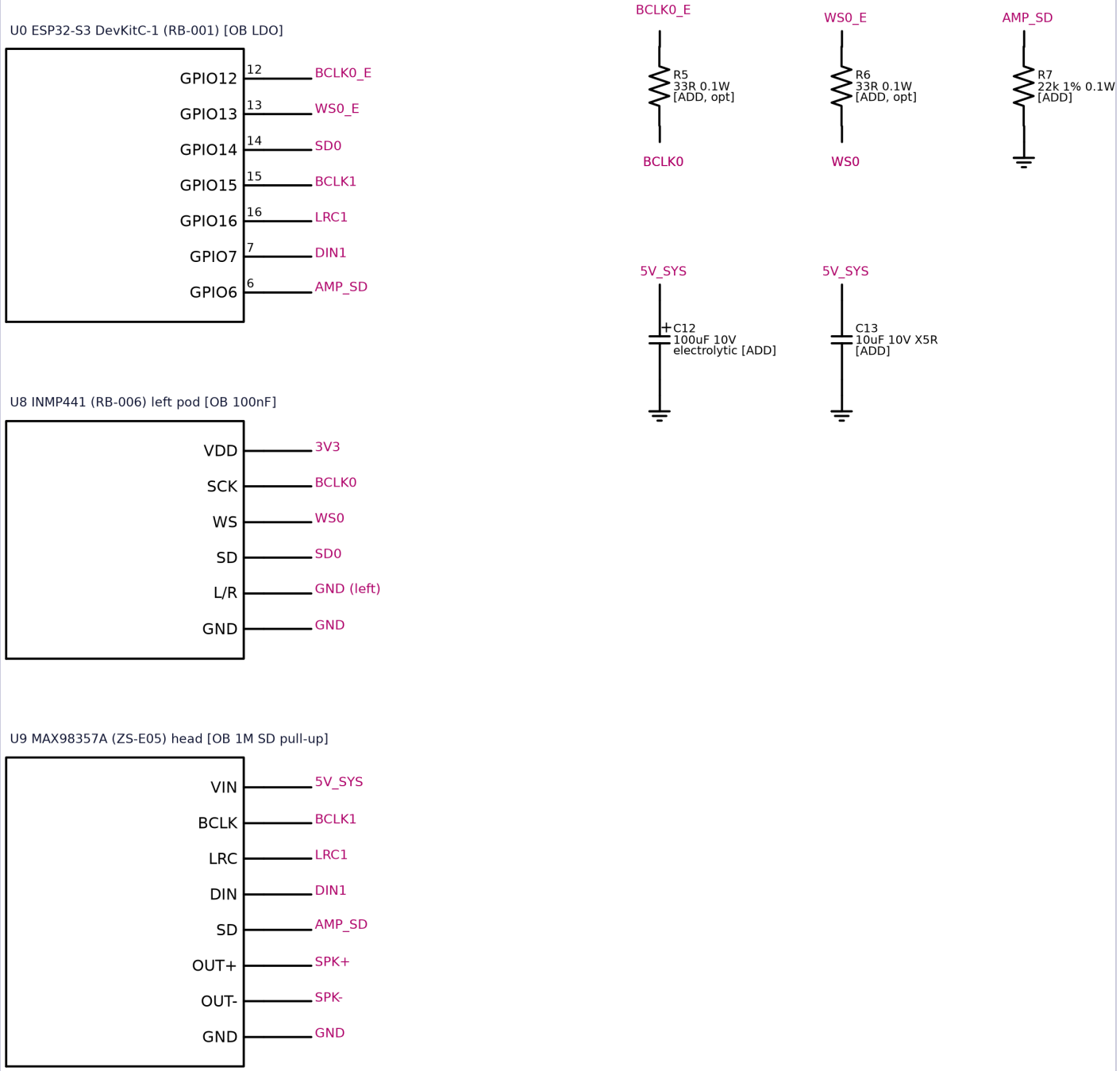
Do:

1. Fix the bulkhead into the back head shell.
2. Mount U9 MAX98357A; fit R7 22 kΩ SD → GND, C12 100 µF + C13 10 µF at VIN.
3. Mount speaker SPK1; mount optional U7 GNSS with antenna side up, C11 10 µF at its VIN.

Wiring: Sheet C (I2S1 GPIO15/16/7, SD GPIO6) and Sheet B (GNSS UART GPIO17/18). Speaker 26 AWG.

Sheet C — Audio: INMP441 mic on I2S0 (REPO 12/13/14) · MAX98357A on I2S1 (DECIDED 15/16/7, SD = GPIO6)

R7 holds the amp in shutdown at boot (see calc in guide). SPK1 = 28 mm 8 ohm ~1 W (ZS-E06): firmware volume cap.



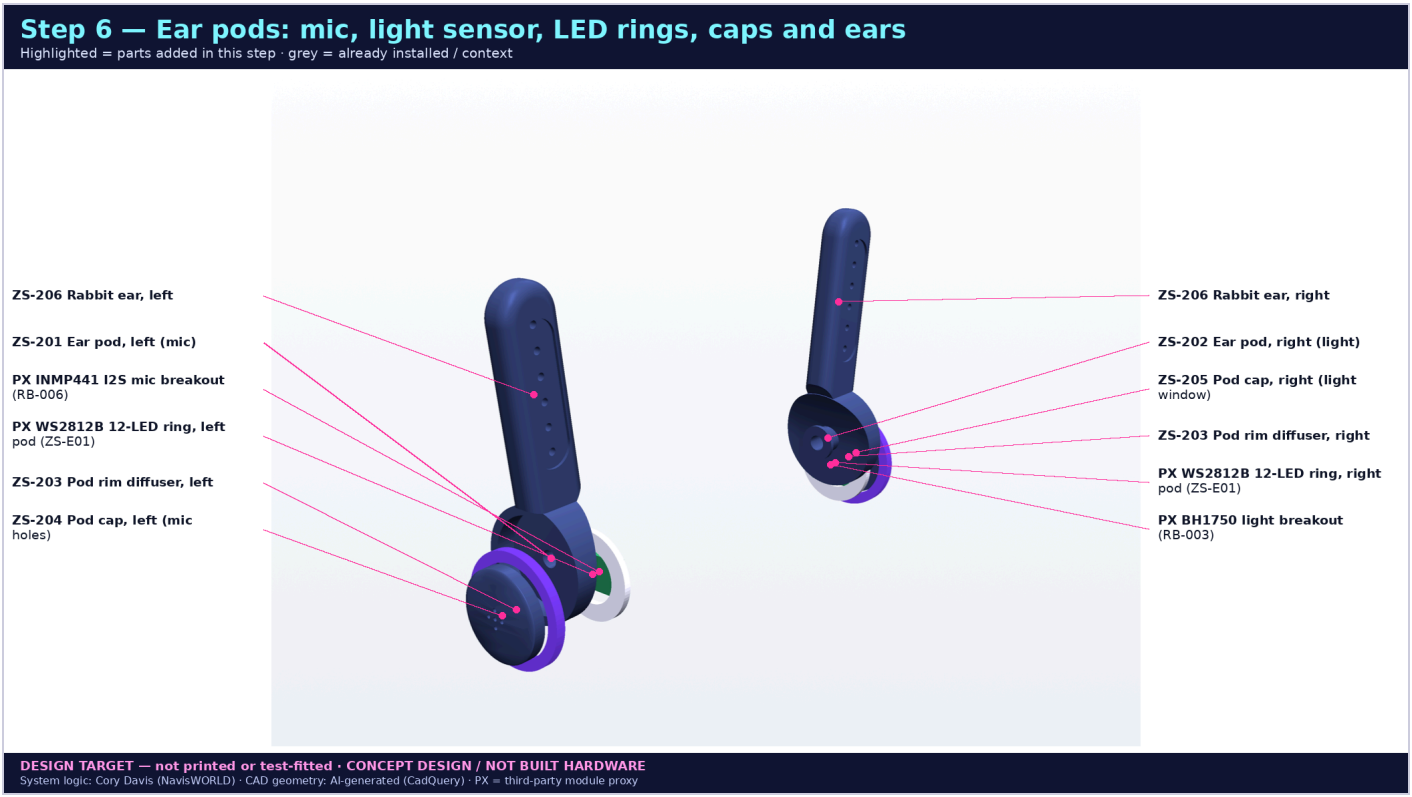
DESIGN TARGET — not built or bench-tested · ESTIMATE values need verification · [OB] = on-board module part · [ADD] = added part

System logic: Cory Davis (NavisWORLD) · schematic drawn by an AI assistant from that logic · same net name = connected

Wiring for this step: docs/schematics/C_audio (pin labels = ESP32-S3 GPIO numbers)

> **CHECK before moving on:** R7 measured 21.8–22.2 kΩ to GND; 8 Ω ±2 across SPK1; no short on 5V_SYS.

Step 6 — Ear pods: mic, light sensor, LED rings, caps and ears



Parts added: ZS-201 Ear pod, left (mic), ZS-202 Ear pod, right (light), ZS-203 Pod rim diffuser, left, ZS-203 Pod rim diffuser, right, ZS-204 Pod cap, left (mic holes), ZS-205 Pod cap, right (light window), ZS-206 Rabbit ear, left, ZS-206 Rabbit ear, right, PX BH1750 light breakout (RB-003), PX INMP441 I2S mic breakout (RB-006), PX WS2812B 12-LED ring, left pod (ZS-E01), PX WS2812B 12-LED ring, right pod (ZS-E01)

Fasteners: Ears to pods: 2 × ZS-F02 M3 × 10 self-tapping. Pod caps: snap / glue into the rims.

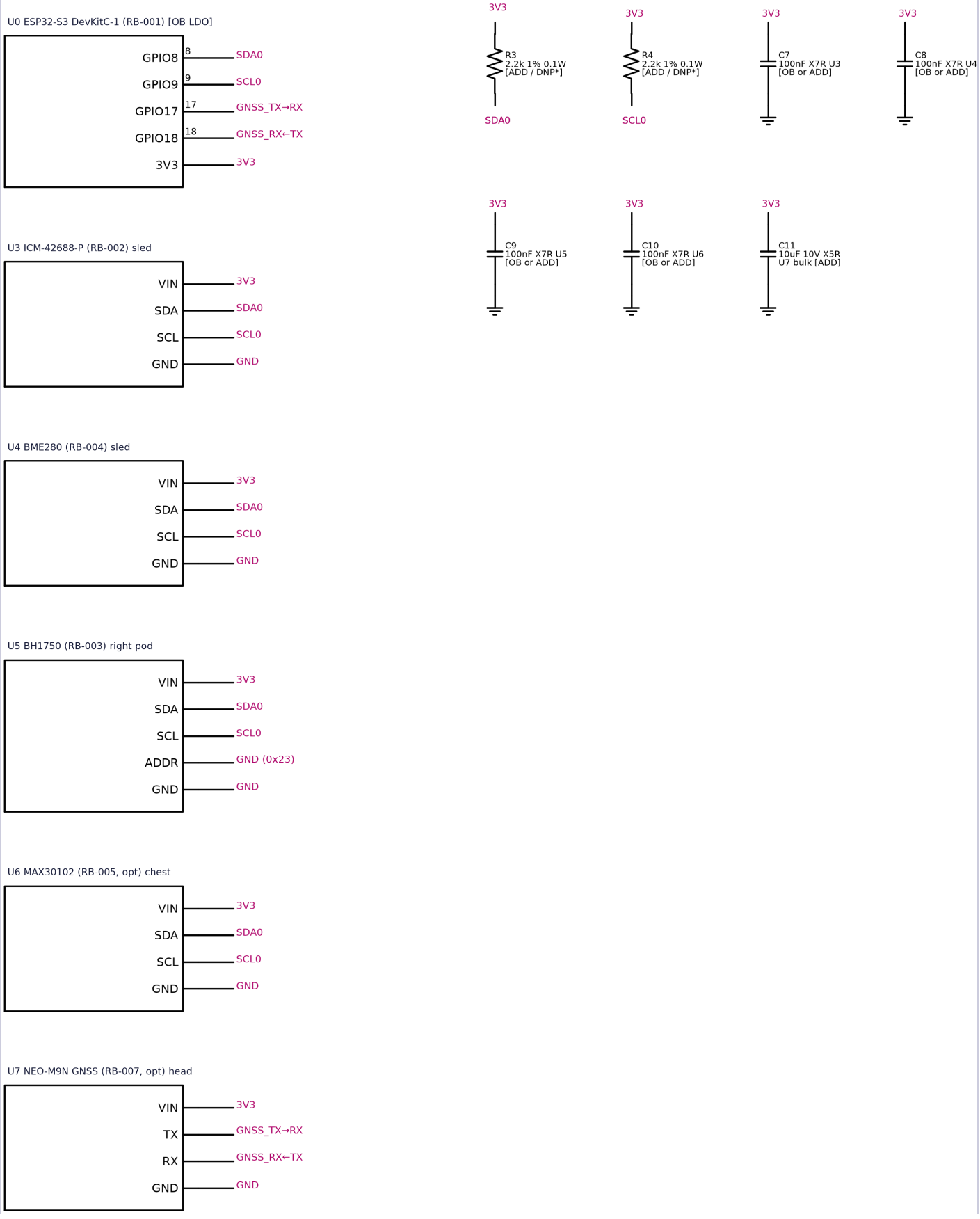
Do:

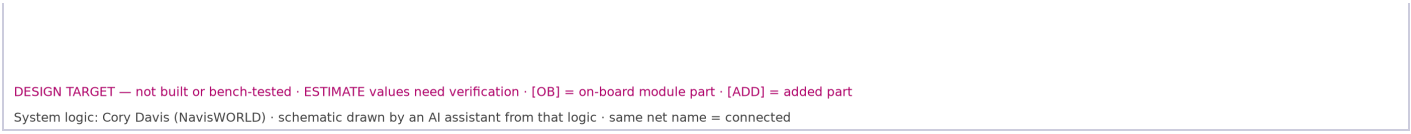
1. Left pod: INMP441 (U8) with its port facing the cap hole, L/R tied to GND.
2. Right pod: BH1750 (U5) behind the window, ADDR → GND for 0x23.
3. Mount both 12-LED rings behind the rim diffusers; C16/C17 100 µF on each ring.

Wiring: Sheet B (BH1750 on I2C0), Sheet C (INMP441 on I2S0 12/13/14), Sheet D (rings). 30 AWG signals, 26 AWG ring power.

Sheet B — I2C0 sensors: GPIO8 SDA0 / GPIO9 SCL0 (REPO pins), 400 kHz

BH1750 0x23 · MAX30102 0x57 · ICM-42688-P 0x68 · BME280 0x76 — no conflicts (AS7341 0x39 is on I2C1). GNSS on UART GPIO17/18 (REPO).





Wiring for this step: docs/schematics/B_i2c0_sensors (pin labels = ESP32-S3 GPIO numbers)

> **CHECK before moving on:** Continuity per pin from the pod connector to the module; mic port unobstructed; window clean.

Step 7 — Close the head

Parts added: ZS-102 Head shell, back, ZS-108 Neck coupler

Fasteners: 4 × RB-008 M3 × 8 through the back-shell clearance bosses into the ZS-F01 inserts. Neck coupler ZS-108 is captured by both shell pairs (no screws).

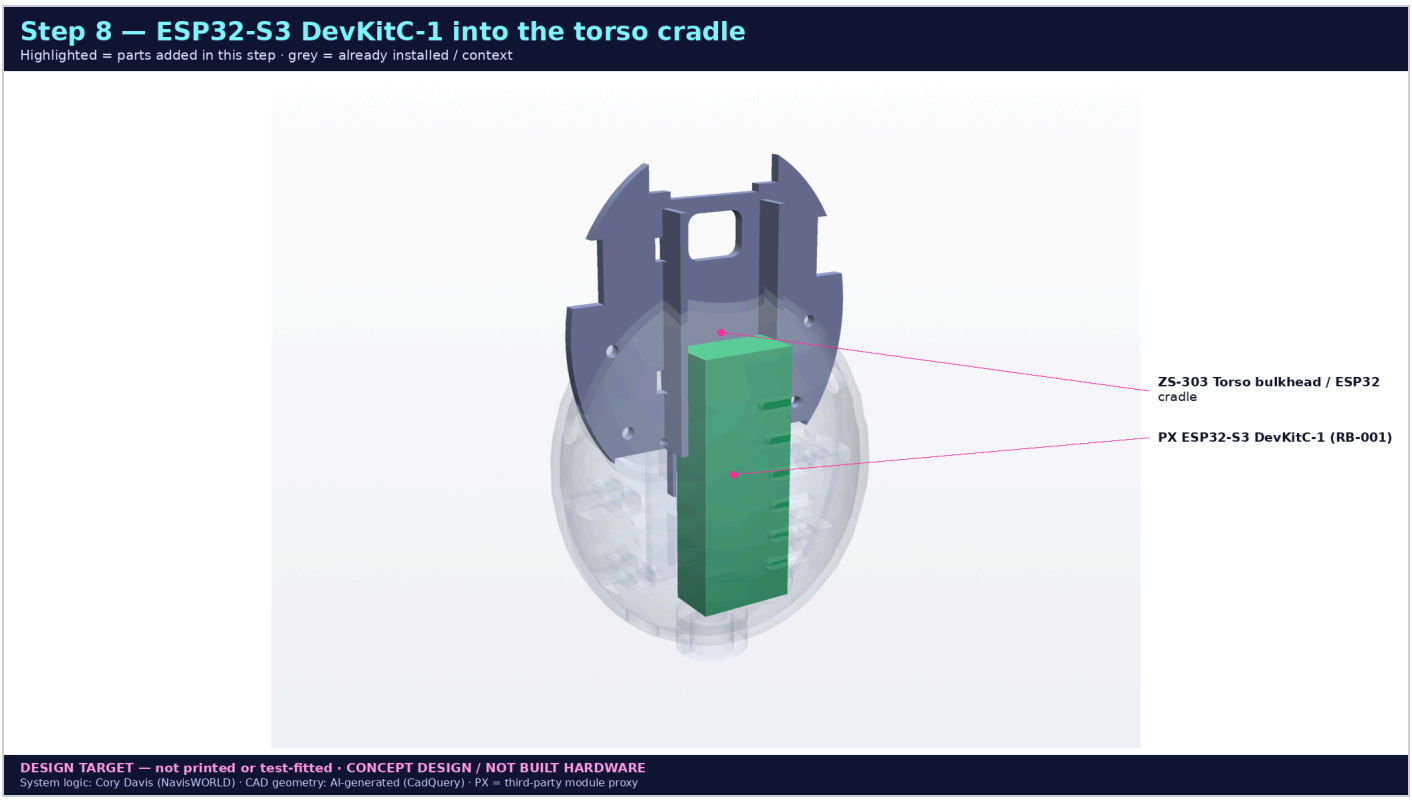
Do:

1. Route the head harness through the neck coupler first.
2. Close the back shell onto the front shell, pod pegs captured by the split.
3. Tighten diagonally, snug only (PLA cracks).

Wiring: Head harness exits the neck: 5V_SYS, GND, VBUS, D+, D−, LED DIN, I2S0 ×3, I2S1 ×3, AMP_SD, SDA0, SCL0, GNSS ×2, VBAT_SENSE (ESTIMATE 17 conductors).

> **CHECK before moving on:** No pinched wire visible at the seam; the head harness pulls freely 5 mm.

Step 8 — ESP32-S3 DevKitC-1 into the torso cradle



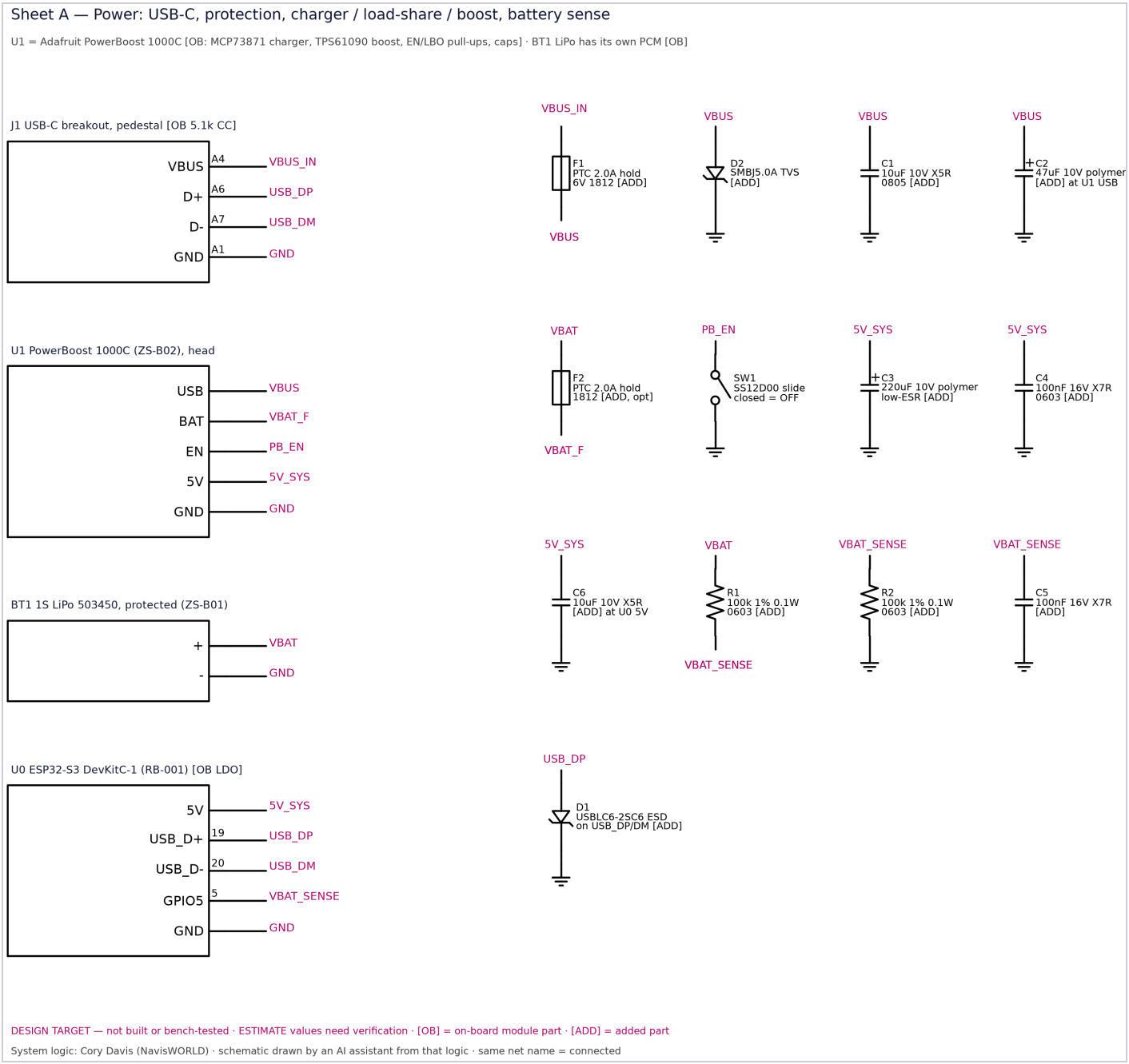
Parts added: ZS-303 Torso bulkhead / ESP32 cradle, PX ESP32-S3 DevKitC-1 (RB-001)

Fasteners: DevKit: 2 × ZS-F05 2.5 mm zip ties in the cradle rails. Cradle ZS-303 is clamped later by the torso RB-008 screws.

Do:

1. Fit C6 10 µF across the DevKit 5V/GND pins.
2. Fit the level-shifter U2 + C14 and R8 330 Ω on the LED data line.
3. Fit R1/R2 100 kΩ + C5 100 nF battery divider at GPIO5.

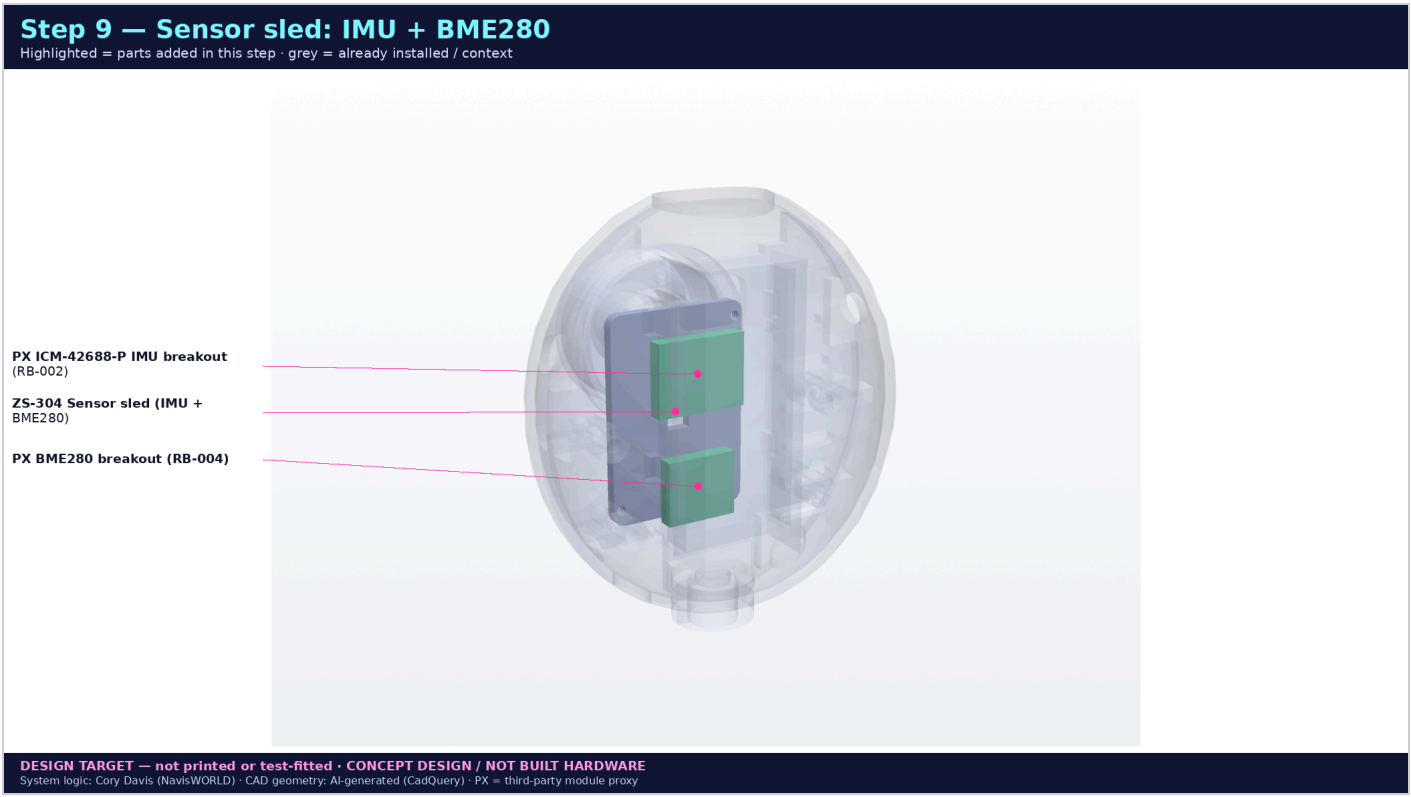
Wiring: Sheets A + D. Use the USB-C port marked USB (native, GPIO19/20) for the pigtail: D+/D−/GND only — do NOT connect pigtail VBUS.



Wiring for this step: docs/schematics/A_power (pin labels = ESP32-S3 GPIO numbers)

> **CHECK before moving on:** DevKit powers from 5V_SYS only; GPIO5 reads VBAT/2 (1.5–2.1 V) with a multimeter.

Step 9 — Sensor sled: IMU + BME280



Parts added: ZS-304 Sensor sled (IMU + BME280), PX ICM-42688-P IMU breakout (RB-002), PX BME280 breakout (RB-004)

Fasteners: Sled: 2 × M2 × 6 PT (ZS-F03) into ZS-303. Breakouts: M2 × 4 or nylon standoffs (ESTIMATE).

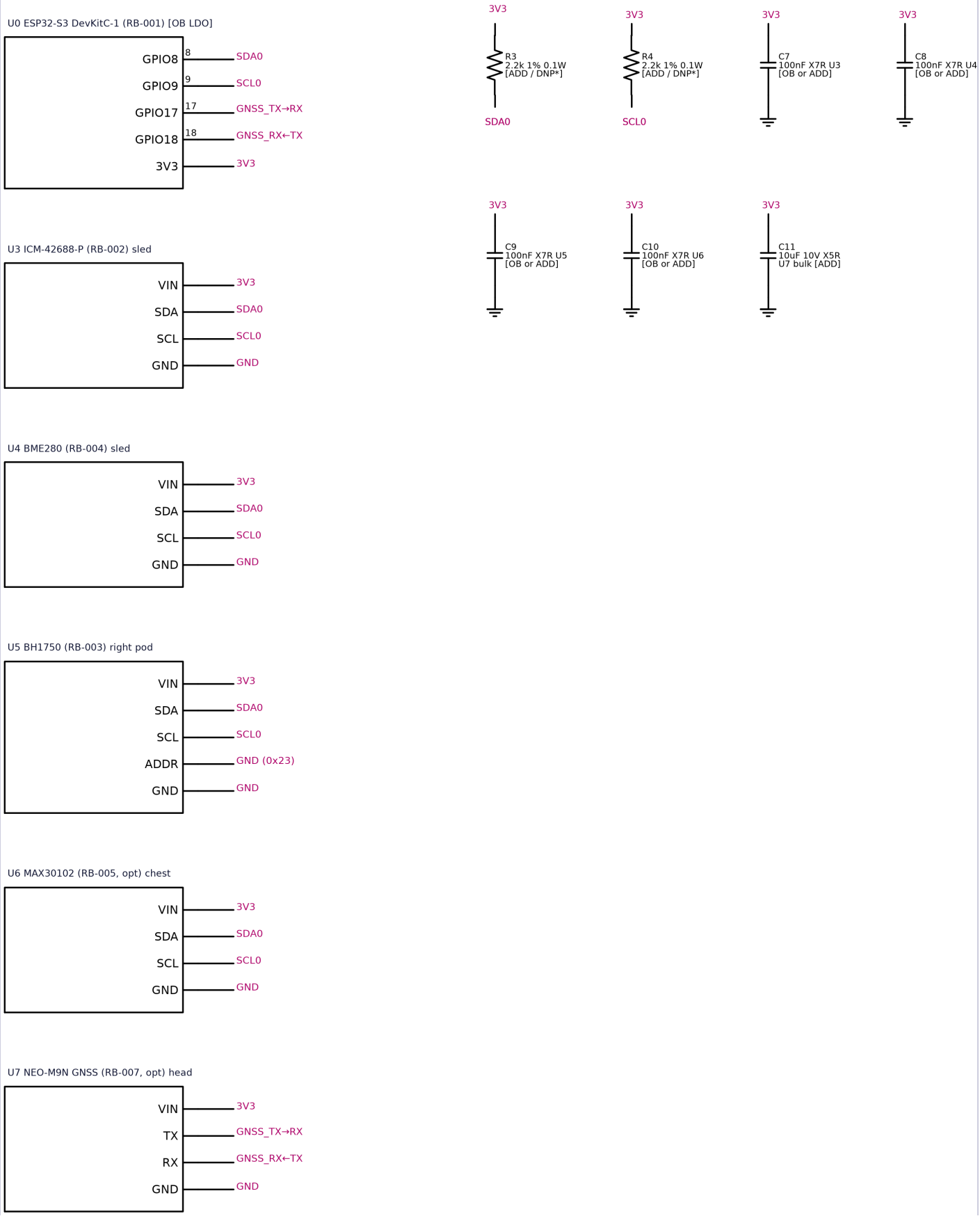
Do:

1. Mount U3 ICM-42688-P flat, axis arrows recorded in your notes (needed for firmware).
2. Mount U4 BME280 near the vent slot, away from the ESP32 (self-heating).
3. Daisy-chain STEMMA-QT / Qwiic cables; fit R3/R4 2.2 kΩ only if needed (see Electrical).

Wiring: Sheet B: I2C0 SDA GPIO8 / SCL GPIO9, 3V3, GND. JST-SH 1.0 mm.

Sheet B — I2C0 sensors: GPIO8 SDA0 / GPIO9 SCL0 (REPO pins), 400 kHz

BH1750 0x23 · MAX30102 0x57 · ICM-42688-P 0x68 · BME280 0x76 — no conflicts (AS7341 0x39 is on I2C1). GNSS on UART GPIO17/18 (REPO).



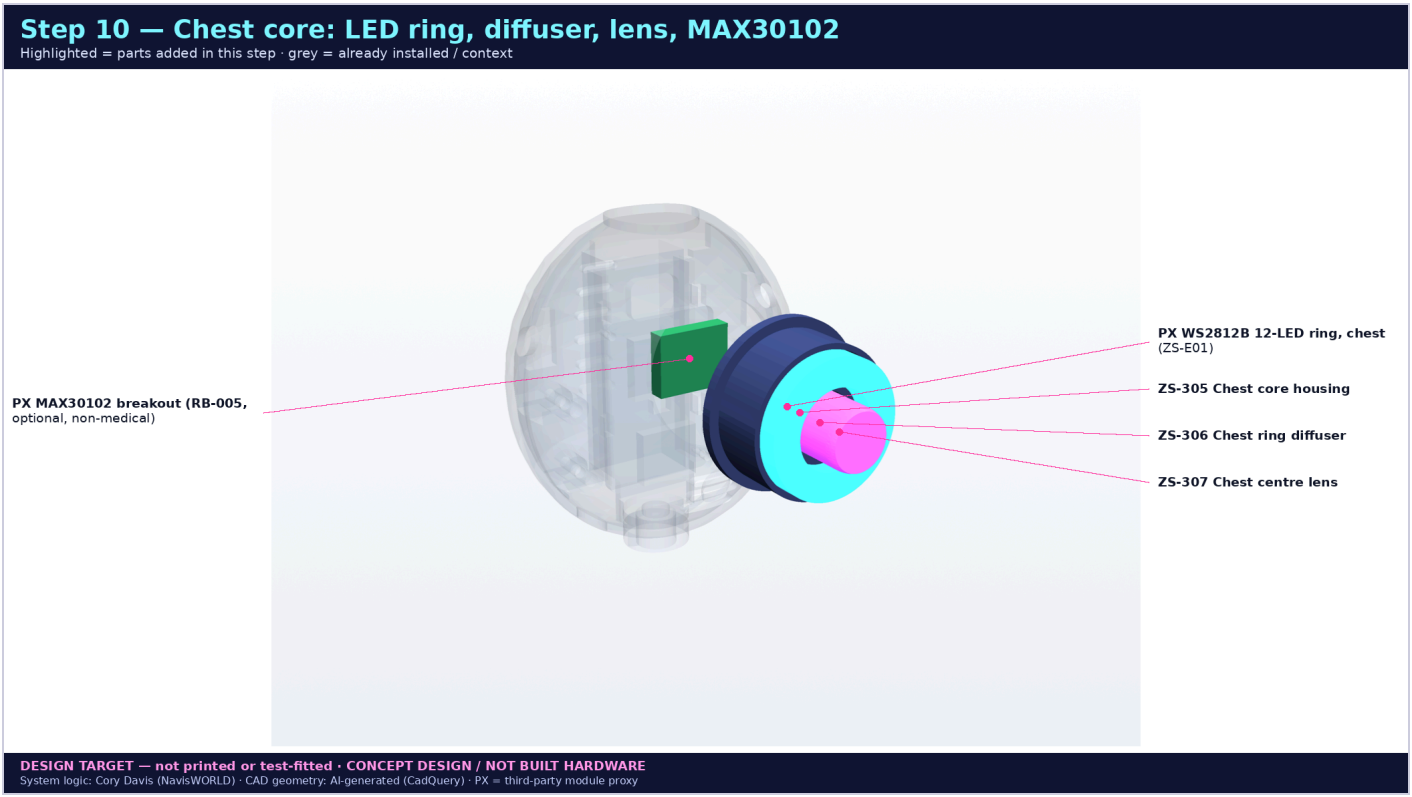
DESIGN TARGET — not built or bench-tested · ESTIMATE values need verification · [OB] = on-board module part · [ADD] = added part

System logic: Cory Davis (NavisWORLD) · schematic drawn by an AI assistant from that logic · same net name = connected

Wiring for this step: docs/schematics/B_i2c0_sensors (pin labels = ESP32-S3 GPIO numbers)

> **CHECK before moving on:** I2C scan on bus 0 shows 0x68 (or 0x69) and 0x76 (or 0x77).

Step 10 — Chest core: LED ring, diffuser, lens, MAX30102



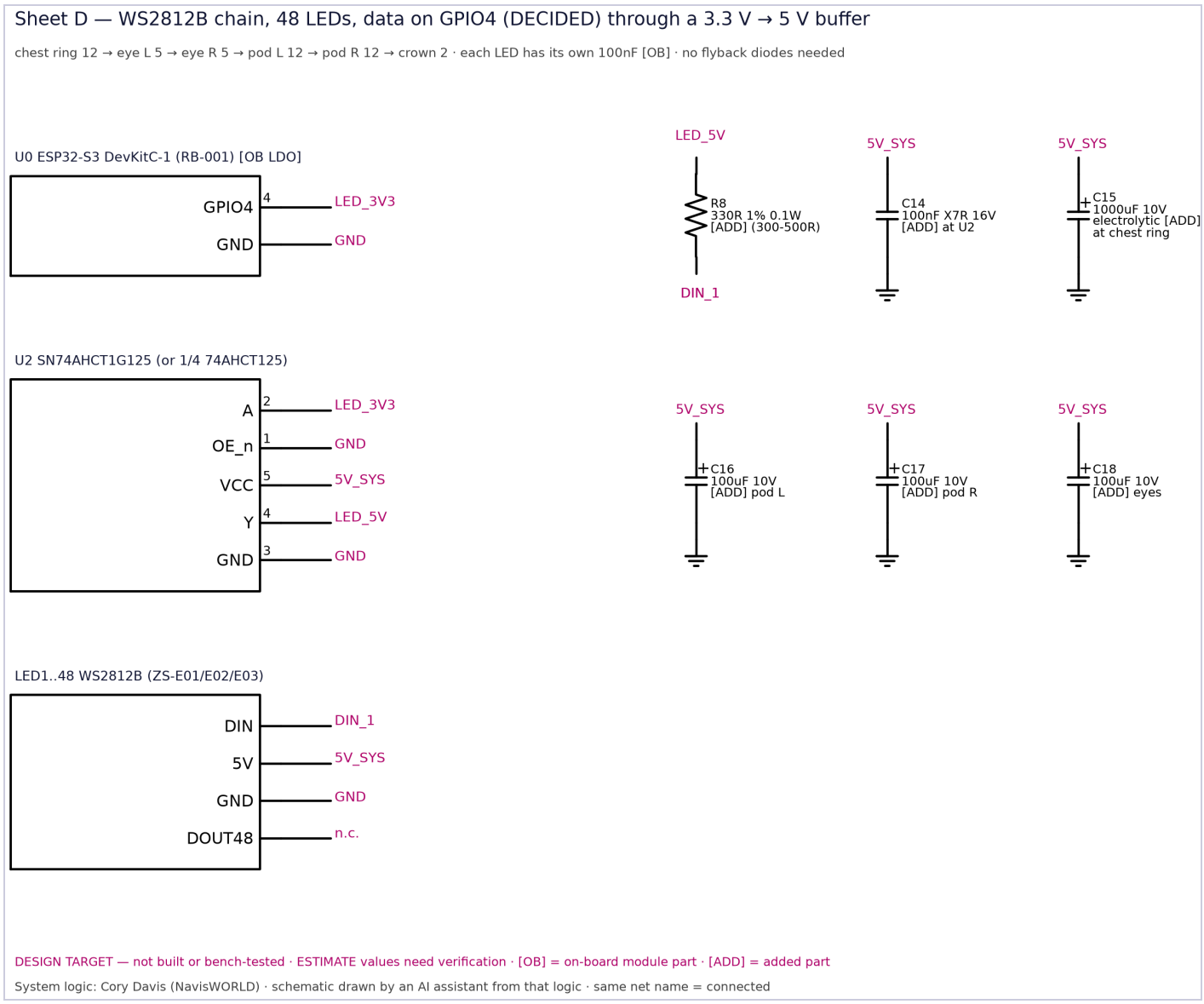
Parts added: ZS-305 Chest core housing, ZS-306 Chest ring diffuser, ZS-307 Chest centre lens, PX WS2812B 12-LED ring, chest (ZS-E01), PX MAX30102 breakout (RB-005, optional, non-medical)

Fasteners: Chest core flange is captured behind the torso front shell (no screws). Diffuser + lens: press-fit.

Do:

1. Seat the 12-LED chest ring; solder C15 1000 µF 10 V across its 5V/GND input (watch polarity).
2. Fit MAX30102 (optional) behind the lens with its window touching the lens.
3. Press in diffuser and lens.

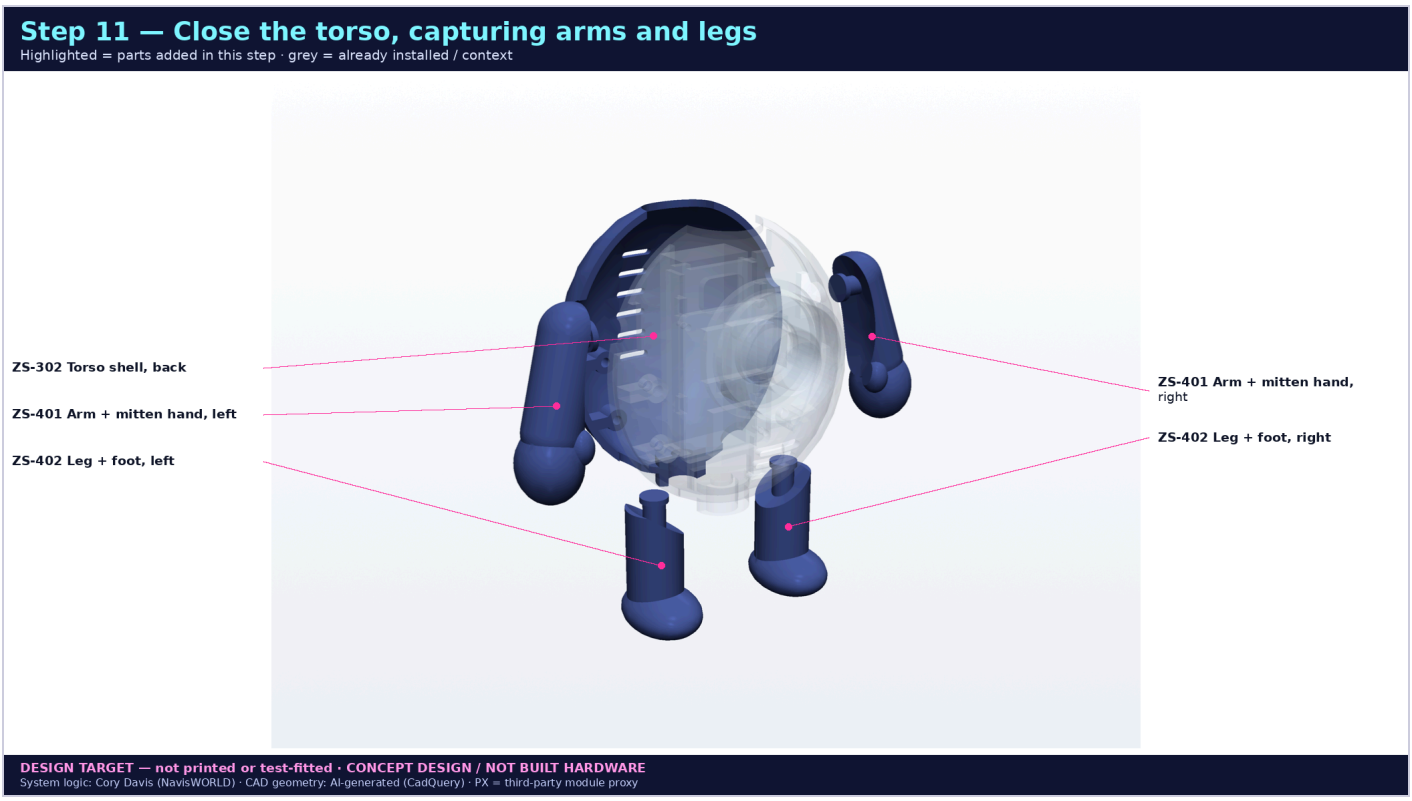
Wiring: Sheet D (chest ring = first LED, DIN from R8). Sheet B (MAX30102 0x57).



Wiring for this step: docs/schematics/D_led_chain (pin labels = ESP32-S3 GPIO numbers)

> **CHECK before moving on:** C15 polarity checked twice; 5V↔GND > 1 kΩ after the cap charges.

Step 11 — Close the torso, capturing arms and legs



Parts added: ZS-401 Arm + mitten hand, left, ZS-401 Arm + mitten hand, right, ZS-402 Leg + foot, left, ZS-402 Leg + foot, right, ZS-302 Torso shell, back

Fasteners: 4 × RB-008 M3 × 8 into the torso ZS-F01 inserts. Limbs are captured by their flanged pegs — no screws.

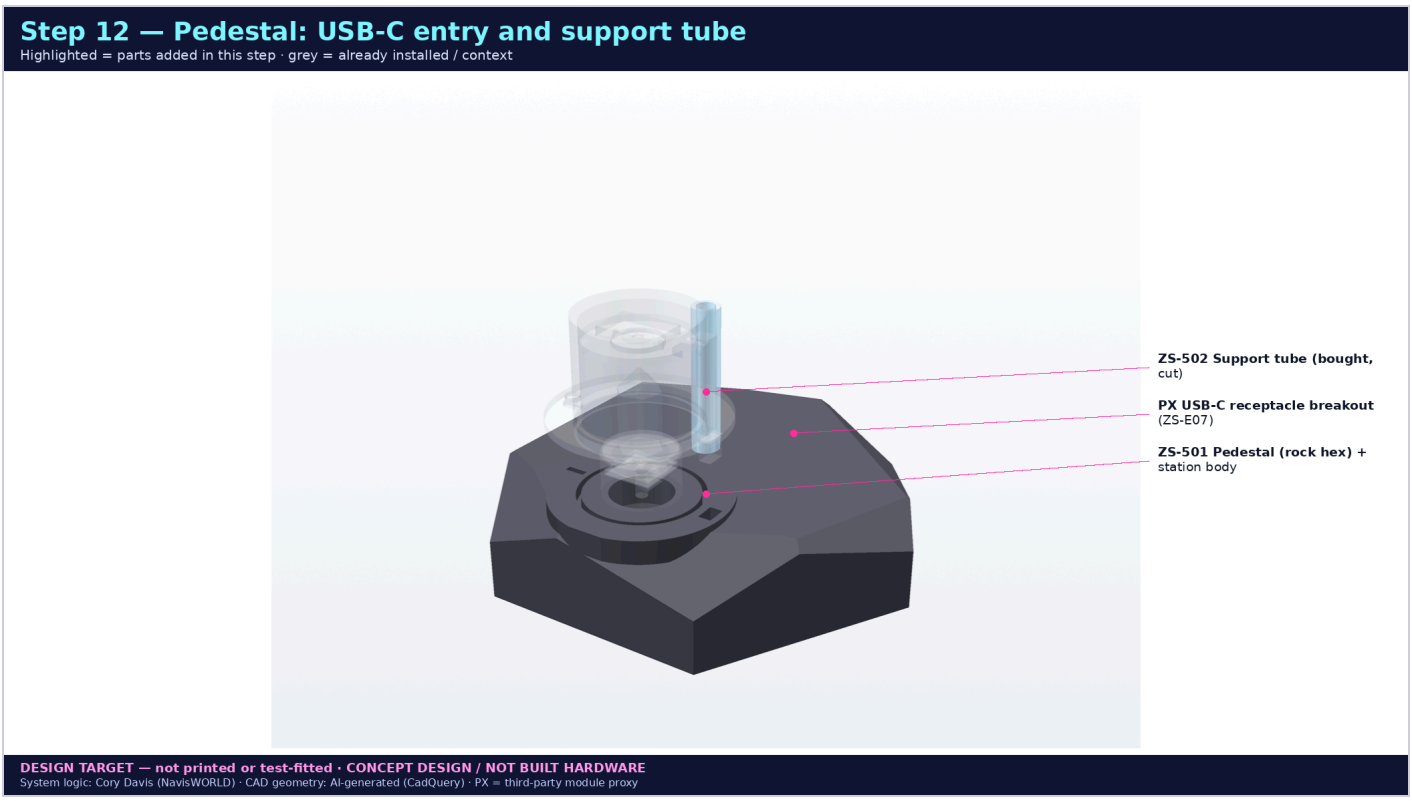
Do:

1. Place all four limb pegs into the front shell sockets.
2. Join head harness to torso harness at the JST-PH joints.
3. Close the back shell; tighten diagonally.

Wiring: Harness table in the Electrical Specifications section: torso ↔ head and torso ↔ tube.

> **CHECK before moving on:** Each limb rotates with light friction and does not fall out.

Step 12 — Pedestal: USB-C entry and support tube



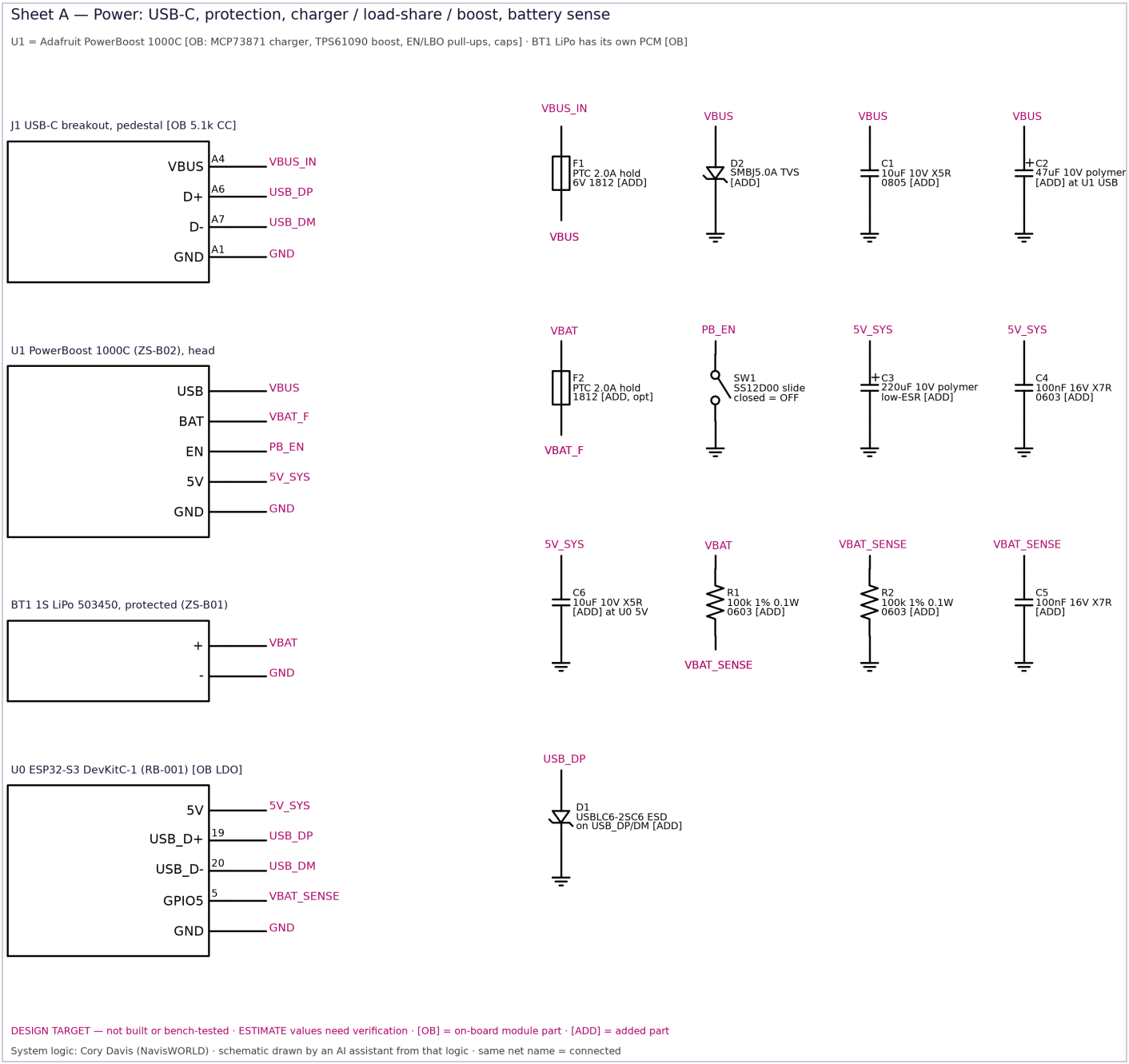
Parts added: ZS-501 Pedestal (rock hex) + station body, PX USB-C receptacle breakout (ZS-E07), ZS-502 Support tube (bought, cut)

Fasteners: USB-C breakout: glued in the pedestal back pocket. Tube ZS-502: glued into the pedestal; slip fit into the torso.

Do:

1. Mount J1, then F1, D2, C1 and D1 right behind it.
2. Feed the 11-conductor station/robot harness up the tube.
3. Press the tube into the pedestal socket.

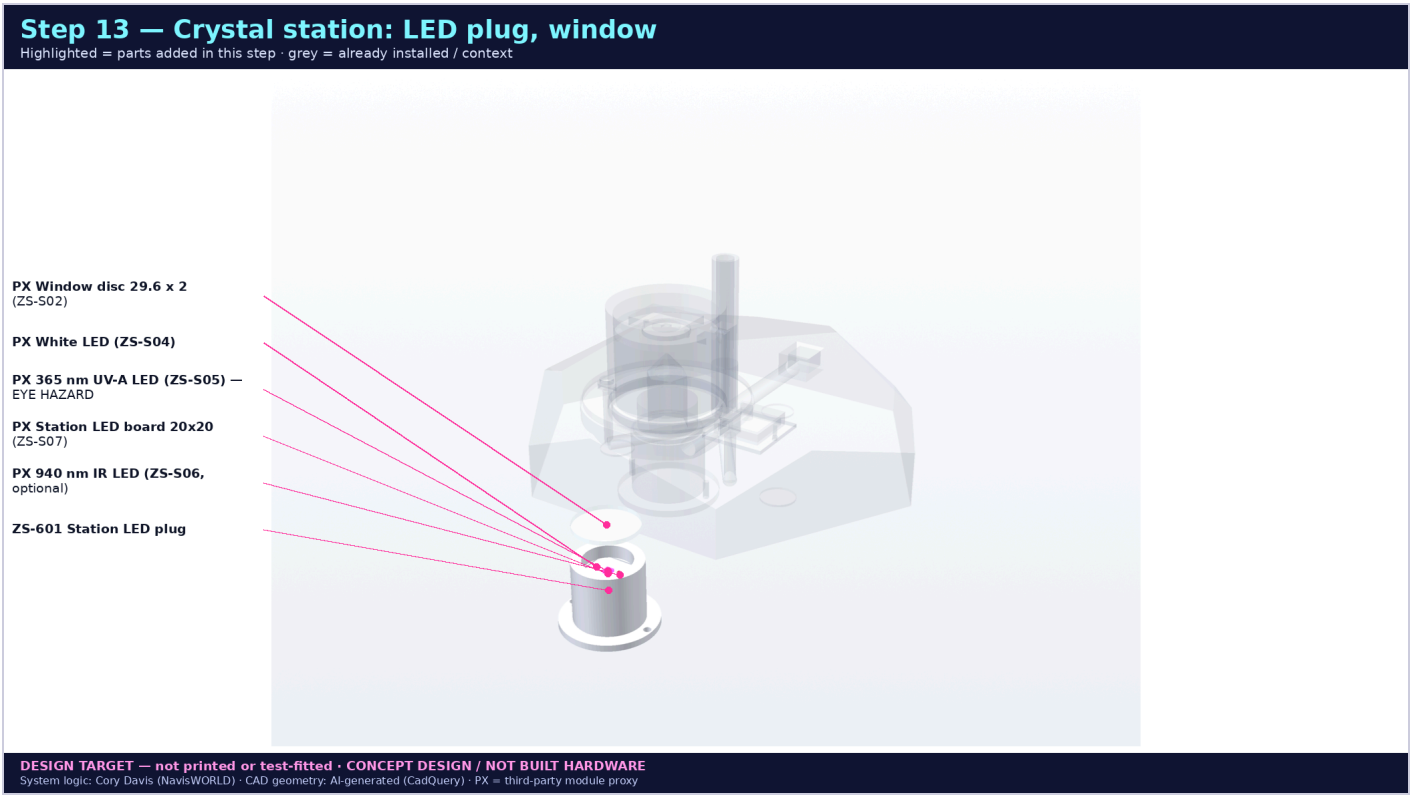
Wiring: Sheet A (VBUS_IN → F1 → VBUS). VBUS/GND 2 × 26 AWG; D+/D− twisted 30 AWG.



Wiring for this step: docs/schematics/A_power (pin labels = ESP32-S3 GPIO numbers)

> **CHECK before moving on:** With a 5 V/3 A adapter: VBUS 4.75–5.25 V at the tube top; VBUS↔GND not shorted.

Step 13 — Crystal station: LED plug, window



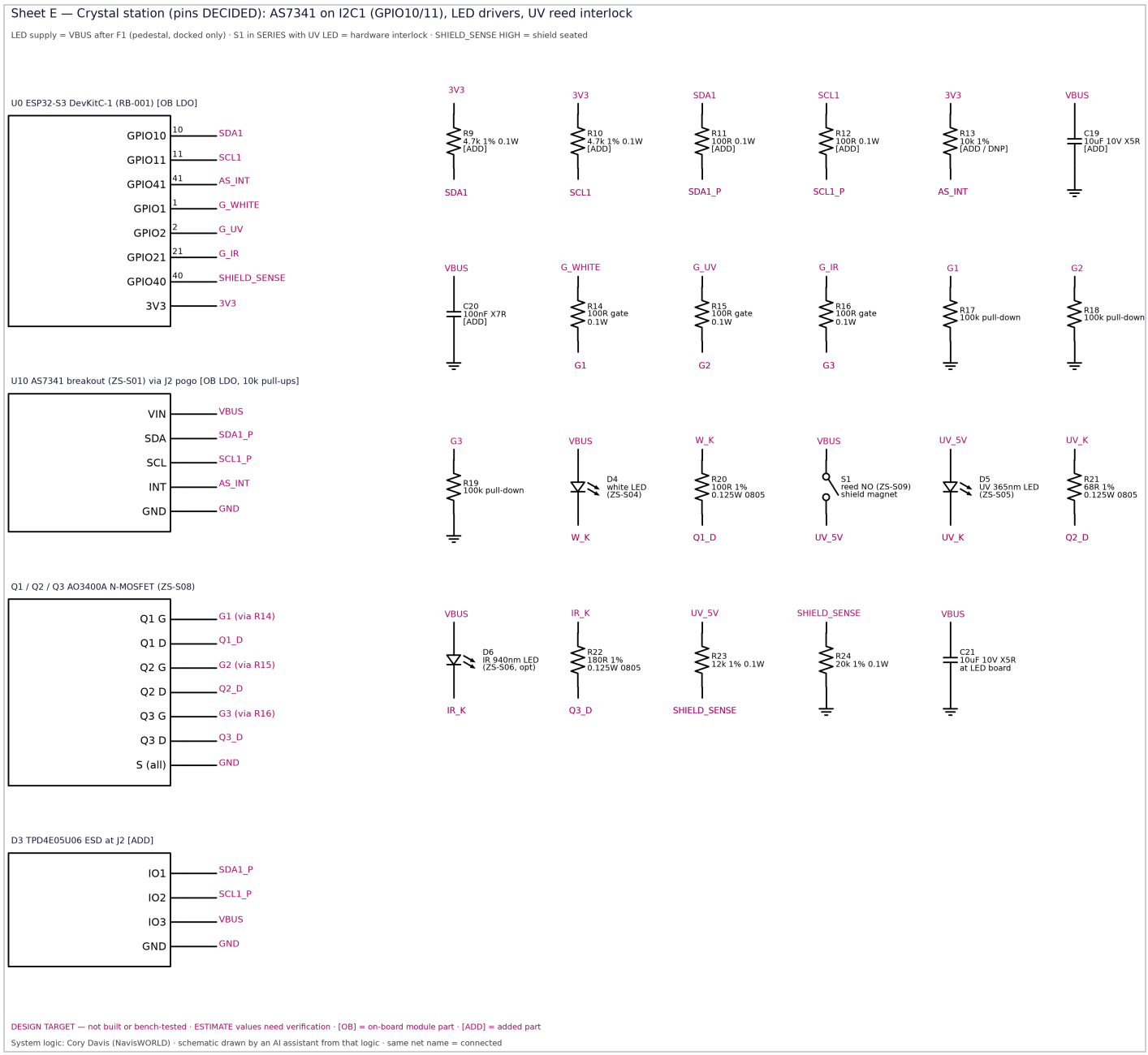
Parts added: ZS-601 Station LED plug, PX Station LED board 20x20 (ZS-S07), PX White LED (ZS-S04), PX 365 nm UV-A LED (ZS-S05) — EYE HAZARD, PX 940 nm IR LED (ZS-S06, optional), PX Window disc 29.6 x 2 (ZS-S02)

Fasteners: LED plug ZS-601: 2 × M3 × 8 PT into the pedestal. Window: inserted from BELOW into its seat (the cup lip holds it from above); no glue on the optical faces.

Do:

1. Solder D4 white + R20 100 Ω, D5 UV + R21 68 Ω, D6 IR + R22 180 Ω (optional), C21 10 µF on the LED board.
2. Clean the borosilicate window with isopropyl alcohol and insert it from below into its seat.
3. Push ZS-601 up behind the window and fix it.

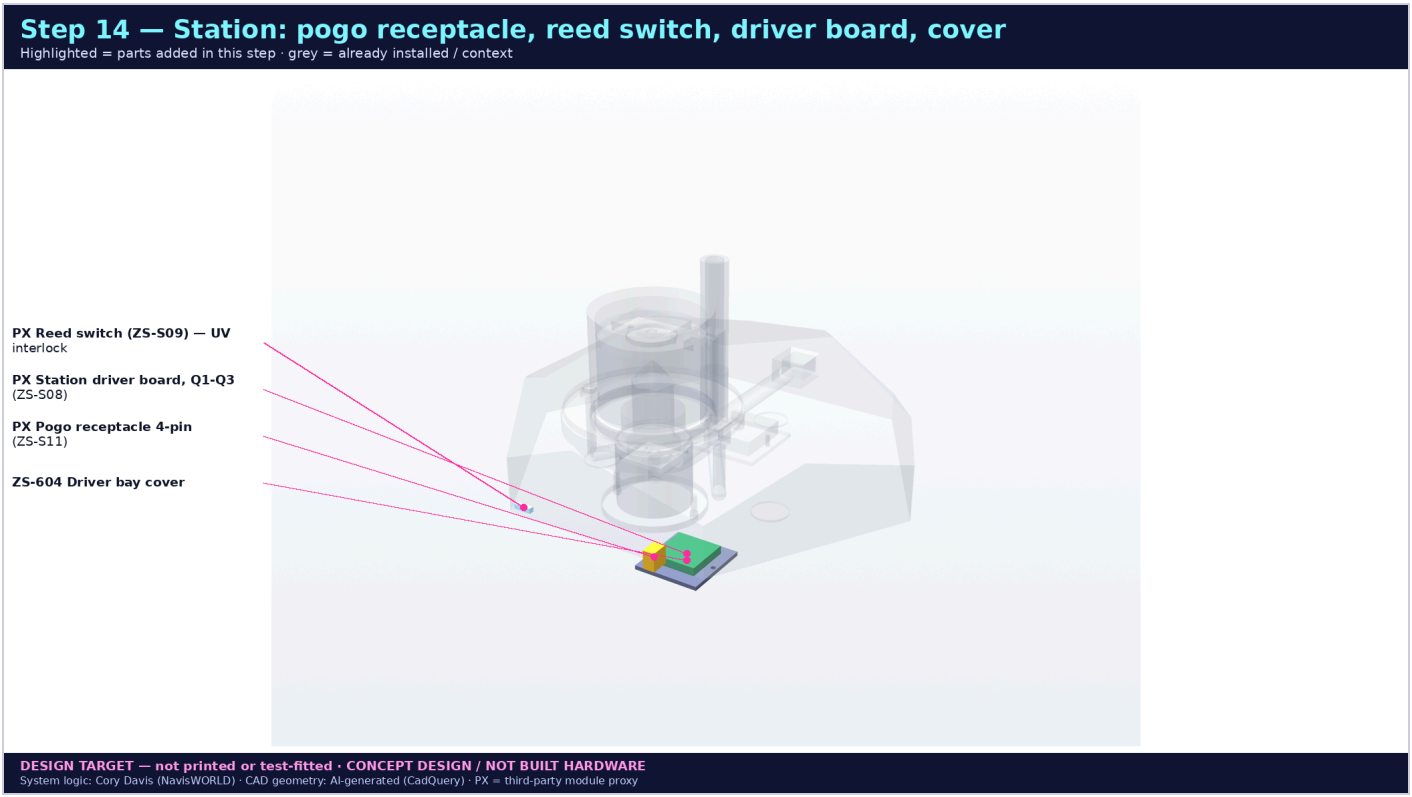
Wiring: Sheet E (anodes to VBUS; UV anode to UV_5V after S1).



Wiring for this step: docs/schematics/E_crystal_station (pin labels = ESP32-S3 GPIO numbers)

> **CHECK before moving on:** Diode-test each LED (white/UV glow faintly; IR seen through a phone camera). Do not look at the UV LED.

Step 14 — Station: pogo receptacle, reed switch, driver board, cover



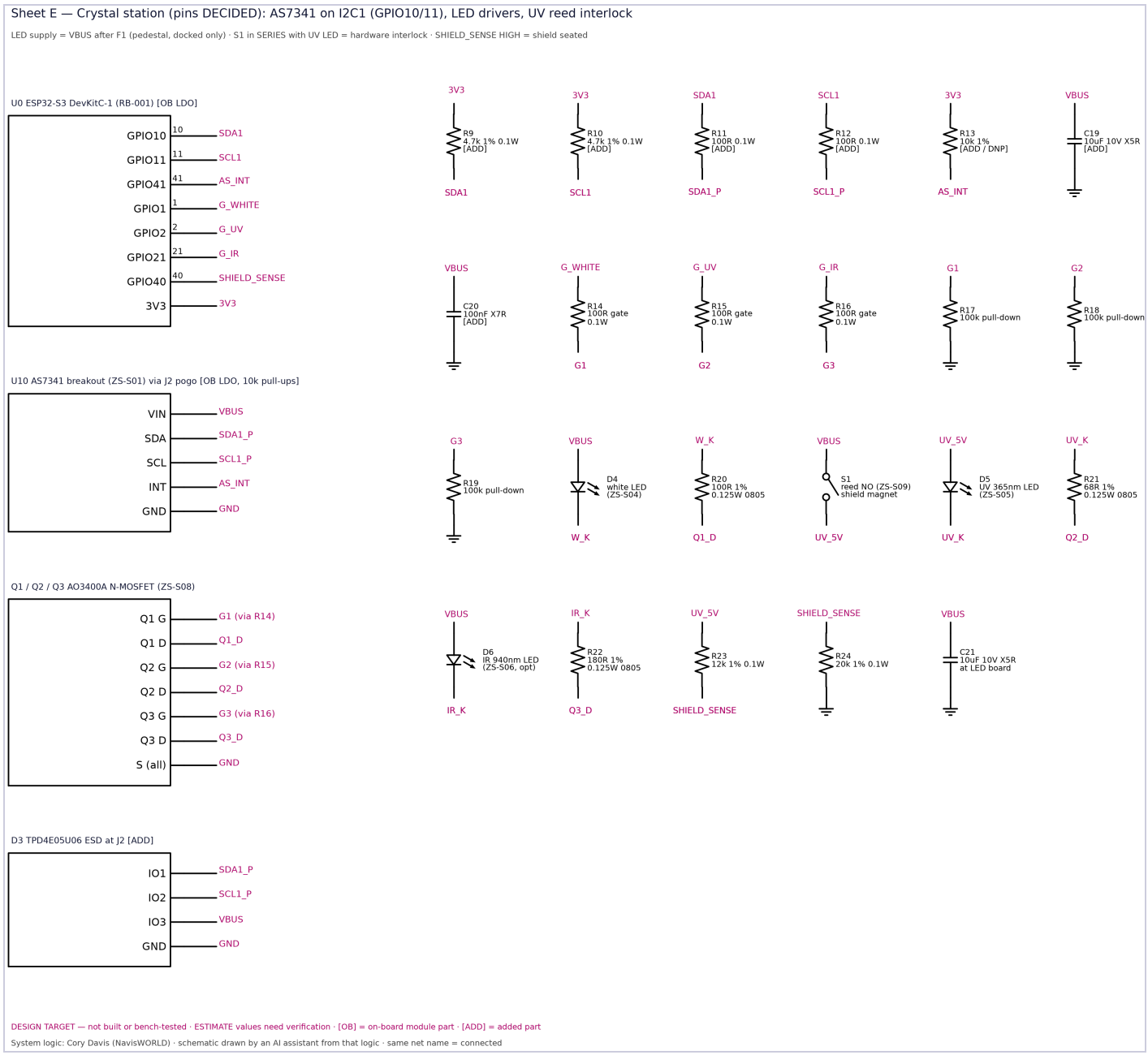
Parts added: PX Pogo receptacle 4-pin (ZS-S11), PX Reed switch (ZS-S09) — UV interlock, PX Station driver board, Q1-Q3 (ZS-S08), ZS-604 Driver bay cover

Fasteners: Driver board sits in the bay under ZS-604; cover: 2 × M2 × 6 PT. Reed: hot glue in its slot (glass — no pressure). Pogo receptacle: pocket at x 29…34, glue.

Do:

1. Build the driver board: Q1–Q3 AO3400A, R14–R16 100 Ω, R17–R19 100 kΩ, R23 12 kΩ / R24 20 kΩ, R9/R10 4.7 kΩ, R11/R12 100 Ω.
2. Mount J2 pogo receptacle + D3 + C19/C20.
3. Reed S1 in SERIES with the UV LED supply.

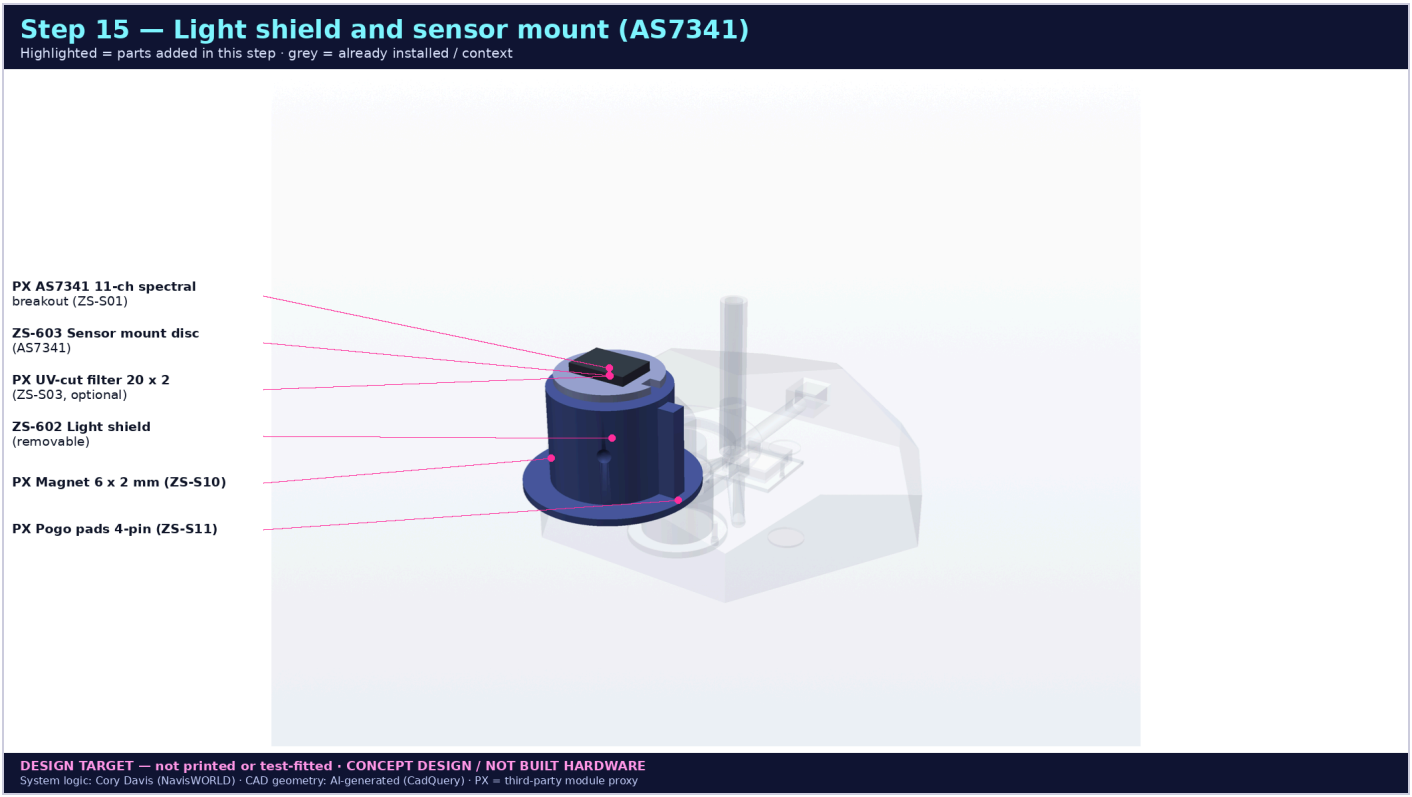
Wiring: Sheet E. Gates GPIO1 / GPIO2 / GPIO21; SHIELD_SENSE GPIO40; I2C1 GPIO10/11; INT GPIO41.



Wiring for this step: docs/schematics/E_crystal_station (pin labels = ESP32-S3 GPIO numbers)

> **CHECK before moving on:** Shield off: SHIELD_SENSE 0 V and UV_5V 0 V. Magnet held at the reed: SHIELD_SENSE ≈ 3.1 V.

Step 15 — Light shield and sensor mount (AS7341)



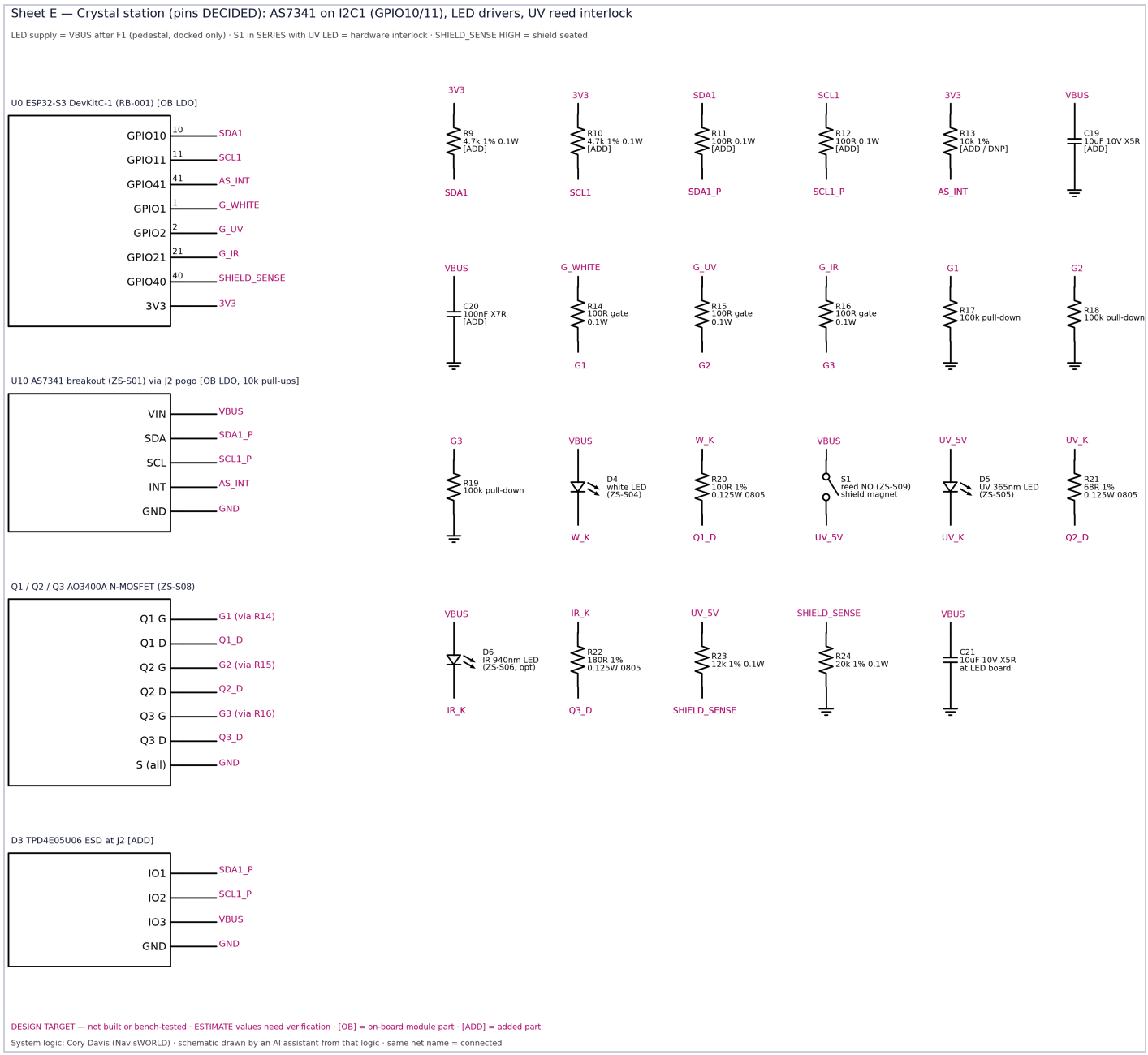
Parts added: ZS-602 Light shield (removable), ZS-603 Sensor mount disc (AS7341), PX AS7341 11-ch spectral breakout (ZS-S01), PX Pogo pads 4-pin (ZS-S11), PX Magnet 6 x 2 mm (ZS-S10), PX UV-cut filter 20 x 2 (ZS-S03, optional)

Fasteners: AS7341 board: 2 × M2 × 5 to ZS-603; ZS-603 glued into the shield top. Magnet Ø6 × 2: glued into the flange pocket above the reed.

Do:

1. Mount the AS7341 breakout in ZS-603, sensor facing down through the aperture.
2. Insert the optional UV-cut filter for fluorescence mode only.
3. Press the magnet into the shield flange above the reed position.

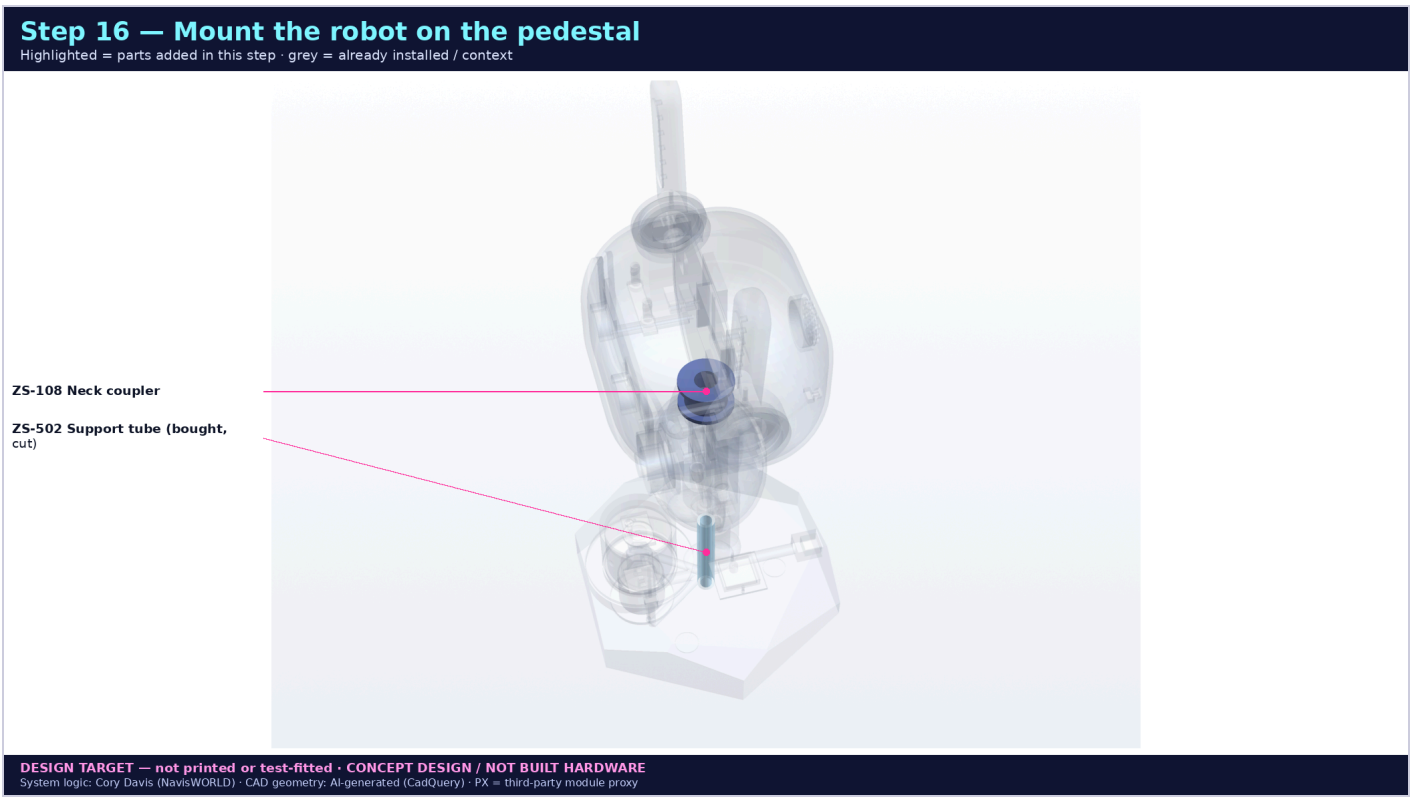
Wiring: Sheet E (pogo: VBUS, GND, SDA1_P, SCL1_P).



Wiring for this step: docs/schematics/E_crystal_station (pin labels = ESP32-S3 GPIO numbers)

> **CHECK before moving on:** Seat the shield: I2C scan on bus 1 shows 0x39 AND GPIO40 reads HIGH. Lift it: both go away.

Step 16 — Mount the robot on the pedestal



Parts added: ZS-108 Neck coupler, ZS-502 Support tube (bought, cut)

Fasteners: Robot slides onto the tube (slip fit, no screws). Optionally 1 drop of removable adhesive (ESTIMATE).

Do:

1. Join the tube harness to the torso harness.
2. Lower the robot onto the tube, front facing the station.
3. Final dress of wires; nothing touching the station window.

Wiring: All sheets. Final continuity check per the harness table.

> **CHECK before moving on:** Robot stands without wobble; total height 314 mm ±2 (DESIGN TARGET).

8. Electrical Specifications

DESIGN TARGET — not built or bench-tested. Values marked **ESTIMATE** are calculated from datasheet typicals, not measured. **[OB]** = part already on a bought module (do not add). **[ADD]** = part you add. **DNP** = do not populate unless the check says so. Designators match `docs/BOM_electrical.csv` and the schematics.

Changed 2026-11-09: LED brightness cap 10 % → **8 %** everywhere (firmware 20/255), decided by Robot X under Cory's delegation (docs/DESIGN_DECISIONS.md D1). Peak 5V_SYS budget 1.03 A → 0.97 A (ESTIMATE).

Pin map — DECIDED (Robot X under Cory's delegation, 2026-11-09)

All pins below are **DECIDED** (docs/DESIGN_DECISIONS.md D3). REPO = Cory's Reality Bridge Probe v0.2 pins, unchanged. The firmware (`firmware/zeref_spark_fw/include/spark_pins.h`, zs-0.4.0) uses exactly this table. Not yet checked on a powered board.

Signal	GPIO	Goes to	Status
I2C0 SDA / SCL	8 / 9	ICM-42688-P, BME280, BH1750, MAX30102	REPO
I2S0 BCLK / WS / SD	12 / 13 / 14	INMP441 mic	REPO
GNSS RX / TX	17 / 18	GNSS (unwired by default, `SPARK_GNSS_WIRED=0`)	REPO
LED data	4	74AHCT125 → 48 × WS2812B (cap 8 %)	DECIDED (approved by Cory)
I2S1 BCLK / LRC / DIN	15 / 16 / 7	MAX98357A (tones + streamed speech)	DECIDED (approved by Cory)
Amp SD (mute)	6	MAX98357A SD, LOW = muted, muted at boot	DECIDED (approved by Cory)
Battery sense	5 (ADC1_CH4)	100 k / 100 k divider, VBAT = 2 × Vadc	DECIDED
I2C1 SDA / SCL	10 / 11	AS7341 (0x39) via pogo, 100 kHz	DECIDED
AS7341 INT	41	AS7341 INT, open drain	DECIDED
White / UV 365 / IR 940 gate	1 / 2 / 21	AO3400A Q1 / Q2 / Q3	DECIDED
Shield sense	40	HIGH = shield seated (after reed S1)	DECIDED
USB D+ / D−	19 / 20	native USB CDC (host link)	DevKit

Voltage rails

Rail	Range	Source	Max load	Notes
VBUS_IN / VBUS	4.75–5.25 V	USB-C J1 → F1 (2 A hold)	≈1.87 A peak (ESTIMATE)	Needs a **5 V / 3 A** USB-C supply. A PC USB2 port (0.5 A) is NOT enough.
VBAT	3.0–4.2 V (3.7 V nom.)	BT1 protected LiPo	≈1.5 A to the boost at 1 A out (ESTIMATE)	Cell PCM cuts at ≈2.5–3.0 V
VS (load-share)	VBUS or VBAT	MCP73871 inside U1	—	internal to the PowerBoost
5V_SYS	5.2 V ±2%	U1 TPS61090 boost	≈1 A	Peak budget ≈0.97 A with the 8 % LED cap — tight, see power budget
3V3	3.3 V	DevKit on-board LDO [OB]	keep external load < 100 mA (ESTIMATE)	feeds I2C0 sensors, mic, GNSS
AS7341 internal	3.3 V / 1.8 V	breakout regulators [OB]	—	fed from VBUS through the pogo
UV_5V	0 V or VBUS	reed S1 (shield seated)	≈21 mA	hardware UV interlock

Power budget (ESTIMATE)

5V_SYS (boost output, ≈1 A limit)

Load	Typical	Peak	Basis
ESP32-S3 + Wi-Fi	0.10 A	0.35 A	datasheet TX burst
WS2812B quiescent (48 × ≈1 mA)	0.05 A	0.05 A	
WS2812B light, 8 % brightness cap (firmware 20/255)	0.08 A	0.23 A	48 × 60 mA × 8 %
MAX98357A + 8 Ω speaker	0.03 A	0.25 A	≈1 W peaks
Sensors + mic (via 3V3 LDO)	0.01 A	0.05 A	MAX30102 LED pulses
GNSS (optional)	0.03 A	0.04 A	acquisition
Total 5V_SYS	**≈0.30 A**	**≈0.97 A**	**tight vs ≈1 A** with the 8 % LED cap (DECIDED, D1); drop GNSS if brown-outs

VBUS (USB-C input)

Load	Typical	Peak
MCP73871 charger + load-share (input limit)	0.6 A	1.8 A
Station LEDs (one at a time, 20 mA)	0.00 A	0.02 A (0.06 A if all on)
AS7341 breakout	<0.01 A	<0.01 A
Total VBUS	**≈0.6 A**	**≈1.87 A**

Battery runtime (ESTIMATE): 1000 mAh × 3.7 V × 0.85 boost efficiency / (5.2 V × 0.30 A) ≈ **2.0 h typical**, ≈0.6 h at sustained peak.

Resistors (every one)

Ref	Sheet	Value	Pkg	Power	Tol.	OB/ADD	ESTIMATE?	Calculation / purpose
R1	A	100 kΩ	0603	0.1 W	±1%	ADD		divider top: VBAT → VBAT_SENSE
R2	A	100 kΩ	0603	0.1 W	±1%	ADD		Vadc = VBAT·R2/(R1+R2) = VBAT/2 → 1.50–2.10 V; I = 4.2/200k = 21 µA
R3	B	2.2 kΩ (DNP if breakouts' parallel pull-ups ≤ 3.3 kΩ)	0603	0.1 W	±1%	ADD / DNP		SDA0 pull-up; tr = 0.847·R·C = 0.847·2.2k·100pF = 186 ns ≤ 300 ns (400 kHz)
R4	B	2.2 kΩ (DNP rule as R3)	0603	0.1 W	±1%	ADD / DNP		SCL0 pull-up; sink current 3.3/2.2k = 1.5 mA ≤ 3 mA
R5	C	33 Ω	0603	0.1 W	±1%	ADD (optional)		I2S0 BCLK series damping (long pod run)
R6	C	33 Ω	0603	0.1 W	±1%	ADD (optional)		I2S0 WS series damping
R7	C	22 kΩ	0603	0.1 W	±1%	ADD	ESTIMATE	boot-mute pull-down: V = 5·22k/(1M+22k) = 0.108 V below 0.16 V shutdown threshold (assumes 1M [OB] pull-up — verify on your board)
R8	D	330 Ω (300–500 Ω range)	0603	0.1 W	±1%	ADD		damps ringing / protects first WS2812 input
R9	E	4.7 kΩ	0603	0.1 W	±1%	ADD		SDA1 pull-up; in parallel with [OB] 10k → 3.2k
R10	E	4.7 kΩ	0603	0.1 W	±1%	ADD		SCL1 pull-up
R11	E	100 Ω	0603	0.1 W	±1%	ADD		SDA1 series (hot-plug current limit)
R12	E	100 Ω	0603	0.1 W	±1%	ADD		SCL1 series

Ref	Sheet	Value	Pkg	Power	Tol.	OB/ADD	ESTIMATE?	Calculation / purpose
R13	E	10 kΩ (DNP if board has INT pull-up)	0603	0.1 W	±1%	ADD / DNP		AS_INT open-drain pull-up
R14	E	100 Ω	0603	0.1 W	±1%	ADD		Q1 gate series
R15	E	100 Ω	0603	0.1 W	±1%	ADD		Q2 gate series
R16	E	100 Ω	0603	0.1 W	±1%	ADD		Q3 gate series
R17	E	100 kΩ	0603	0.1 W	±1%	ADD		Q1 gate pull-down (LED OFF at boot / floating)
R18	E	100 kΩ	0603	0.1 W	±1%	ADD		Q2 gate pull-down
R19	E	100 kΩ	0603	0.1 W	±1%	ADD		Q3 gate pull-down
R20	E	100 Ω	0805	0.125 W	±1%	ADD	ESTIMATE	(5.0−3.0)/0.020 = 100 Ω → 20 mA; P = 0.02²·100 = 0.04 W
R21	E	68 Ω	0805	0.125 W	±1%	ADD	ESTIMATE	(5.0−3.6)/0.020 = 70 Ω → E12 68 Ω → 20.6 mA; P = 0.029 W
R22	E	180 Ω	0805	0.125 W	±1%	ADD	ESTIMATE	(5.0−1.3)/0.020 = 185 Ω → 180 Ω → 20.6 mA; P = 0.076 W
R23	E	12 kΩ	0603	0.1 W	±1%	ADD		SHIELD_SENSE = 5·20k/32k = 3.13 V (HIGH) when S1 closed; I = 0.16 mA
R24	E	20 kΩ	0603	0.1 W	±1%	ADD		pulls SHIELD_SENSE to 0 V when S1 open. If VBUS = 5.25 V → 3.28 V (≤ 3.6 V abs max OK)

LED current-limit resistors, worked: R = (V − Vf) / If, P = If² · R

LED	V supply	Vf (ESTIMATE)	If target	R calc	Chosen (E12)	Actual If	P in R
D4 white	5.0 V	3.0 V	20 mA	(5.0−3.0)/0.020 = 100 Ω	R20 100 Ω	20.0 mA	0.040 W
D5 UV 365 nm	5.0 V	3.6 V	20 mA	(5.0−3.6)/0.020 = 70 Ω	R21 68 Ω	20.6 mA	0.029 W
D6 IR 940 nm	5.0 V	1.3 V	20 mA	(5.0−1.3)/0.020 = 185 Ω	R22 180 Ω	20.6 mA	0.076 W

Use 0805 0.125 W parts (≥ 1.6× margin). Check your LEDs' real Vf at 20 mA and recompute; AO3400A drop (≈0.7 mV at 20 mA) is negligible.

Other resistor math

- **Battery sense (R1/R2 100 k / 100 k, C5 100 nF):** Vadc = VBAT × 100k / 200k → 1.50 V (3.0 V) … 2.10 V (4.2 V). Inside the ESP32-S3 ADC 11 dB range (≈0–2.9 V usable, ESTIMATE). Divider current 4.2 V / 200 kΩ = 21 µA (≈0.5 mAh/day). Firmware: VBAT = 2 × Vadc (calibrate with a multimeter once).
- **I2C0 pull-ups (R3/R4 2.2 kΩ):** rise time tr = 0.847 × R × Cbus = 0.847 × 2.2 kΩ × 100 pF ≈ 186 ns (I2C fast mode limit 300 ns). Most breakouts already carry 10 kΩ (or 4.7 kΩ) each — measure SDA→3V3 with power off: if ≤ 3.3 kΩ already, leave R3/R4 out (DNP).
- **I2C1 pull-ups (R9/R10 4.7 kΩ ‖ on-board 10 kΩ = 3.2 kΩ):** longer pogo + tube run (≈30 cm, ESTIMATE ≈150 pF): tr ≈ 0.847 × 3.2k × 150 pF ≈ 0.41 µs → run I2C1 at **100 kHz** (limit 1 µs). R11/R12 100 Ω series limit hot-plug current.
- **MOSFET gates:** R14–R16 100 Ω limit GPIO edge current into the gate; R17–R19 100 kΩ hold the LEDs OFF while the ESP32 boots (pins float).
- **Shield sense (R23 12 kΩ / R24 20 kΩ):** shield seated → S1 closed → SHIELD_SENSE = 5.0 × 20/(12+20) = 3.13 V (HIGH; 3.28 V at 5.25 V — below 3.6 V abs. max). Shield off → R24 pulls to 0 V (LOW). Current 0.16 mA.
- **Amp boot-mute (R7 22 kΩ):** with the breakout's 1 MΩ SD pull-up [OB] to VIN: V = 5 × 22k / 1.022M = 0.108 V < 0.16 V shutdown threshold → amp OFF until GPIO6 drives HIGH. ESTIMATE — verify your breakout's pull-up value.
- **NeoPixel data (R8 330 Ω):** in the 300–500 Ω range; sits at the first LED's DIN to damp ringing on the 3.3→5 V buffered line.

Capacitors (every one)

Ref	Sheet	Value	Type	Pkg	Voltage	Tol.	OB/ADD	Location / purpose
C1	A	10 µF	MLCC X5R	0805	10 V	±10%	ADD	Pedestal after F1 — VBUS local decoupling
C2	A	47 µF	Polymer / low-ESR electrolytic	radial 6.3 mm	10 V	±20%	ADD	Head, at U1 USB pad — absorbs cable inductance at charger input
C3	A	220 µF	Polymer / low-ESR electrolytic	radial 6.3 mm	10 V	±20%	ADD	At U1 5V out — boost output reservoir for LED + amp peaks
C4	A	100 nF	MLCC X7R	0603	16 V	±10%	ADD	At U1 5V out — HF decoupling
C6	A	10 µF	MLCC X5R	0805	10 V	±10%	ADD	At U0 5V pin — DevKit input decoupling
C5	A	100 nF	MLCC X7R	0603	16 V	±10%	ADD	At GPIO5 — ADC sample-hold reservoir; τ = 50k·100n = 5 ms
C7	B	100 nF at U3 IMU	MLCC X7R	0603	16 V	±10%	OB (usually) / ADD	U3 — add only if breakout lacks it
C8	B	100 nF at U4 BME280	MLCC X7R	0603	16 V	±10%	OB / ADD	U4
C9	B	100 nF at U5 BH1750	MLCC X7R	0603	16 V	±10%	OB / ADD	U5
C10	B	100 nF at U6 MAX30102	MLCC X7R	0603	16 V	±10%	OB / ADD	U6 — MAX30102 LED pulses need local cap
C11	B	10 µF at U7 GNSS	MLCC X5R	0805	10 V	±10%	ADD (optional)	U7 — GNSS acquisition current bursts
C12	C	100 µF	Aluminium electrolytic	radial 5 mm	10 V	±20%	ADD	At U9 VIN — class-D supply reservoir
C13	C	10 µF	MLCC X5R	0805	10 V	±10%	ADD	At U9 VIN
C14	D	100 nF	MLCC X7R	0603	16 V	±10%	ADD	At U2
C15	D	1000 µF	Aluminium electrolytic	radial 8 mm	10 V	±20%	ADD	Chest ring 5V/GND — LED bulk — inrush & surge protection
C16	D	100 µF	Aluminium electrolytic	radial 5 mm	10 V	±20%	ADD	Pod ring L
C17	D	100 µF	Aluminium electrolytic	radial 5 mm	10 V	±20%	ADD	Pod ring R
C18	D	100 µF	Aluminium electrolytic	radial 5 mm	10 V	±20%	ADD	Eye boards
C19	E	10 µF	MLCC X5R	0805	10 V	±10%	ADD	At J2 (pedestal) — pogo VIN bulk
C20	E	100 nF	MLCC X7R	0603	16 V	±10%	ADD	At J2
C21	E	10 µF	MLCC X5R	0805	10 V	±10%	ADD	LED board — LED switching decoupling

Rule: voltage rating ≥ 2× rail for MLCC (DC-bias derating), ≥ 1.6× for electrolytics/polymer. Watch polarity on C2, C3, C12, C15–C18.

Diodes, fuses, switches, ICs, connectors

Ref	Sheet	Part	Type	Pkg	Rating	OB/ADD	Note
J1	A	USB-C 16-pin receptacle breakout (5.1k CC pull-downs on board)	Connector	breakout	5 V / 3 A	ADD (module)	CC 5.1k = [OB]; only VBUS/GND/D+/D− used
D1	A	USBLC6-2SC6	ESD array	SOT-23-6	5.25 V working	ADD	on D+/D−; VBUS pin to VBUS

Ref	Sheet	Part	Type	Pkg	Rating	OB/ADD	Note
D2	A	SMBJ5.0A	TVS diode	SMB (DO-214AA)	5.0 V standoff, 600 W pk	ADD	clamps hot-plug spikes on VBUS
F1	A	PTC resettable fuse 2.0 A hold / ~4 A trip	Polyfuse	1812	6 V	ADD	VBUS peak ≈1.87 A (charger limit 1.8 A + station LEDs 60 mA)
U1	A	Adafruit PowerBoost 1000C (ZS-B02)	Module	PCB 36×23 mm	in 5 V, out 5.2 V ≈1 A	ADD (module)	MCP73871 + TPS61090; on-board caps, EN & LBO pull-ups [OB]
BT1	A	1S LiPo 503450 ≈1000 mAh, with PCM (ZS-B01)	Battery	50×34×5 mm	3.0–4.2 V	ADD	PCM (over-charge/discharge/short) is [OB] in the cell
F2	A	PTC 2.0 A hold	Polyfuse	1812	6 V	ADD (optional)	backup to cell PCM
SW1	A	SS12D00 SPDT slide switch	Switch	through-hole 2.54 mm	50 V / 0.5 A	ADD	EN to GND = OFF (switch closed = OFF). Charging still works when OFF.
U2	D	SN74AHCT1G125 (or 1 gate of 74AHCT125)	Logic buffer	SOT-23-5 / DIP-14	VCC 5 V	ADD	3.3 V → 5 V level shift (VIH = 2.0 V at 5 V)
LED1–48	D	WS2812B (100 nF per LED on ring PCBs)	Smart LED	5050	5 V, 60 mA max each	OB (ring modules)	48·60 mA·8% brightness cap = 0.23 A peak (cap 10→8 % on 2026-11-09
U10	E	AS7341 breakout (ZS-S01)	Sensor module	breakout	VIN 3.3–5 V (on-board LDO)	ADD (module)	I2C 0x39 fixed; on-board 10k pull-ups and 1.8 V/3.3 V regs [OB]
J2	E	4-pin magnetic pogo pair	Connector	2.54 mm pitch	2 A per pin (typ.)	ADD	VBUS, GND, SDA1_P, SCL1_P
D3	E	TPD4E05U06	ESD array	USON-10 / SOT	5.5 V working	ADD	ESD on exposed pogo pins
Q1	E	AO3400A (white)	N-MOSFET logic level	SOT-23	30 V, 5.7 A, Rds 33 mΩ @ 4.5 V	ADD	Vgs(th) ≤ 1.45 V → fully on at 3.3 V for 20 mA
Q2	E	AO3400A (UV)	N-MOSFET	SOT-23	30 V	ADD	
Q3	E	AO3400A (IR, optional)	N-MOSFET	SOT-23	30 V	ADD	gate on GPIO21 (v0.3)
D4	E	White LED 3 mm / 3528 (ZS-S04)	LED	3 mm	Vf ≈3.0 V @ 20 mA	ADD	
D5	E	365 nm UV-A LED (ZS-S05)	LED	3535	Vf ≈3.6 V @ 20 mA	ADD	check IEC 62471 risk group
D6	E	940 nm IR LED (ZS-S06, optional)	LED	3 mm	Vf ≈1.3 V @ 20 mA	ADD (optional)	
S1	E	Glass reed switch NO ~12 mm (ZS-S09)	Reed switch	axial	≥100 V / 0.5 A	ADD	in SERIES with UV LED supply: VBUS → S1 → UV_5V (hardware interlock)

Flyback diodes: none needed — there are no motors, relays or other inductive loads; the speaker is driven by a class-D bridge with its own protection. **ESD/TVS:** D1 (USB data), D2 (VBUS surge), D3 (exposed pogo pins). **Fuses:** F1 on VBUS, F2 (optional) on the battery lead, plus the cell's own PCM.

Connectors and pitches

Where	Connector	Pitch	Pins / note
I2C breakouts (U3–U6, U10 side)	JST-SH (STEMMA QT / Qwiic)	1.0 mm	4: GND, 3V3, SDA, SCL
Battery BT1 → U1	JST-PH	2.0 mm	2 (check polarity — no standard!)

Where	Connector	Pitch	Pins / note
LED rings / eye boards	JST-PH	2.0 mm	3: 5V, DIN/DOUT, GND
Head ↔ torso, torso ↔ tube harness joints	JST-PH (or JST-GH for locking)	2.0 mm (1.25 mm GH)	split into ≤ 6-pin groups
DevKit headers	Dupont female	2.54 mm	or solder directly
Station shield ↔ pedestal J2	4-pin magnetic pogo	2.54 mm	VBUS, GND, SDA1_P, SCL1_P
Pedestal power/data in J1	USB-C receptacle breakout	—	5.1 kΩ CC [OB]
Speaker	JST-PH or solder	2.0 mm	2

Wire gauge per run

Run	Current (peak)	Wire	Length (ESTIMATE)
VBUS / GND: J1 → tube → torso → head (U1)	≤ 1.87 A	2 × 26 AWG silicone in parallel each (or 1 × 22 AWG if it fits the tube)	≈30 cm
5V_SYS / GND: U1 → torso → LEDs, DevKit	≤ 1.03 A	2 × 26 AWG in parallel	≈15 cm
BAT: BT1 → U1	≤ 1.5 A	24 AWG	≤ 75 mm
LED ring 5V/GND branches	≤ 0.15 A each	26 AWG	≤ 15 cm
Speaker	≤ 0.4 A	26 AWG	≤ 10 cm
All signals (I2C, I2S, gates, sense, LED data)	< 10 mA	30 AWG silicone	—
USB D+ / D−	data	30 AWG twisted pair, ≈ 1 twist/cm	keep < 30 cm
Station LED supply (VBUS in pedestal)	≤ 60 mA	28–30 AWG	≤ 10 cm

26 AWG silicone ≈ 0.134 Ω/m: 30 cm at 1.87 A on a single wire ≈ 75 mV drop each way — doubled wires halve it.

Harness: tube (pedestal ↔ torso) — 11 conductors (ESTIMATE fit in 8 mm ID)

#	Signal	Gauge
1	VBUS	26 (×2 if room)
2	GND	26 (×2 if room)
3	USB D+	30 twisted
4	USB D−	30 twisted
5	SDA1	30
6	SCL1	30
7	G_WHITE (GPIO1)	30
8	G_UV (GPIO2)	30
9	G_IR (GPIO21)	30
10	AS_INT (GPIO41)	30
11	SHIELD_SENSE (GPIO40)	30

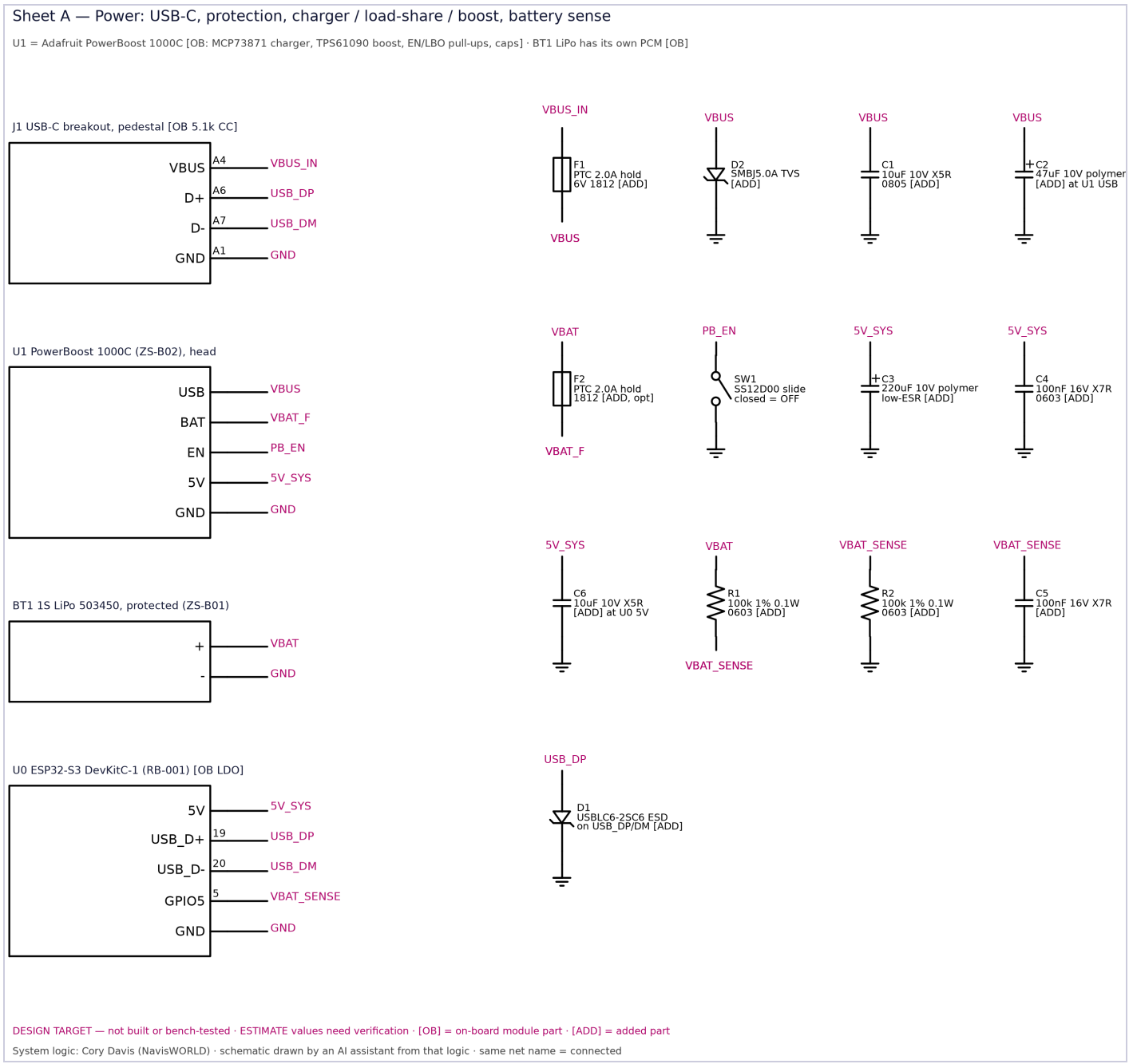
Per-module specifications

Module	Supply	Current	I2C address	Interface / pins	Datasheet (verify revision)
U0 ESP32-S3 DevKitC-1 (RB-001)	5 V in (on-board LDO → 3V3)	≈100 mA typ, ≈350 mA Wi-Fi TX peak	—	USB (GPIO19/20), GPIO	Espressif ESP32-S3 datasheet · DevKitC-1 guide
U3 ICM-42688-P (RB-002)	3.3 V	≈1 mA	0x68 (0x69 if AD0 high)	I2C0	TDK ICM-42688-P
U4 BME280 (RB-004)	3.3 V	≈1 mA peak, µA avg	0x76 (0x77)	I2C0	Bosch BME280
U5 BH1750 (RB-003)	3.3 V	0.12–0.19 mA	0x23 (ADDR low)	I2C0	ROHM BH1750FVI
U6 MAX30102 (RB-005, opt)	3.3 V logic (LED 5 V on most breakouts)	≈0.6 mA + LED pulses up to 50 mA	0x57	I2C0	ADI MAX30102
U7 NEO-M9N GNSS (RB-007, opt)	3.3 V	≈30–40 mA acquiring	—	UART GPIO17/18	u-blox NEO-M9N
U8 INMP441 (RB-006)	3.3 V	1.4 mA	—	I2S0 GPIO12/13/14	TDK INMP441
U9 MAX98357A (ZS-E05)	5 V (2.5–5.5 V)	≈3 mA idle, ≈250 mA at ~1 W/8 Ω (ESTIMATE)	—	I2S1 GPIO15/16/7, SD GPIO6	ADI MAX98357A
U10 AS7341 breakout (ZS-S01)	VIN 3.3–5 V (on-board regs)	≈0.3 mA active	0x39 (fixed)	I2C1 GPIO10/11, INT GPIO41	ams-OSRAM AS7341 DS000504
LED1–48 WS2812B	5 V (VIH ≥ 0.7·VDD → needs U2)	≈1 mA idle, 60 mA max each	—	1-wire 800 kHz, GPIO4	WS2812B (Adafruit PDF)
U2 SN74AHCT1G125	5 V	< 0.1 mA	—	buffer	TI SN74AHCT1G125
U1 PowerBoost 1000C	in 5 V USB / 3.7 V LiPo; out 5.2 V	≈1 A out (TPS61090), charge 1 A	—	EN, LBO pins	Adafruit guide · MCP73871 · TPS61090
Q1–Q3 AO3400A	Vgs(th) 0.65–1.45 V	5.7 A max (20 mA used)	—	GPIO1/2/21	AOS AO3400A
D1 USBLC6-2SC6 · D3 TPD4E05U06 · D2 SMBJ5.0A	5 V lines	—	—	protection	ST USBLC6-2 · TI TPD4E05U06 · Littelfuse SMBJ

I2C address map: **bus 0** 0x23, 0x57, 0x68, 0x76 — no conflicts; **bus 1** 0x39 alone. All datasheet links were correct when written; always check the current revision.

Schematics

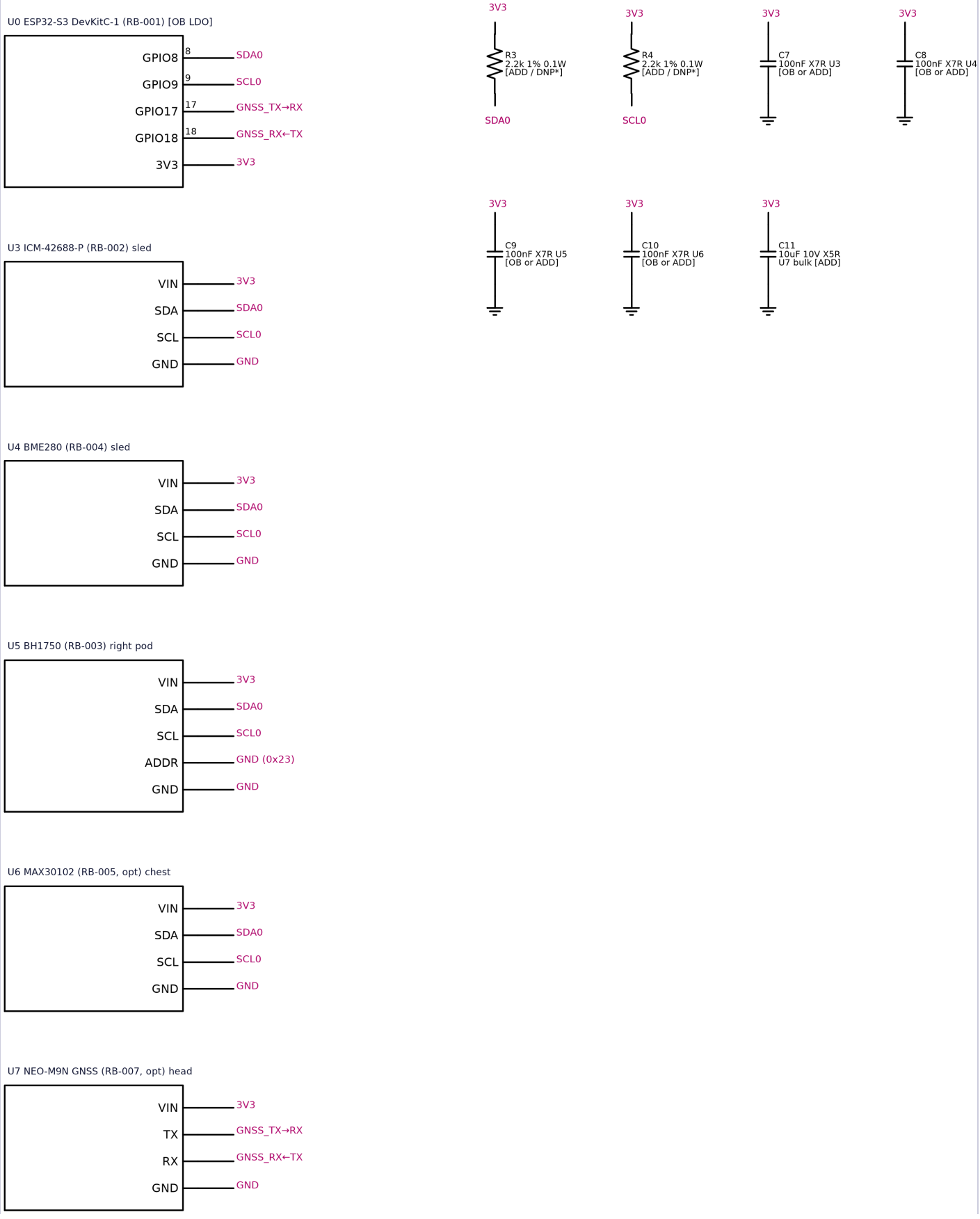
Drawn in net-label style: **same net name = connected**. [OB] on-board, [ADD] added.



Sheet A — Power (SVG: docs/schematics/A_power.svg)

Sheet B — I2C0 sensors: GPIO8 SDA0 / GPIO9 SCL0 (REPO pins), 400 kHz

BH1750 0x23 · MAX30102 0x57 · ICM-42688-P 0x68 · BME280 0x76 — no conflicts (AS7341 0x39 is on I2C1). GNSS on UART GPIO17/18 (REPO).



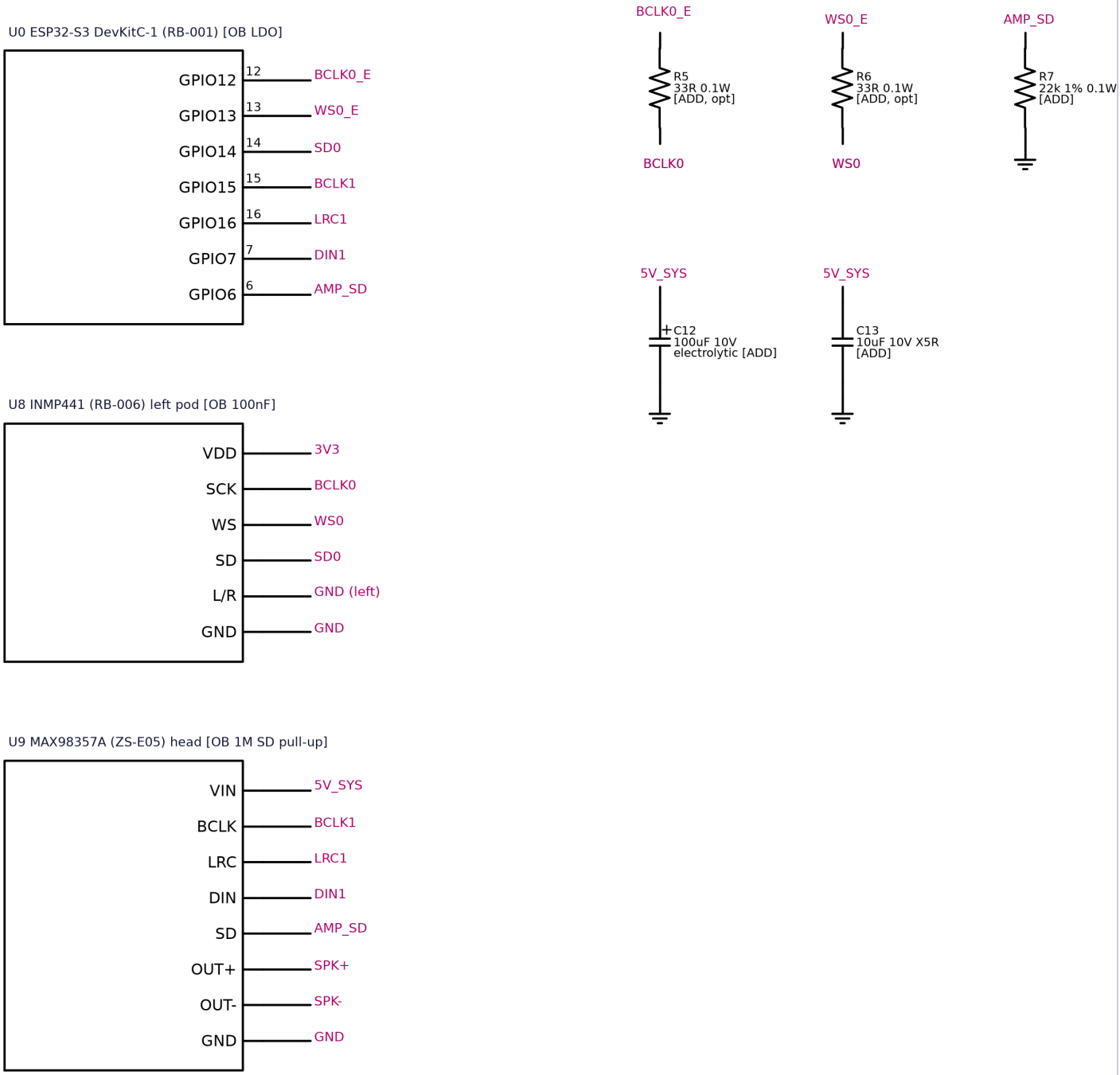
DESIGN TARGET — not built or bench-tested · ESTIMATE values need verification · [OB] = on-board module part · [ADD] = added part

System logic: Cory Davis (NavisWORLD) · schematic drawn by an AI assistant from that logic · same net name = connected

Sheet B — I2C0 sensors + GNSS (SVG: docs/schematics/B_i2c0_sensors.svg)

Sheet C — Audio: INMP441 mic on I2S0 (REPO 12/13/14) · MAX98357A on I2S1 (DECIDED 15/16/7, SD = GPIO6)

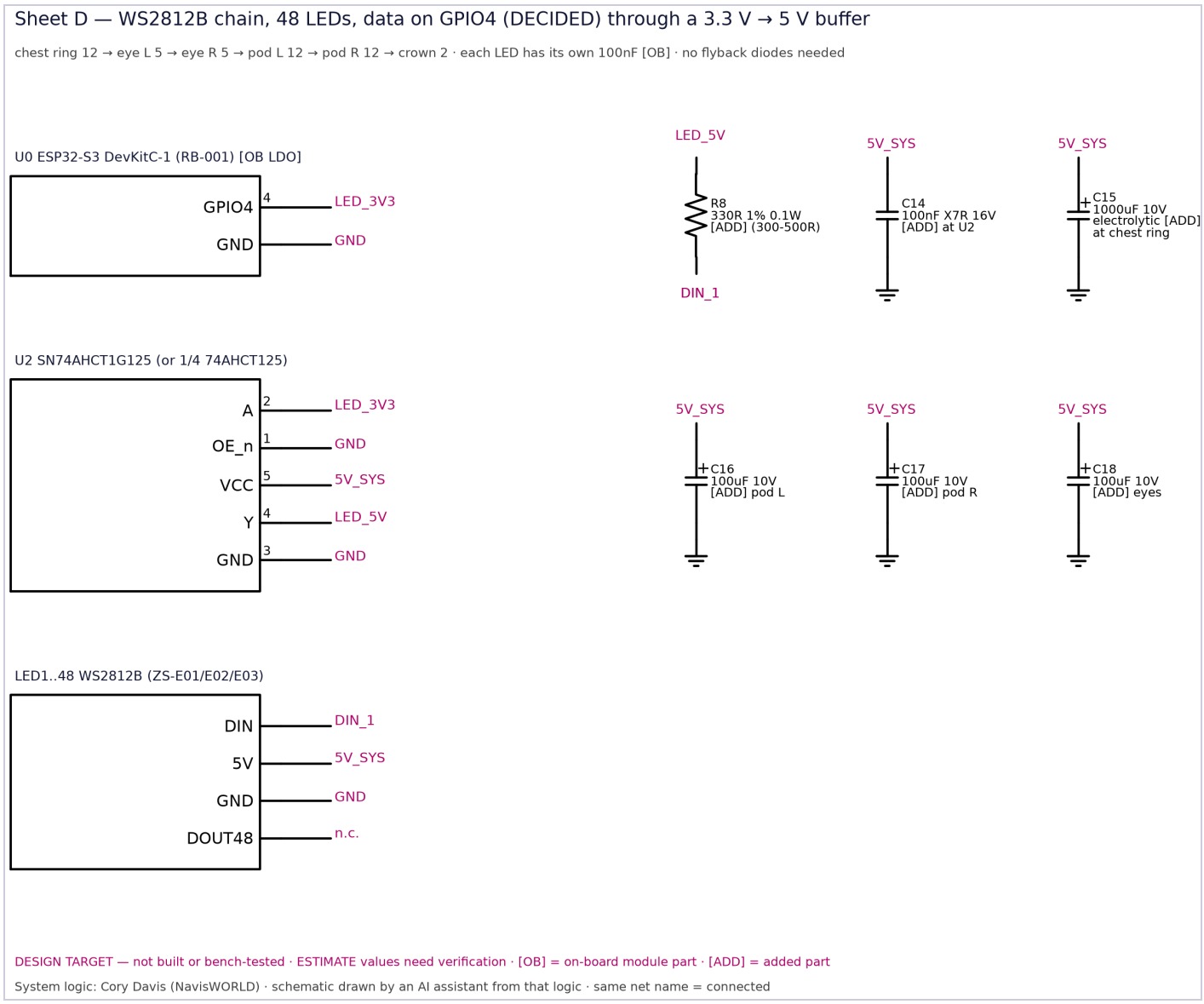
R7 holds the amp in shutdown at boot (see calc in guide). SPK1 = 28 mm 8 ohm ~1 W (ZS-E06): firmware volume cap.



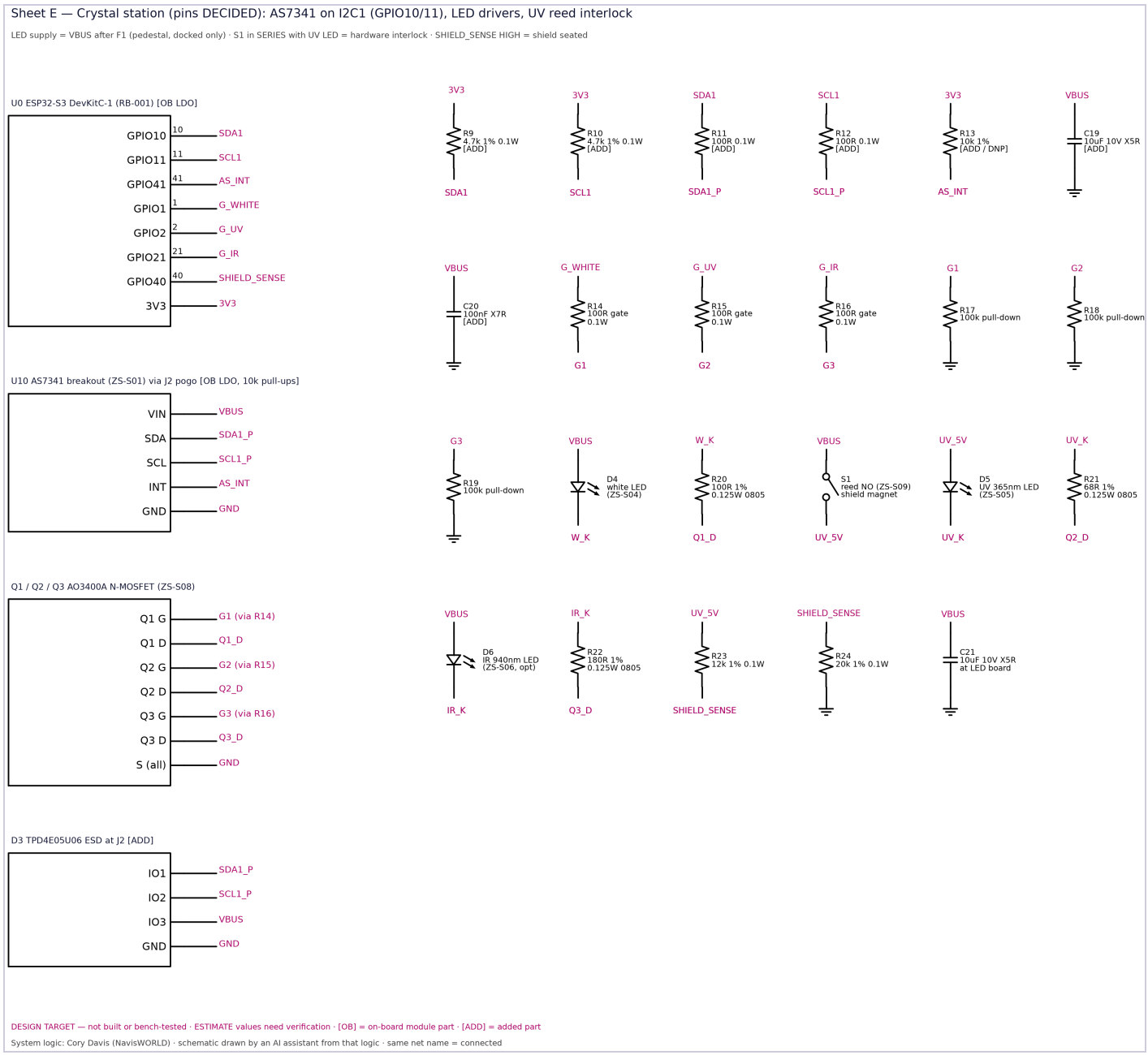
DESIGN TARGET — not built or bench-tested · ESTIMATE values need verification · [OB] = on-board module part · [ADD] = added part

System logic: Cory Davis (NavisWORLD) · schematic drawn by an AI assistant from that logic · same net name = connected

Sheet C — Audio (SVG: docs/schematics/C_audio.svg)



Sheet D — LED chain (SVG: docs/schematics/D_led_chain.svg)



Sheet E — Crystal station (SVG: docs/schematics/E_crystal_station.svg)

9. Stage gates 0–3 (PROPOSED — reconstructed)

> The original brief's Stage 0–3 text was not available to this build; these gates are a **reconstruction** from the system logic and must be confirmed by Cory.

Stage	Build state	Gate (all must pass)
0 — bare DevKit	ESP32-S3 on USB only, no wiring	Firmware flashes; native USB prints the boot line + self-test JSON; host `spark_brain` receives frames (Wi-Fi build only: softAP visible when SPARK_AP_PASS is set, WebSocket :81 streams JSON)
1 — sensor bench harness	U3–U8 on a breadboard / JST-SH chain	I2C0 scan = 0x23, 0x57, 0x68, 0x76 (present ones); every sensor value plausible and labelled MEASURED; mic RMS rises when you speak; GNSS NMEA (if fitted)
2 — power + LEDs + audio in the body	Steps 1–11	SW1 OFF → 5V_SYS < 0.3 V; ON → 5.1–5.25 V; battery reading within 0.05 V of a meter; 48 LEDs light in order at ≤ 8 % (firmware cap); amp silent at boot, tone plays when unmuted; no brown-out during LED + audio + Wi-Fi together (log min 5V_SYS)
3 — full assembly + station	Steps 12–16	Shield interlock: shield off → UV_5V = 0 V and SHIELD_SENSE LOW; shield on → 0x39 on I2C1 + HIGH; dark/white calibration passes repeatability (§11); robot height 314 mm ±2

10. Firmware flash and bring-up tests

Spark firmware **zs-0.4.0** is in this package at `firmware/zeref_spark_fw/` (PlatformIO, Arduino framework). It is built on Cory's Reality Bridge Probe v0.2, which is kept unchanged in `firmware/reality_bridge_probe_v0.2_original/` (his repo is not edited). It covers every DECIDED pin: LEDs (GPIO4, cap 8 %), amp I2S1 15/16/7 + mute 6 (tones and streamed speech), battery GPIO5, station I2C1 10/11 + INT 41, gates 1/2/21, shield sense 40, UV guard. Compile status: USB build and Wi-Fi build both **compile**; **never run on a board yet**.

1. Install VS Code + PlatformIO. Open `firmware/zeref_spark_fw`.
2. Connect the DevKit's **USB** (native) port — the firmware talks over native USB CDC (needed for speech audio). If upload fails: hold BOOT, tap RESET, release BOOT.
3. `pio run -e zeref-spark -t upload` then `pio device monitor`. Expect the line `Zeref Spark zs-0.4.0 (Reality Bridge Probe 0.2.0 base)` and a self-test JSON.
4. On the host: `cd host && python -m spark_brain serial /dev/ttyACM0` (the brain lives there — the robot is the nervous system). Optional Wi-Fi: build with `SPARK_AP_PASS='8+ chars' pio run -e zeref-spark-wifi` (Wi-Fi accepts speech lines only; UV/SCAN stay USB-only).

Test	How	Pass
I2C0 scan	scanner sketch / firmware self-test	expected addresses only
I2C1 scan	`Wire1.begin(10, 11, 100000)`	0x39 only when shield seated
Battery	read GPIO5 ×2, compare to meter	±0.05 V
LEDs	chain test at 4 % then 8 % (`BRIGHT 10`, then `BRIGHT 20`)	48 LEDs, correct order: chest → eyes → pods → crown
Amp mute	boot, measure SPK1	silent until GPIO6 HIGH
Speech	host: `python -m spark_brain serial PORT --speak --say 'hello'` after sending `MUTE 0`	clear speech (local piper TTS, or labelled stub tones); amp mutes about 60 ms after the end; `speech` frames show no underruns
Gates	toggle GPIO1/2/21 with shield seated	each LED alone; UV never with shield off
Shield sense	lift / seat	GPIO40 LOW / HIGH
Brown-out	LEDs 8 % + speech audio + Wi-Fi TX for 60 s	no reset; 5V_SYS ≥ 4.9 V

Firmware honesty labels: raw counts **MEASURED**, ratios **DERIVED**, any classification **ESTIMATE** (with confidence). No metaphysical claims.

11. Crystal station calibration

Fix the settings first and record them with every reading: gain (start at 64×), ATIME/ASTEP (start at ~50 ms), LED warm-up 2 s, and a temperature log (BME280 / internal). 1. **Warm-up**: robot on for ≥ 10 min, shield on. 2. **Dark reference D**: ZS-605 dark plug in the cup, shield on, **all LEDs off**. Take 10 readings and store the mean and standard deviation for each channel. This is sensor offset plus any ambient leak. 3. **Stray-light check**: dark plug still in, each LED on in turn. The counts must stay within ~3 standard deviations of D. If they don't, light is leaking around the plug or the shield (fix with paint or tape before going further). 4. **Ambient leak check**: empty cup, shield on, LEDs off, room lights switched on and off. Any change of more than ~3 standard deviations means the shield or groove leaks. 5. **White reference W** (per LED mode): PTFE disc ZS-S12 in the cup, shield on, LED on. Take 10 readings. Adjust gain so the highest channel is 30–80 % of full scale (no saturation; full scale is set by ATIME/ASTEP, max 65535). Store W for each mode. 6. **Empty-cup reference E** (optional): window only, for absolute transmission. 7. **Crystal reading S**: same mode, gain and timing. Take 5 readings and keep the raw counts (**MEASURED**). 8. **Derived values**: `T_ch = (S − D) / (W − D)` for transmission vs PTFE. In fluorescence mode (UV on, UV-cut filter in), `F_ch = (S − D) / (W_white − D)` is a **relative** emission index only. Both are **DERIVED**. 9. **Repeatability gate**: remove and re-seat the crystal 3 times. If any T_ch changes by more than ~5 %, flag the reading "unrepeatable" (placement, chips and surface dirt all matter). 10. Any "looks like amethyst / clear quartz / ..." label is an **ESTIMATE** from nearest-match against samples Cory has labelled himself. Show it with its confidence and the raw data, and never call it MEASURED. No energy or metaphysical claims are made.

12. Troubleshooting

Symptom	Likely cause	Fix
Nothing powers with SW1 ON	EN shorted / switch wired inverted; cell PCM tripped	check SW1 (closed = OFF); plug USB to reset PCM
Resets when LEDs flash or audio plays	5V_SYS peak ≈ 1 A limit; missing C3/C15	cap LEDs at 8 %, lower volume, check bulk caps, check 2× 26 AWG power wires
First LED flickers / random colours	no level shift, R8 missing, GND not shared	fit U2 + R8, run GND alongside data
I2C0 device missing	wrong address jumper, swapped SDA/SCL, no pull-ups	scan, check ADDR pins, measure pull-up resistance
I2C bus hangs	too-strong pull-ups stacked (< 1 kΩ)	remove R3/R4 or breakout pull-ups
AS7341 not found	shield not seated; pogo dirty; I2C1 too fast	reseat, clean pads, run at 100 kHz
UV never lights	reed open (magnet misplaced) — this is the interlock working	check SHIELD_SENSE ≈ 3.1 V with shield on
UV lights with shield OFF	**STOP** — interlock wiring wrong	power off; S1 must be in series with the UV supply
Station readings drift	warm-up, ambient leak, temperature	10 min warm-up, repaint shield inside, log temperature
Hum / hiss in speaker	ground loop / I2S edges	shorten wires, fit R5/R6, star ground at U1
Battery % jumps	ADC noise / no C5	fit C5, average 16 samples, calibrate
Brown-out from PC USB	PC port 0.5 A	use a 5 V / 3 A USB-C supply

Known gaps (honest list)

- Nothing has been printed, test-fitted, powered or measured. All ESTIMATE values need bench confirmation.
- PX proxies are simplified envelopes — measure your boards.
- Stage 0–3 gates and the brief are reconstructions.
- All pins are DECIDED (Robot X under Cory's delegation, 2026-11-09), incl. IR gate 21 and shield sense 40 (HIGH = seated) — still unverified on a powered board.
- 5V_SYS peak ≈ 0.97 A vs ≈ 1 A boost limit with the 8 % LED cap — tight.
- Firmware zs-0.4.0 (LEDs, amp + streamed speech, battery, station, UV guard) compiles (USB and Wi-Fi builds) but has never run on a board.
- AS7341 gains, lux map, finger threshold and battery calibration are ESTIMATE defaults until checked on hardware.